Default risk surprises and banks' loan pricing

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ABSTRACT

Using one million loans matched to firms' public credit ratings, I show that interest rate differentials unexplained by observables at issuance contain information about future changes in firms' default probability. On average, firms whose default probability deteriorates by 1 p.p. one year after loan issuance were charged an otherwise unexplained 0.04 p.p. rate premium at issuance; the premium doubles for younger and smaller firms and strengthens in crises. Information that banks know at loan issuance arises as a potential driver of interest rate dispersion and of the positive relationship between rates and default risk. Yet, it is not necessary: the same pattern can emerge in a market with search frictions. I bring these forces together in a search model of loan pricing with asymmetric information in which banks see an imperfect public signal on firms' risk types. In equilibrium, the positive rate-risk relationship can arise either when the signal is sufficiently precise to separate borrowers, or, even when it is uninformative, from search frictions that lead banks to randomize between pooling borrowers and inducing self-selection through pricing. I use a calibrated version of the model to rationalize the empirical findings. I find that banks are informed and separate borrowers in good credit rated markets, and they are uninformed and pool borrowers in bad credit rated ones.

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1 Introduction

Credit markets exhibit substantial interest rate variation across firms and banks. Most of this variation is explained by macroeconomic conditions and observables at loan issuance, including observable credit risk, contract characteristics and bank-firm relationships - see Altavilla et al. (2024). However, interest rate differentials persist across similar firms and loan products; this fact has been documented in different settings and countries: Portugal (Ferreira and Kozeniauskas (2026)), the US (Amiti et al. (2026)), Italy (Bocola et al. (2026)), Brazil (Cavalcanti et al. (2023)). These differentials translate into economically meaningful differences in borrowing costs. Using French data, I estimate that firms facing an unexplained interest rate increase of one standard deviation pay rates approximately 30% higher than the average for their credit rating class. For firms, particularly those that rely heavily on bank financing, such differences can have persistent consequences for investment and growth. In this paper, I show that both imperfect competition among banks and the information about borrowers embedded in rates are necessary to understand unexplained rate differentials.

I start by documenting new empirical facts: *interest rate differentials across similar firms borrowing similar loan products from the same bank contain information on future changes in firms' default risk*. In addition, this predictive content is heterogeneous across firms and contracts: it doubles for firms in ex-ante safer rating classes, for younger and smaller firms, and at longer maturities. I measure future changes in firms' default risk using public rating scores assigned to firms by Banque de France, which, as I will show, are a sufficient statistic for creditworthiness. The novelty of the approach is that I map firms' rating revisions after loan issuance directly into changes in firms' default probability using historical default frequencies in each rating class. Looking at revisions delivers more variation than using actual defaults only, and compared to banks' internal ratings that report the subjective expectation of the bank, public ratings are a realized outcome: they capture fundamental firm information, and are not biased by banks' regulatory considerations and capital requirements.

To tease out the forces at play in the data, I develop a model of loan pricing with search frictions, which generate imperfect competition between banks, and asymmetric information, in which banks see an imperfect public signal on firms' risk types. The signal allows banks to sort borrowers, while search limits firms' ability to arbitrage away differences in borrowing costs. In equilibrium, the positive rate-risk relationship can arise either when the signal is sufficiently precise to separate borrowers, or, even without information, from search frictions that lead banks to randomize between pooling borrowers and inducing self-selection through pricing. By calibrating the model to the data, I show that *both information and search are necessary to account for the empirical patterns*. I find that,

in safer markets, banks are informed and separate borrowers; in riskier ones, they are uninformed and pool borrowers. This suggests that the riskiness of worse rating classes is already well summarized by the rating.

I start by merging granular administrative datasets on business loans, bank-firm relationships and public credit ratings assigned by Banque de France, spanning from 2006 to 2024. For each observed loan, this dataset combines all firm-contract information available at issuance, as well as the evolution of firms' ratings in subsequent years. First, I measure the unexplained component of rates by regressing the loan rate on observables at issuance; I show that the standard deviation of rate residuals is substantial, around 1 percentage point, and varies over time and across firm characteristics; I use standard measures of concentration, such as the HHI, to show that in less competitive markets unexplained rate dispersion is lower.

I exploit the evolution of each firm's default probability after issuance as a proxy for the firm's true type at issuance. In particular, I use Banque de France public rating data: I compute the historical three-year default frequency in each rating class and I use it to map each rating category to a measure of default probability; then, whenever a firm is upgraded/downgraded to a new rating category, I attribute the corresponding change in default probability as the difference between the new and the old rating class.

Using public credit ratings has several advantages. First, Banque de France's ratings are a direct measure of firms' creditworthiness, based on balance-sheet data, firms' history and qualitative data (interviews); consequently, to assess firms' risk outcomes, they are more accurate than other proxies, for example ex-post returns ratios, or growth outcomes. Second, firms are reviewed every year and I observe multiple firm-level rating transitions over my sample; consequently, this generates much more variation than using actual defaults only, which are generally less frequent in the data. Finally, this is a realized outcome rather than the reported subjective expectation of the bank. When available, other studies include banks' internal ratings to explain interest rates (for example, Amiti et al. (2026)), but typically they do not test whether they effectively predict future default. This is an important validation exercise, as banks' internal assessments may be biased by regulatory considerations and capital requirements.

I regress the loan rate on later changes in the firm's default probability; controlling for a wide set of observables at loan issuance ensures that predictable changes in firms' default probability are already absorbed, and that the regression coefficient will capture the surprises. The coefficients are positive and significant, stronger in the first year after loan issuance and fading at longer horizons: on average, a firm pays 0.04pp more at issuance if default probability is revised up by 1pp one year later, 0.02pp more if the revision comes three years later, and the effect disappears beyond seven years. Moreover, the information content of interest rates is heterogeneous across market segments: it is

positive and significant for firms in ex-ante safer rating classes and absent in riskier ones. Also, it almost doubles for firms aged 0–4 years relative to firms aged 10 years or more, it is higher for firms with fewer than 10 employees and at longer maturities, and it peaks in 2009–2012. These coefficients likely underestimate the effect: they are measured only on firms whose rating is revised, which is only a fraction of the firms that banks identified as riskier. This overall evidence suggests that the correlation is stronger when there is more to learn about the borrower.

I test whether paying a relatively higher rate can itself induce a rating downgrade, but I exclude this mechanism by comparing firms borrowing at fixed versus variable rates and by showing that those borrowing at a variable rate are not more likely to be downgraded when market reference rates increase. Specifically, I test whether firms borrowing at variable rates are more likely to be downgraded after three years when the reference rates increase relative to firms borrowing at fixed rates. I run the test on the entire sample, and also restricting to firms borrowing at the end of 2021, for which the rate hike of 2022 was unexpected. I find no evidence that firms facing increased debt-servicing costs are more likely to be downgraded, thus I rule out this mechanism.

Overall, this evidence suggests that banks are anticipating firms' outcomes: information that banks know at loan issuance is a driver of interest rate dispersion and of the positive relationship between rates and risk. However, from a theoretical perspective, information is not necessary to deliver this pattern. I show this in a model of loan pricing with imperfect competition, modelled via Burdett and Judd (1983) search frictions, and asymmetric information, in which banks observe an imperfect public signal on firms' risk type. Specifically, banks and firms match in the credit market: a fraction of firms matches with two banks and can compare their offers, the remaining fraction with one bank only; moreover, firms can be of two types differing in risk. Both search frictions and the signal shape the equilibrium. Search frictions deliver dispersion even among identical types: banks compete for borrowers, but search frictions prevent them from perfectly undercutting each other, and a mixed-strategy equilibrium arises. When there are two types, banks choose either to pool safe and risky borrowers at low rates, or to induce self-selection of risky types by offering high rates only, depending on the signal received and on the degree of adverse selection; when the signal is informative enough, banks sort types and charge on average lower rates to safer types.

In equilibrium, the positive rate-risk relationship can arise in two cases: either when the signal is sufficiently precise to separate borrowers, or, even without information, when search frictions lead banks to randomize between pooling borrowers and inducing self-selection through pricing. Information also has implications for surplus sharing: more information increases the surplus of banks and safe firms relative to riskier firms. In fact, when adverse selection is mild and in the absence of information, both good types

and banks cross-subsidize bad types in the pooling equilibrium; when adverse selection is severe, banks offer only high rates and safe firms may be pushed out of the market altogether.

I run a calibration exercise to assess which of the equilibria predicted by the model is consistent with the empirical evidence. I repeat the calibration separately in each Banque de France rating class: the rating is a summary statistic of creditworthiness, so firms in the same rating category are observationally equivalent in terms of risk. I calibrate other parameters from the data, separately in each rating class: the maximum rate residual observed as the reservation rate of bad types; the share of firms improving relative to those worsening as the share of good types; the default probabilities as the average default probability of worsening firms for bad types, and of improving firms for good types. I pin down three parameters: the share of firms observing two offers π, which governs the competitiveness of the market; the signal precision q; the reservation rate of good types, which cannot be calibrated from the data, but is important to assess the degree of adverse selection. I target three data moments: the mean and the total variance of rate residuals, and the correlation between rates and riskiness. I find that the parameter for competition is stable and slightly decreasing as I move to riskier rating classes. On the contrary, the parameter for signal precision varies considerably. It is almost one in ex-ante safer classes, consistent with a separating equilibrium. It is almost 0.5, that is, completely uninformative, in ex-ante riskier rating classes, and with lower adverse selection within the class, predicting that banks are little informed and pooling. I use the calibrated parameters to simulate how the surplus is shared across banks and heterogeneous types in different classes.

I then depart from the main model, particularly from the public signal assumption, to outline under which necessary conditions the calibrated precision could be sustained if the signal was costly and private. I ask under which conditions a bank would profit from deviating from the estimated precision, focusing on the extreme cases of perfect or no signal which emerge from the calibration. I find that a signal that only sorts borrowers cannot sustain a full information equilibrium, and this is due to the randomization induced by the search frictions. Consider the case in which the informed banks' rate distributions for safe and risky types overlap; a deviating bank that stops screening and posts any rate inside the overlapping region earns the same expected profits as informed banks on both types, but saves the cost of the signal. Full information is therefore sustainable only by introducing an additional punishment for the deviator, for example the case in which the signal also cuts the cost of lending - a *monitoring* effect. Assuming a common cost of acquiring precision across classes and the calibrated parameters, I show that all eight calibrated precisions are equilibria if the monitoring benefit is at least nine times stronger in the safest rating classes than in the riskiest.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. In Section 3 I show that interest rate differentials across similar firms borrowing similar loan products from the same bank contain information on future changes in firms' default risk. In Section 4 I develop a search model of loan pricing with asymmetric information in which banks see an imperfect public signal on firms' risk types. In Section 5 I run a calibration on each rating class separately, and I show that in different rating classes the degree of information embedded in loan rates is high in best rating classes, low in worse ones. Section 6 asks whether the estimated signal precision is what a bank would choose. Section 7 analyzes the implications for surplus sharing. Section 8 concludes.

2 Related Literature

This paper connects several strands of literature: the empirics of interest rate dispersion, the theory of price dispersion under imperfect competition, the theory of adverse selection and screening, and the work on information acquisition and market power in banking.

Rate dispersion in loan markets. In credit markets, rate dispersion after controlling for observable fundamentals is the object of many papers leveraging loan-level data in different contexts. Cavalcanti et al. (2023) document dispersion across observationally equivalent firms in Brazil and focus on the consequences for capital misallocation and economic development; they show how this mechanism can contribute to differences in economic development across countries. Altavilla et al. (2024) decompose the variation in rates on Euro Area loans into macroeconomic fundamentals, bank identity, firm and loan characteristics, which overall account for 80–85% of total rate variation. I obtain a similar share of explained variation, but I focus on the unexplained component of loan pricing. Using French data, Mazet-Sonilhac (2025) provides evidence on the existence of search frictions in the French loan market, which motivates a framework where search and information frictions interact. I engage in the ambitious exercise of disentangling the relative contribution of both frictions to observed dispersion. Amiti et al. (2026) use US large-loan data to document dispersion within narrow bins of observationally equivalent firms and quarters; measures of banks' private information about default risk (internal ratings) fail to explain rate variation, and they conclude that search frictions account for most of the unexplained variation. In my setting, lacking direct measures of banks' private information, I will find an innovative way to extract the information component: I use the future evolution of firms' credit risk, as summarized by external credit ratings. Ferreira and Kozeniauskas (2026) similarly documents "idiosyncratic noise" in financing rates arising from heterogeneous search abilities, with default risk explaining little of the residual variation. In line with these papers, I find that search frictions play a predominant role in explaining rate dispersion, calling for policy intervention. However, I depart from these papers by showing that the low contribution of banks'

private information about default risk to rate dispersion does not imply that banks are not screening at all. On the contrary, I will find that banks have strong screening incentives, particularly in buckets of firms with ex-ante low risk; omitting this component can mislead the understanding of the role of screening in loan pricing, and leads to overstating the contribution of search frictions to observed dispersion.

Price dispersion in mortgage markets. A closely related body of evidence comes from mortgage markets, where residual dispersion has been traced to price discrimination, borrower riskiness and switching costs. Woodward and Hall (2012) and Bhutta et al. (2021) attribute a large share of it to shopping intensity: borrowers who solicit few quotes, or who are less financially sophisticated, systematically overpay for otherwise comparable loans, which is the household-market counterpart of the search friction I model. Coen et al. (2024) document price dispersion in the UK mortgage market and show that the price differences they measure are not driven by borrower risk, as gauged by realized default outcomes, but by heterogeneity in the menus of offers that lenders construct: lenders price-discriminate against borrowers who are likely to choose poorly, while competing for those who shop more effectively. Menu heterogeneity is therefore a third source of dispersion, distinct from search and from screening. I believe I can abstract from it for two reasons. First, I compare loans that are similar in both firm and contract characteristics, assuming that the rate varies linearly in the characteristics I control for, so that differences in contract design are absorbed rather than left in the residual. Second, within a given lender–month the loan terms I observe follow rather similar characteristics, which limits the scope for menu-based discrimination within my comparison groups. In line with this literature, I find that search frictions play an important role in explaining dispersion; I depart from it in also finding evidence of an unobserved risk component in dispersed rates, which I recover from the forward information contained in rate residuals.

Price dispersion as an equilibrium outcome. My model builds on the literature showing that identical sellers can charge different prices in equilibrium. Diamond (1971) establishes the benchmark result that this literature has to overturn: if all sellers are identical and all buyers pay the same small search cost, the only equilibrium is the monopoly price. Dispersion therefore requires that buyers differ in what they observe. Varian (1980) shows that when some consumers know all prices and the others know only one, sellers have no pure-strategy best response: cutting the price wins the informed buyers, raising it exploits the uninformed ones. Sellers then randomize, and prices differ even though costs and quality do not. Burdett and Judd (1983) obtain the same result when buyers differ in how many quotes they sample, and Stahl (1989) shows that dispersion also arises under sequential search, with monopoly pricing and competitive pricing emerging as limiting cases when the share of shoppers goes to zero or to one. Baye et al. (1992) describe the full set of mixed-strategy equilibria in this class of models. I adopt this structure to

explain why banks quote different rates to firms that are observationally equivalent, but my environment departs from it in two respects. First, the seller's cost is buyer-specific and unobserved: the return on a given quote depends on the borrower's hidden type, so the equilibrium rate distribution has to reconcile banks' indifference over the rates in the support with the participation constraints of both types. Since the two types differ in their reservation rate, the support is partitioned at those thresholds, and whether the lower segment is offered to both types depends on the severity of adverse selection, which governs the existence of a pooling region. Dispersion is therefore not only within-type, as in the papers above, but also between-type. Second, the precision of banks' information is an equilibrium object rather than a primitive: banks pay for a signal, and its value is governed by the profits they expect to earn, which in turn depend on the intensity of competition. The two frictions thus shape the rate distribution jointly, and the extent of dispersion responds to the return on screening as much as to the search friction itself. Finally, I bring the resulting distribution to loan-level data and recover its structural parameters, so that the relative weight of the two forces becomes a quantitative question rather than a theoretical possibility.

Adverse selection, screening and the cost of pooling. The second ingredient is the theory of markets where quality is hidden, which delivers between-type dispersion and my results on surplus sharing. Akerlof (1970) shows that asymmetric information can unravel a market. Rothschild and Stiglitz (1976) give the standard screening framework: an uninformed seller can separate types only by distorting the contract offered to the good type, so competitive screening leads to separating contracts, although an equilibrium may not exist. Stiglitz and Weiss (1981) bring this logic to credit markets: because higher rates worsen borrower selection and reduce expected returns, banks may ration credit rather than increase rates, and under pooling good borrowers subsidize bad ones. Bester (1985) shows that rationing disappears if lenders can screen, in his case by offering menus of collateral and rates. My surplus-sharing results are the quantitative version of this argument. When the signal is imprecise, my model produces the same two costs: good types pay a pooled rate that subsidizes bad types, and when adverse selection is severe their offer exceeds their reservation rate and they leave the market. A more precise signal reduces both. The difference is in how types are separated: not through the contract (collateral, quantity or menus) but through costly information, so separation is a matter of degree and can be estimated from the data. Lester et al. (2019) are the closest link between these literatures, as they add screening through menus to a market with search frictions and adverse selection. I find no evidence of the rate-quantity relation that menus would imply in my sample, so I model screening as information acquisition instead; the implications for surplus sharing and for cross-subsidization from good to bad firms are similar.

Information acquisition and market power in credit markets. *Disentangling search frictions and information* is challenging as they most likely interact in my setting. Banks' private information is a source of market power, and search frictions have consequences for the birth of new lending relationships and banks' screening ability. Carletti et al. (2024) provide a comprehensive review of how market power and informational advantages interact in banking, and Dell'Ariccia (2001) shows that the informational advantage of incumbent banks works as a barrier to entry, so market structure is itself shaped by information frictions. The interaction between competition and information acquisition has been directly theorized in credit markets: Hauswald and Marquez (2006) show that banks invest in information technology to screen borrowers, but stronger competition reduces the return on this investment and leads to under-screening; Dell'Ariccia and Marquez (2006) show that lending booms driven by intensified competition generate looser lending standards, as banks cannot recoup screening costs when rivals can poach identified good borrowers. Both papers provide theoretical grounding for the competition-information trade-off at the heart of my model. On the empirical side, Petersen and Rajan (1994) show that lending relationships improve credit availability and terms for small firms, evidence that banks hold information that is not public; Claessens et al. (2025) use internal bank ratings to isolate the private information component in loan terms, finding that it significantly affects rates, collateral, and maturity; Agarwal and Hauswald (2010) show that geographic proximity to the borrower reduces informational asymmetries and improves pricing accuracy. Central to my interpretation is the *hold-up problem*: Sharpe (1990) shows how relationship lending generates informational rents, as a bank that learns about its borrower gains an informational advantage that weakens the borrower's outside option, and Rajan (1992) builds on this to study the borrower's choice between informed and arm's-length debt; Cahn et al. (2024) show that the public disclosure of credit ratings attenuates this hold-up by reducing banks' informational monopoly. Ioannidou and Ongena (2010) document that firms switching banks receive rate discounts, consistent with incumbents extracting rents from informationally captive borrowers. Flanagan (2023) estimates that screening costs are embedded in loan rates and are larger for more opaque borrowers, directly linking information acquisition to pricing. Related sources of market power include *switching costs and relationship rents*: Drechsler et al. (2017) demonstrate that concentrated deposit markets allow banks to maintain wide spreads; Petersen and Rajan (1995) show that greater competition reduces relationship lending;[1] Degryse and Ongena (2005) document spatial price discrimination, with banks charging higher rates to distant borrowers who are less likely to search; and Boot and Thakor (2000) show that intensified competition erodes banks' incentives to invest in relationship-based screening. The paper most closely related to mine is Crawford et al.

[1]For a survey of relationship lending, see Boot (2000); Berger and Udell (1995) and Berger and Udell (2002) provide foundational empirical evidence on how relationship banking benefits small firms.

(2018), who jointly estimate asymmetric information and lender market power in Italian mortgages and find both frictions quantitatively important; I differ in two ways: first, I model active information acquisition by banks rather than screening via menus, as I do not find a positive correlation between loan take-up and rates in my data; second, I also find important interactions, but while they find that market power can mitigate the effects of adverse selection (prices increase by less when banks have market power and adverse selection worsens), I find that market power gives banks enough incentive to screen, as they would recover most of the surplus loss induced by adverse selection. In a mortgage market setting, Agarwal et al. (2024) develop a structural model combining search frictions with lender screening, finding that both jointly determine credit allocation and calling for extensions to the business loan market, which this paper pursues. In my setting, I assume for simplicity that search frictions are exogenous to market conditions; however, banks' screening incentives will depend on the degree of competition in the market via their effect on expected profits.

Search frictions and information in other financial markets. The search friction framework has been widely applied *beyond bank credit*. Duffie et al. (2005) develop the foundational search model for over-the-counter (OTC) financial markets, showing how search costs generate bid-ask spreads and dealer rents. Farboodi et al. (2025) show that intermediaries extract rents precisely because search frictions give them market power over their clients. In the interbank money market, Bech and Monnet (2015) apply a random search model to explain the dispersion of overnight rates and trading volumes in OTC reserve markets. On the theoretical side, Guerrieri et al. (2010) characterize equilibrium in competitive search markets with adverse selection, where high-quality sellers separate through the prices they post, generating a positive relationship between prices and quality in equilibrium. This mechanism is closely related to the rate-risk correlation I document in my data, and to the separating equilibria that arise in my model under information acquisition.

Implications for the pass-through of policy rates. Monetary policy is not the focus of this paper, but my decomposition speaks to the literature connecting the structure of the loan market to the pass-through of policy rates. The deposit market power documented by Drechsler et al. (2017) lets banks keep spreads wide when policy rates move; Scharfstein and Sunderam (2016) find that market power in mortgage origination dampens pass-through; and Wang et al. (2022) find that bank market power weakens the transmission to corporate investment. Closer to my setting, Puglisi (2023) documents dispersion in the US auto loan market and shows that it generates asymmetric pass-through across borrowers, and Delis et al. (2025) show that tighter credit conditions fall mostly on weaker borrowers. Since I find that most of the residual dispersion comes from imperfect competition rather than from risk-based pricing, my results suggest that such

effects may operate through a large part of the rate distribution; the pooling regions of my model, where banks quote the same rate schedule to both types, would add a further source of rigidity. I leave a full treatment of transmission to future work.[2] I also contribute to the literature that exploits the interest rate structure to evaluate the effects of monetary policy on firms' liabilities and outcomes; for example, Gürkaynak et al. (2022) use stock market equity prices to show that investors price-in policy rate changes more when firms have a higher share of floating rate liabilities; Core et al. (2025) also exploit the difference between floating and fixed rate loans to show that the effect of a monetary policy tightening on inflation is weakened if firms facing higher borrowing costs adjust prices.

3 Empirics

I have combined multiple administrative datasets on business loans, bank-firm relationships and public credit ratings assigned by the French central bank. The resulting loan-level dataset collects all loan terms, firm financial and demographic information and the evolution of default probability after loan issuance. I use this dataset to compute interest rate residuals after controlling for observables at loan issuance, and I show that they contain forward information about unexpected changes in firms' default probability each year after loan issuance. This correlation is positive and significant, particularly at shorter horizons. Moreover, it is heterogeneous across market segments, particularly strong for young, small firms, during financial crises and at longer maturities. I test whether the direction of causality runs from rates to rating worsening in a quasi-experimental setting. Overall, this evidence suggests that forward-looking information embedded in rates reflects information anticipated by banks at the time of loan issuance.

3.1 Data

Loan level data. The main dataset is a quarterly survey of all new loan contracts and credit lines to businesses in the first month of each quarter from 2006 Q1 to 2024 Q2 ("*Nouveaux Crédits*", provided by Banque de France). It includes anonymized bank identifiers, at the bank unit level, a wide range of loan terms (rate, amount, maturity, loan purpose, fixed or variable rate, pledged guarantees), firm-level identifiers and information on firms' demographics (age, sector, location, group affiliation), and financial and credit information (including balance-sheet size, debt structure, credit rating score, short-term due repayments, proportion of drawn and undrawn credit lines, revenues).

Bank-firm relationships. I recover information on bank-firm relationships from the

[2]The shock series reported in the Appendix (Fig. B.9) is built from the high-frequency Euro Area monetary policy surprises of Altavilla et al. (2019).

French credit register ("Service Central des Risques", SCR). It collects monthly data on outstanding loans of at least €25,000 and credit lines granted by credit institutions and investment firms to companies. I use it to track the length of the existing relationship at loan issuance, whether the firm has multiple bank relationships, and to compute concentration indexes at the department-quarter level.

Public credit ratings. The dataset "COTA" contains the credit rating assigned annually by Banque de France to firms and it represents a forward-looking assessment of creditworthiness over a three-year period. Firms are reviewed when their turnover exceeds €750K (€1250K from 2024). As shown in Fig. A.1, the rating scale has 12 rating categories, including one entry for liquidation (P) and one for when there is not enough information to assign a rating (0). A firm is evaluated on the basis of: the latest balance-sheet data available; data on default/delay events in payments (CIPE); sector-specific information; qualitative information, including interviews with firm managers on an annual basis, to assess the risks and difficulties the firm is facing. This scale was replaced by a more granular one from January 2022; to ensure continuity, I use Banque de France's conversion table to convert the new scale to the old one.

Cleaning and sample. I focus on fixed-term loan products to private companies, so I exclude credit lines/other types of credit with undefined maturity; I exclude syndicated and regulated loans. I also recover whether a firm is granted multiple loans or other products in the same month. Finally, I set to missing loan-level observations for which total firm assets fall below the 0.1st or above the 99.9th percentile, and express every balance-sheet control and the loan-to-assets ratio as a percentage of total assets. After cleaning, the dataset has 1.9 million observations and 0.9 million unique firms. I report a descriptive table of the main loan and firm characteristics for the regression sample, by rating class. The average loan is around €1 million, with substantial heterogeneity between rating classes, and it represents between one fourth and one third of the firm's total bank debt.

Rating Scale	**N obs.**	**rate** (mean)	**rate** (s.d.)	**Quantity** (mean, K-Eur)	**Maturity** (mean, months)	**Var. Rate** (in %)	**Age** (mean, years)	**Loan/Assets** (mean, in %)	**Bank Debt/Assets** (mean, in %)
3++	7948	2.1	1.8	3754.5	59.0	35	39.0	5.4	19.6
3+	22681	2.2	1.8	2919.3	53.5	37	33.7	5.0	20.6
3	43073	2.3	1.8	2271.5	42.4	37	31.7	5.3	22.0
4+	71629	2.8	2.1	1092.2	34.8	39	28.1	6.2	23.0
4	112974	2.8	2.0	670.2	27.9	43	26.1	7.1	25.0
5+	106418	2.9	2.0	545.5	23.9	48	23.4	7.7	30.3
5	46510	3.5	2.2	781.8	17.9	51	23.1	8.3	31.0
6	13625	3.7	2.2	923.4	16.6	52	21.4	10.4	34.2
7	2657	4.2	2.3	134.7	14.8	62	19.7	8.6	32.7
8	1592	5.0	2.6	107.8	13.2	61	18.0	8.4	31.2
9	268	5.5	2.7	134.0	11.9	48	16.5	10.0	31.7
P (liquidation)	3716	5.0	2.0	114.5	3.4	26	21.8	5.7	33.4
0 (no rating)	24777	3.2	2.1	1895.8	45.0	40	16.1	12.4	27.2
Total	457868	2.9	2.1	1098.1	30.5	44	25.8	7.3	26.5

Table 1: Descriptive table of the loan-level regression sample. Source: Banque de France (Nouveaux Credits, COTA, SCR, FIBEN). Rate: total effective rate (including all fees).

3.2 Interest Rate Residuals

Baseline approach. I measure interest rate residuals by regressing each loan's interest rate on observables at loan issuance (see Altavilla et al. (2024), Amiti et al. (2026) for the closest approach): I gradually include fixed effects at the macro- (month, lender) and micro- (loan and firm characteristics) levels, and all observed loan and firm characteristics. For each firm-bank-loan-time observation $\text{Rate}_{i,b,l,t}$, I run:

$$r_{ilbt} = \alpha + \mathbf{x}'_{ilbt}\gamma + \mu_{bt\tau s} + \varepsilon_{ilbt}, \tag{1}$$

where t is the month, b the bank, i the firm and l the loan. The vector $\mathbf{x}_{ilbt}$ holds what the bank observes at issuance: log loan size, loan over assets, maturity, variable rate dummy, personal loan dummy, individual entrepreneur dummy, firm age and age squared, young firm dummy, new bank-firm relationship dummy, dummies for firms taking several products or several loans in the month, and, as a share of assets, short-term debt, turnover, pledged guarantees, undrawn and drawn credit, bank and non-bank debt. Fixed effects include Month × Bank × Loan type × 2-digit Sector, which captures the supply schedule for similar products for each bank; then I include region, size and rating fixed effects, and a group-membership dummy. The regression sample includes 450 thousand observations and 120 thousand firms.

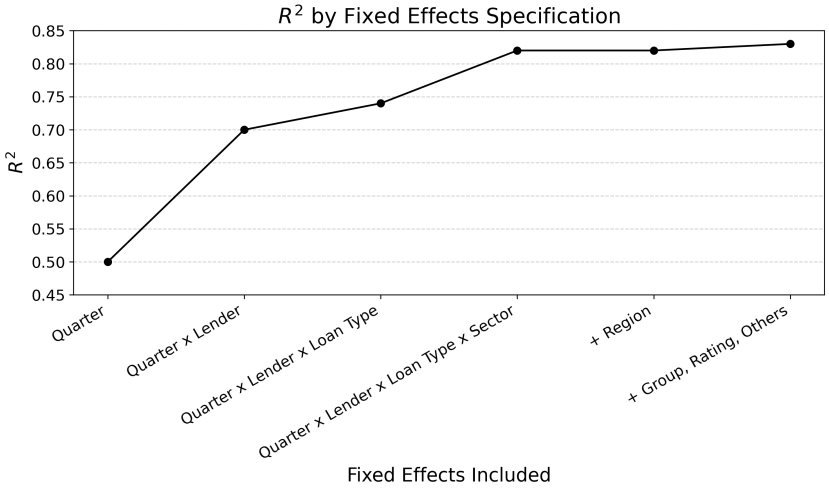


Figure 1: The figure shows the increase in R^2 of the baseline regression after gradually including fixed-effects interactions and the other observables. Observables explain up to 83% of the total variation in rates.

Fig. 1 summarizes the share of variation explained as I progressively add interacted fixed effects and other observables. Overall, observables explain 83% of the total variation in interest rates. Month-bank fixed effects alone explain 70%, while adding loan type times sector accounts for an additional 10 percentage points. The remaining observables

contribute little additional explanatory power. The full regression table is reported in Table B.1 in the appendix.

I run alternative specifications in Appendix B: I experiment with other regression models, to account for the fact that by construction there are no rates below zero (Tobit, truncated regression), but with worse explanatory performance (Appendix B.2). I explore two other dimensions: within-firm and within-bank. I include a specification with firm fixed effects (Table B.2 and Fig. B.4), which shows that some within-firm variation persists, although much lower than the comparison across firms; interestingly, the time-series in this case confirms increasing dispersion also within firm in crisis periods. I also run the baseline specification one bank at a time: conditional on observables and loan terms, the distribution of the R^2 of the regression of rates on observables is itself dispersed (Fig. B.5).

Time and cross-sectional distribution of rate residuals. I compute the standard deviation of residuals over time and across firm characteristics. Rate dispersion is substantial, 1pp on average over the time period - a meaningful gap in the context of loan pricing and external financing in general - and varies over the time period, as shown in Fig. B.2.

Rate residuals and competition. I study the distribution of unexplained rate residuals in relation to standard measures of competition. Using credit registry data, I aggregate lending information across geographical areas and time. Specifically, I sum the amount of drawn credit for each firm at the bank-department-quarter level to assess the relative exposure of each bank in that market, and I compute a standard Herfindahl-Hirschman Index (HHI) of concentration. Then I compute the standard deviation of residuals by quarter-department. I find that in more concentrated markets firms pay higher rates, but the distribution of rates is less dispersed (Fig. B.8).

Rate residuals and monetary policy. Using high-frequency monetary policy shocks, I show that overall dispersion decreases after a monetary policy tightening: the standard deviation of rate residuals across each lender's loans falls after a rate shock (Table B.3). Appendix B.5 describes how the shocks are recovered and the specification.

3.3 From public credit ratings to default probabilities

Throughout the sample, I observe firms' annual rating reviews, with firms staying in the same class or transitioning to a different one. When a firm is upgraded to a safer rating class, it moves to a class in which firms have a lower expected probability of default. The objective now is to turn a rating update into a change in default probability.

Approach. I compute historical 3-year ahead default frequencies by rating class and year: I take all firms in the same rating category at 01/01/x and compute the share that defaulted by 31/12/x+2. The resulting default frequencies are reported in Fig. 2, which

shows two important facts. First, ratings do very well at discriminating across firms' default risk: safer classes have consistently lower default frequencies; they are a reliable measure of credit risk. Second, default frequencies feature time variation, but the relative distance in default frequencies between classes is stable over time. This stability allows me to focus on changes in default probabilities arising from rating transitions, rather than also incorporating changes due to time-varying macroeconomic conditions, by fixing the default probability associated with each rating class to a specific year (I take 2021, the most recent before the rating reform). Specifically, for each firm i, I assign a default probability $\mathbf{P_{i,t}}$ at time t as the default frequency associated with that class in 2021 and $\Delta P_{i,t+k} = \left(P_{i,t+k} - P_{i,t+k-1}\right)$ is the annual default probability change, in percentage points, of firm i between $k-1$ and k years after issuance.

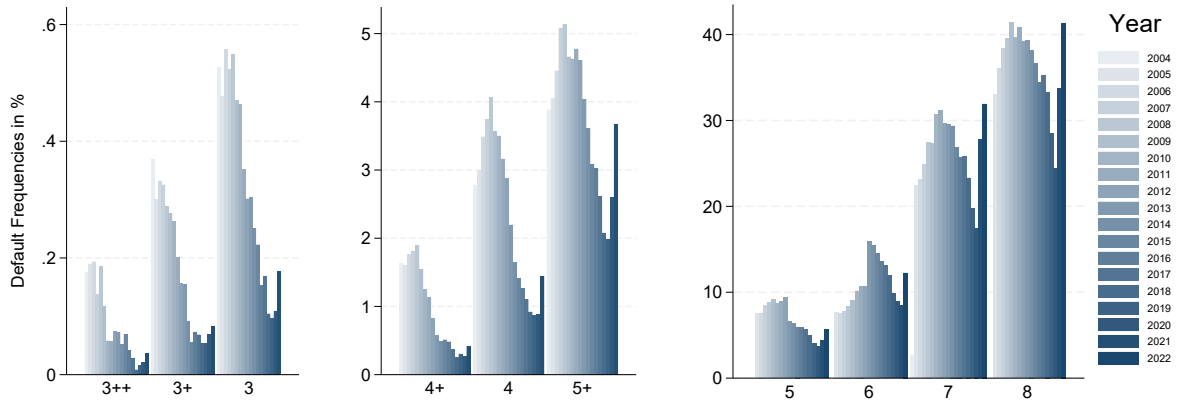


Figure 2: 3-year ahead default frequencies, by rating class and year (2004-2022).

Using rating transitions has three advantages over other measures of realized or perceived riskiness. First, it is an assessment of firms' default risk over a three-year horizon, a more direct measure of creditworthiness than ex-post balance-sheet indicators or growth outcomes. Second, rating changes are far more frequent than defaults: firms are reviewed every year and can change class several times; for robustness, Appendix B.6.1 in the appendix repeats the test with defaults. Finally, this is a realized outcome rather than the reported subjective expectation of the bank. Other papers include banks' internal ratings to explain interest rates, but typically they do not test whether they effectively predict future default. This is an important validation exercise, as banks' internal assessments may be biased by regulatory considerations and capital requirements.

3.4 Rate residuals and changes in default probability

Consider the following regression:

$$r_{ilbt} \;=\; \alpha \;+\; \sum_{k=1}^{K} \beta_k \, \Delta P_{i,t+k} \;+\; \mathbf{x}'_{ilbt} \, \gamma \;+\; \mu_{bt\tau s} \;+\; \varepsilon_{ilbt}, \tag{2}$$

where r_{ilbt} is the rate on loan l granted by bank b to firm i in month t, $\mathbf{x}_{ilbt}$ and $\mu_{bt\tau s}$ are the same controls included when measuring rate residuals; $\Delta P_{i,t+k}$ is the annual default probability change, in percentage points, of firm i between k-1 and k years after issuance - this term is zero if the firm does not change rating class. I set the horizon $K = 10$ and use a balanced sample, so the same firms identify every β_k, and I exclude loans to firms that are already in default when the loan is granted ($P_{i,t} \neq 1$).

The main focus is to assess whether rates contain forward information on changes in default probability, as summarized by $\beta_k \neq 0$. Two features of Eq. (2) matter for the reading of β_k. All K surprises enter jointly, so β_k is the correlation with the revision at horizon k holding the revisions at every other horizon fixed. And since the controls and the fixed effects sit in the same regression, β_k equals the coefficient of a regression of the rate residual on the rating surprise residual: it is a statement about the part of the rate the observables do not explain.

I plot the β_k in Fig. 3: the coefficients are positive and significant, stronger in the first year after loan issuance and fading at longer horizons. This suggests that rates contain forward-looking information about the borrower's creditworthiness that is not captured by observable characteristics at the time of loan issuance. The fact that the correlation is stronger in the first year may indicate that banks have better information about short-term risks or that the impact of unobserved factors diminishes over time. I repeat the exercise using actual defaults in Appendix B.6.1. I reverse the regression by putting the rating revision on the left and the rate on the right, focusing on predicting the annual change, with and without a balanced panel, Appendices B.6.2 and B.6.3, or the cumulative change from loan issuance, Appendix B.6.4.

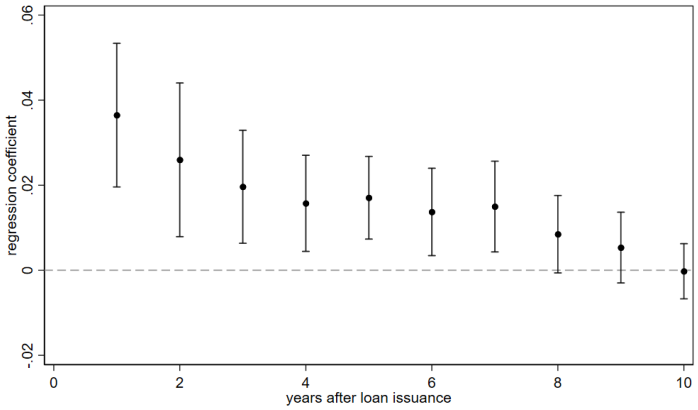


Figure 3: Correlation between rate residuals and unexpected changes in default probability, by year. Balanced sample.

3.4.1 Correlation across firms' demographics

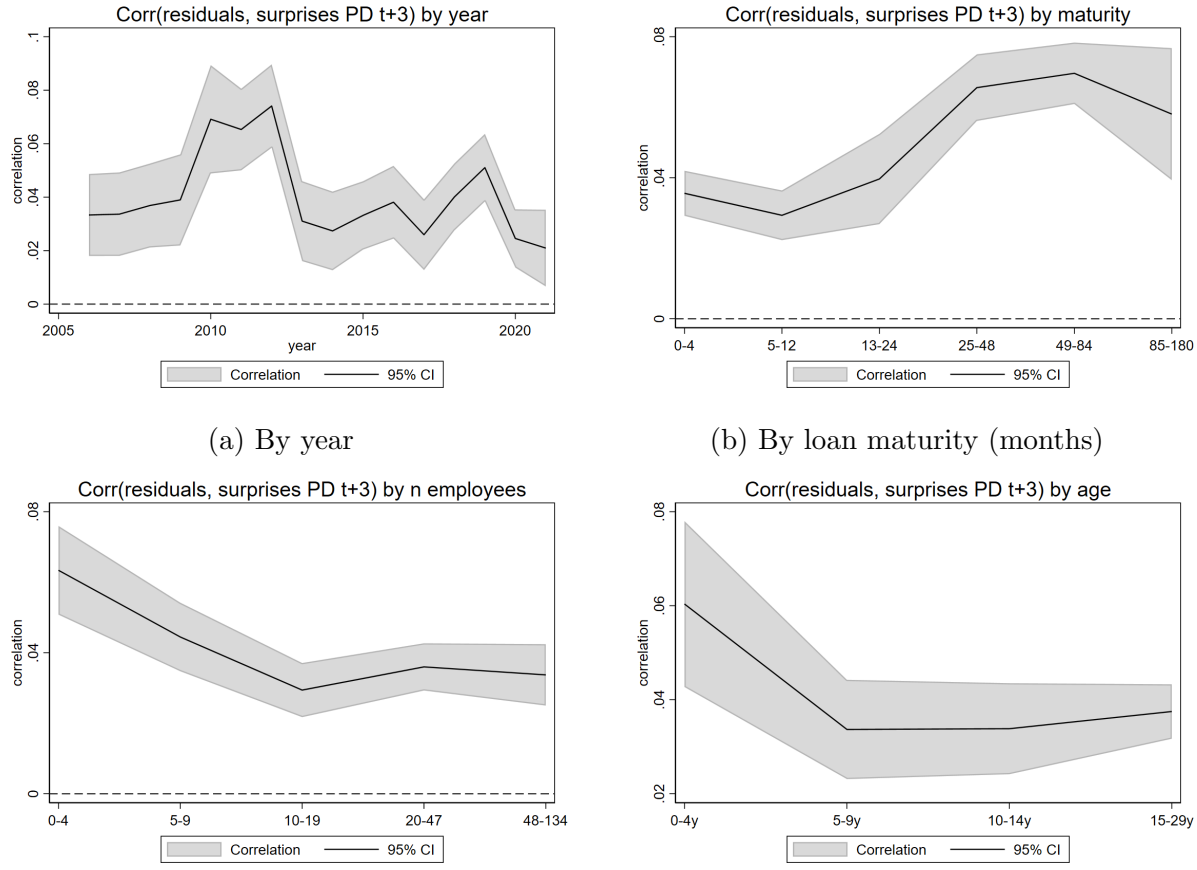
I further explore how the forward information contained in rates varies across borrowers' demographics. I separate firms by their Banque de France **rating** at loan issuance into three broad categories, and I document that the positive correlation is driven by firms with a relatively good rating, while it is not significant for relatively worse firms (see Fig. B.18). This finding might be counterintuitive, as riskier firms are closer to the default threshold.

One possibility is that interest rates may induce a worsening of the credit rating by mechanically increasing firm leverage, and in this case I should find a stronger effect when the loan is relatively large. By adding to Eq. (2) interaction terms between changes in default probabilities and **loan share of assets**, I find that the correlation coefficients when comparing the 10th and 90th percentiles of loans over assets are not significantly different (Fig. B.19). I repeat the exercise with the share of bank debt, in case the residual was informative about the rates on other loans, and results are robust. I move to **age interactions**, and I find that the correlation for younger firms (below the 10th percentile of age in my sample) is significantly larger than for older ones (above the 90th percentile), as shown in Fig. B.20. Age is both a proxy for risk/growth prospects (Haltiwanger et al., 2013) and a proxy for time in the credit market (Cloyne et al., 2023), so this finding suggests that the correlation may be stronger when information is more valuable.

The above methods require many observations, so I rely on a different approach to examine the time dimension and other cross-sectional dimensions: I compute the rate residuals $\tilde{r}_{ilbt}$ and the surprises in rating changes $\widetilde{\Delta P}_{ilbt}$ for the entire population:

$$r_{ilbt} = \alpha^r + \mathbf{x}'_{ilbt}\,\gamma^r + \mu^r_{bt\tau s} + \tilde{r}_{ilbt},$$
$$\Delta P_{i,t\to t+3} = \alpha^P + \mathbf{x}'_{ilbt}\,\gamma^P + \mu^P_{bt\tau s} + \widetilde{\Delta P}_{ilbt},$$

and I extract the pure correlation within each group of firms $\text{Corr}\left(\tilde{r}_{ilbt}, \widetilde{\Delta P}_{ilbt}\right)$.



(a) By year

(b) By loan maturity (months)

(c) By number of employees

(d) By firm age

Figure 4: *Informational content of rate residuals across borrower and loan characteristics.* Each panel shows the correlation $\hat{\rho}_g$ between the unexplained rate component and the unexpected 3-year change in default probability (ΔPD_{t+3}), both computed in the entire population and correlated by group g. Shaded bands are 95% confidence intervals based on the Fisher z-transformation applied to $\hat{\rho}_g$.

Fig. 4 shows the results. The correlation varies over **time** (Fig. 4a), peaking around 2009–2012 at the height of the global financial crisis and sovereign debt crisis, a period of heightened uncertainty, and declining during the zero lower bound period. It rises with **loan maturity** up to the 25–84 month range, which also suggests that the value of information tends to be stronger at longer horizons (Fig. 4b). Finally, the correlation is decreasing in firm size, as measured by **number of employees** (Fig. 4c) and age (Fig. 4d): the correlation for firms aged 0–4 years (≈ 0.06) is roughly twice that for firms aged 10 years or more, and similarly higher for firms with fewer than 10 employees relative to larger ones. Younger and smaller firms have shorter credit histories and less public information available.

Overall, these cross-sectional patterns suggest that the informational content of unexplained rate residuals is stronger when information is more valuable: for young and small

firms, during the crisis and for longer contracts. In other words, the residuals may reflect *banks' information* about the evolution of firms' creditworthiness. The alternative explanation is that it is the result of a *leverage effect*, with higher rates inducing higher ex-post riskiness through the worsening of firms' fundamentals. Loan-size interactions in Fig. B.19 provide evidence against the leverage hypothesis. However, to rule it out, I adopt a quasi-experimental design, by comparing the effect of a rise in rates between firms borrowing at fixed and variable rates.

3.5 Test: do higher rates induce rating worsening?

I directly rule out that higher rates cause a credit quality deterioration by testing the transmission of interest rate changes to firms' default probabilities in a quasi-experimental setting, exploiting the interest rate structure of the contract, fixed versus floating. In particular, I compare floating- and fixed-rate loans to test whether firms with variable-rate loans are more likely to transition to a worse rating class when reference rates increase. I find no significant impact of increased debt-servicing costs on subsequent rating worsening.

Baseline. I regress the change in default probability from loan origination t in year 1 to the end of the third year on cumulative increases in reference rates in the first two years from loan issuance. Variation comes from two channels: the change in reference rates is zero for firms borrowing at fixed rates and positive for firms borrowing at variable rates, and for the latter firms it specifically depends on which reference rate the loan is indexed to. In addition, as the intensity of the effect depends on the individual firm's exposure to variable rates, I add interaction terms of the reference rate increase with the loan-to-assets and total bank debt-to-assets ratios, following Gürkaynak et al. (2022) and Core et al. (2025). I restrict the sample to loans with a maturity of at least 30 months, so that refinancing does not drive the results.

$$\Delta P_{i,t\to t+3} \;=\; \beta\,\Delta\text{Ref}_{l,t\to t+2} \;+\; \delta\left(\Delta\text{Ref}_{l,t\to t+2}\times\text{Share}_{i,t}\right) \;+\; \mathbf{x}'_{ilbt}\,\gamma \;+\; \mu_{bt\tau s} \;+\; \varepsilon_{ilbt}, \quad (3)$$

where $\Delta P_{i,t\to t+3}$ is the change in firm i's default probability over the three years after loan issuance, in percentage points; note that the rating review may happen in any month of year three. $\Delta\text{Ref}_{l,t\to t+2}$ is the overall change in loan l's own reference rate until the end of year two: each floating loan is tied to a specific index (EONIA, Euribor 1m, 3m or 1y, TMO), and a fixed-rate loan is assigned a change of zero. $\text{Share}_{i,t}$ is either loan-over-assets or bank debt-over-assets, proxying the firm's overall exposure to floating rates. $\mathbf{x}_{ilbt}$ holds what the bank observes at issuance: log loan size, loan over assets, maturity, dummies for a variable rate, a personal loan and an individual entrepreneur, firm age and its square, a dummy for a firm younger than three years, and, each scaled by total assets, short-

term debt, undrawn credit, pledged guarantees, revenues, bank debt, non-bank debt and mobilized credit, plus region, turnover-class, rating-class and group-status fixed effects. $\mu_{bt\tau s}$ absorbs lender $\times$ month $\times$ loan-type $\times$ two-digit-sector effects. Standard errors are two-way clustered by lender and by month.

The coefficients β and δ compare how default probability varies for the group of firms experiencing a rate increase against the group of firms that did not experience it, depending also on their exposure to variable rates. As shown in Table B.4 (column 5), neither β nor δ is significant. I report seven specifications, using: a floating-rate dummy (column 1), the cumulative rate increase (column 2), the annual increases in reference rates entered separately (column 3), interactions with the share of the loan over assets (columns 4–6) and interactions with the share of the firm's total bank debt over assets (column 7), as a proxy for the firm's total exposure to variable-rate loans.

I check the identifying assumptions in Appendix B.7.1. The main issue is self-selection into fixed- or variable-rate contracts. To mitigate this risk, I take two approaches. First, I show in Table B.5 that the contract choice is predominantly driven by supply-side and sectoral factors: the lender-month dimension alone accounts for approximately 60% of the variation, and the full specification explains up to 91%. Rating transitions for firms taking variable-rate loans do not differ significantly from those of firms taking fixed-rate loans (Fig. B.21). Second, I repeat the exercise on loans issued in 2021q3–q4, at the end of the ZLB period, when firms did not expect a rate hike: I again find no significant effect of reference rate increases on rating worsening (Table B.6), including when the reference rate change accumulates through 2023 (Table B.7).

3.6 From the empirics to the model

Paying a higher interest rate does not worsen the credit rating; this fact, along with the predictive power of residuals being stronger in contexts when information is more valuable, suggests that the correlation between rate residuals and future default documented above reflects information that banks know at issuance. However, this cannot be the sole explanation. Interest rate differentials may be unrelated to firms' risk, and due to other market frictions, as in Amiti et al. (2026), Ferreira and Kozeniauskas (2026). I show indeed that rate dispersion varies with the degree of market concentration, increasing in less concentrated markets. Consequently, I move to a model of loan pricing where information and imperfect competition interact and determine loan rate offers. I model imperfect competition via search frictions, modelled as in Burdett and Judd (1983); Amiti et al. (2026) use this modelling choice to rationalize interest rate dispersion and quantify search costs. I extend this framework by adding asymmetric information, with firms differing in riskiness. In the context of loan pricing under adverse selection, a common modelling choice is screening via price-quantity menus (Stiglitz and Weiss (1981), Craw-

ford et al. (2018)). Lester et al. (2019) combine search frictions, asymmetric information and screening via menus in a general market setting. In the French loan market, I do not find a positive relationship between rates and quantities[3], so I replace the menu with a public signal of the firm's type; I interpret the signal as the information that banks can learn about a borrower in that market and that they price, which is not observed by the econometrician.

4 Model

In this section, I develop a model of loan pricing that combines search frictions, which generate imperfect competition between banks, with asymmetric information about firms' risk types. I introduce an imperfect public signal of firms' risk types into the price dispersion framework of Varian (1980) and Burdett and Judd (1983), building on the approach of Lester et al. (2019). I model banks as sorting borrowers through the interest rates they offer, the instrument I observe in the data, and a public signal that provides an imperfect indication of firms' types, rather than relying on price-quantity menus. This formulation allows me to incorporate asymmetric information while retaining a framework that can be taken directly to the data. The model generates a mixed-strategy equilibrium in interest rates and allows me to study how search frictions and the public signal jointly shape the relationship between interest rates and default risk. In the next section, I bring the model to the data by calibrating it to match the empirical findings.

4.1 Environment

The market consists of a continuum of firms and a continuum of banks, both of measure one. Firms need one unit of capital to undertake projects. A project succeeds with probability p[4] and, when successful, firms get a return $\bar{r}$. I assume *limited liability*, so firms only repay the loan if the project is successful. Firms' outside option when not undertaking the project is an infinitely small, strictly positive value $\varepsilon > 0$. Consequently, firms' reservation rate is equal to $\bar{r}$. Banks offer loans of fixed quantity one at an interest rate r. The opportunity cost of bank capital is set by the central bank's policy rate, r^{CB}. Therefore, the competitive break-even rate for banks, c, is defined by the condition that the expected repayment equals the opportunity cost of capital, $c = \frac{r^{CB}}{p}$.

Search Frictions. As in Burdett and Judd (1983) (BJ), I assume that each bank meets a fraction π of firms ("shoppers") who compare its offer with one from a competitor

[3] I find that, conditional on all observable characteristics in my sample, there is a negative relationship between rates and quantity (Table B.8, column 2); the relationship between rates and loan size relative to assets is not significant (column 1).

[4] It is possible to generalize the model such that a leverage effect is at play, namely $p(r)$ and $p'(r) < 0$, and the main results go through. Given the fixed-versus-floating test of Section 3.5, which supports that higher rates do not raise firms' subsequent default risk, I present the version with constant p.

bank, while the remaining fraction $1-\pi$ ("captives") only observe its offer. However, banks cannot distinguish shoppers from captives, and the specific number of offers a firm has sampled is private information to the firm. I assume that π is constant across firm types, although this assumption can be relaxed. This structure leads to a mixed-strategy equilibrium in interest rates.[5]

Asymmetric Information. The population of firms is divided into two types. A type is denoted as $\theta \in \{g, b\}$, with population shares μ_g and μ_b, where $\mu_b + \mu_g = 1$. Types differ in riskiness, $p_b < p_g$. Consequently, for banks the cost of lending to a bad firm is strictly higher than that of lending to a good firm: $c_g < c_b$, with $c_i = r^{CB}/p_i$. In the following, I also assume that the reservation rate of good firms is lower than that of bad firms: $\bar{r}^g < \bar{r}^b$. This assumption is consistent with thinking that risky projects are high-risk, high-yield. Finally, I look at the case in which $c_b < \bar{r}^g$; this is a sufficient condition to ensure that the cost of pooling both types is always below $\bar{r}^g$, and that a pooling region exists. As we will see, this structure leads to lending to bad-types only or partially pooling equilibrium in interest rates, depending on the severity of the adverse selection problem.[6]

Signal. Banks observe a public, imperfect signal of firms' type. The signal is binary, $s \in \{g, b\}$

$$\Pr(s = \theta \mid \theta) = q_\theta \geq 0.5, \quad \forall \theta \in \{g, b\},$$

where q_θ is the signal precision, i.e. the probability of correctly identifying a θ type. The value $q = 0.5$ means the signal is completely uninformative, $q = 1$ means it reveals the type. I assume $q_g = q_b = q$. The signal is *public*, so each bank matching with a firm observes the same realization of s.

The share of types obtaining a signal σ_θ and banks' posterior beliefs about facing a good type α^{s_θ}, conditional on obtaining a signal, follow from Bayes' rule:

$$\begin{aligned}
\sigma_g &\equiv \Pr(s = g) = q_g\mu_g + (1 - q_b)\mu_b, \\
\sigma_b &\equiv \Pr(s = b) = q_b\mu_b + (1 - q_g)\mu_g, \\
\alpha^{s_g} &\equiv \Pr(\theta = g \mid s = g) = \frac{q_g\mu_g}{q_g\mu_g + (1 - q_b)\mu_b}, \\
\alpha^{s_b} &\equiv \Pr(\theta = g \mid s = b) = \frac{(1 - q_g)\mu_g}{(1 - q_g)\mu_g + q_b\mu_b}.
\end{aligned}$$

[5] The same dispersed equilibrium can be obtained with informed and uninformed buyers as in Varian (1980), or under sequential search as in Stahl (1989); see Baye et al. (1992) for the full set of mixed-strategy equilibria in this class of models.

[6] The pooling region and the exit of good types are the counterparts, in my setting, of the two costs of hidden quality identified by Rothschild and Stiglitz (1976) and Stiglitz and Weiss (1981); here, however, separation is obtained through the market's public signal rather than through a contractual screening device as in Bester (1985).

By construction, $\sigma_g + \sigma_b = 1$. Note that $\alpha^{s_g} > \mu_g > \alpha^{s_b}$: a good signal increases the belief that the firm is good, while a bad signal decreases it.

Timing. π firms are randomly matched with two banks, $1 - \pi$ with one bank. Firms know their type θ and the number of quotes they receive. Each bank matches with a population of firms of measure one. Banks know all parameters $(\mu_g, p_g, p_b, \bar{r}^g, \bar{r}^b, \pi, q, r^{CB})$, but cannot observe the true type θ, the offers sampled by the firm, nor the quotes of the other bank; each bank reads each firm's signal and quotes a rate, and all banks quote simultaneously. Each firm sees its quotes, accepts the lowest quote if below $\bar{r}^\theta$, rejects otherwise. A shopper with two equal quotes picks each bank with probability 1/2. Finally, projects succeed or fail, and firms repay.

Rate offers and payoffs. Banks maximize their expected profits over rates. $\Delta(\mathbb{R}_+)$ is the set of probability measures on it. The banks' strategy is to offer a (cumulative) distribution over rates for each signal $F = (F^{s_g}, F^{s_b}) \in \Delta(\mathbb{R}_+) \times \Delta(\mathbb{R}_+)$. Firms maximize their expected surplus. A firm of type $\theta \in \{g, b\}$ that borrows at rate r obtains expected payoff $p_\theta(\bar{r}^\theta - r)$, where $p_\theta > 0$ is its project's success probability and $\bar{r}^\theta$ is its reservation rate. A firm that does not borrow obtains zero. Let $R \subseteq \mathbb{R}_+$ denote the set of rate offers received by a firm, with $|R| \in \{1, 2\}$, depending on whether the firm is captive or a shopper. Given R, the firm chooses the lowest offered rate, if lower than their reservation rate, and rejects all offers otherwise. A firm receiving two equal offers chooses each bank with probability one half. A firm indifferent between borrowing and not borrowing accepts the offer.

Definition of Equilibrium. For a given parameter vector $\{\mu_g, p_g, p_b, \bar{r}^g, \bar{r}^b, r^{CB}, \pi, q\}$, a symmetric equilibrium consists of a pair of bank rate distributions $F^* = (F^{s_g,*}, F^{s_b,*}) \in \Delta(\mathbb{R}_+)^2$ and a firm strategy a^* such that:

1. **Firm optimality.** For every firm type $\theta \in \{g, b\}$ and every set of received offers $R \subseteq \mathbb{R}_+$, with $|R| \in \{1, 2\}$, the firm chooses an option that maximizes its expected surplus:

$$a^*(\theta, R) \in \arg\max_{a \in R \cup \{\varnothing\}} U_\theta(a),$$

   where $U_\theta(r) = p_\theta(\bar{r}^\theta - r)$ for $r \in R$ and $U_\theta(\varnothing) = 0$ denotes the payoff from not borrowing.

2. **Bank optimality.** For each signal $s_i \in \{s_g, s_b\}$, every rate in the support of $F^{s_i,*}$ maximizes the bank's expected profit, given the strategies of competing banks and the firm's optimal strategy:

$$\text{supp}(F^{s_i,*}) \subseteq \arg\max_{r \geq 0} \Pi^{s_i}(r; F^*, a^*).$$

Every rate in the support earns the same profit, and no rate outside it earns more. This formulation also embeds firms' choice, as the bank's expected profits are conditional on the firm's acceptance of the offer, which is a function of the firm's type and the number of quotes it has sampled. The equilibrium is symmetric in the sense that all banks post the same distribution for each signal, and all firms follow the same acceptance strategy.

4.2 Properties of Equilibria

Conditional on observing each signal, banks face σ_g firms with good signals and σ_b firms with bad signals, and for each signal the bank updates its beliefs about the shares of types in the population. The bank also takes into account the probability that the offer is accepted, when it is competing over the π shopper firms with another bank. Conditional on receiving signal s_i, the bank's expected profits from offering a rate r are:

$$\Pi^{s_i}(r) = \begin{cases} \left[1 - \pi + \pi(1 - F^{s_i}(r))\right](r - c^{avg,s_i}) & \text{if } r \leq \bar{r}^g, \\ (1 - \alpha^{s_i})\left[1 - \pi + \pi(1 - F^{s_i}(r))\right](r - c_b) & \text{if } \bar{r}^g < r \leq \bar{r}^b, \\ 0 & \text{if } r > \bar{r}^b. \end{cases}$$

Profits are region specific: on the pooling region, below $\bar{r}^g$, both types accept, and the bank's expected cost of lending is $c^{avg,s_i} \equiv \alpha^{s_i} c_g + (1 - \alpha^{s_i}) c_b$, where $c_\theta = r^{CB}/p_\theta$. On the separating region, above $\bar{r}^g$, only bad types accept, which the bank expects to meet with probability $(1 - \alpha^{s_i})$.

I show a set of properties of the equilibrium. The equilibrium distributions F^{s_i} are continuous, with no mass points (Lemma 4.1), and the highest rate in the support is either $\bar{r}^g$ or $\bar{r}^b$ (Lemma 4.2). If banks lend on both regions, the support of F^{s_i} is disconnected, with a gap between the two regions (Lemma 4.3). Finally, the support of F^{s_g} lies below the one of F^{s_b} (**??**).

Lemma 4.1 (No mass points)**.** *In a symmetric equilibrium F^{s_i} is continuous: no rate is posted with strictly positive probability.*

The intuition behind the lemma is that no bank can optimally post a rate with strictly positive probability. If a rate r were posted with probability $m > 0$, a bank charging r would split shoppers equally with its rival whenever both banks post the same rate. By undercutting r by an arbitrarily small amount ε, the bank would attract an additional fraction $m/2$ of shoppers, while the loss in its margin vanishes as ε goes to zero. Since this deviation increases the probability of acceptance without reducing profits in the limit, a distribution with an atom cannot be an equilibrium. The same logic applies at the lower bound of the separating region: undercutting a rate just above $\bar{r}^g$ brings good types

back into the market and reduces the expected cost of lending from c_b to c^{avg,s_i}, making the deviation profitable there as well. Hence, the equilibrium distribution F^{s_i} must be continuous. The proof is in Appendix C.1.

Lemma 4.2 (The top of the support is a reservation rate)**.** *In a symmetric equilibrium the highest rate in the support of F^{s_i} is either $\bar{r}^g$ or $\bar{r}^b$. Banks earn their equilibrium profit on captive firms alone,*

$$\Pi^{s_i} = \max\left\{(1-\pi)\left(\bar{r}^g - c^{avg,s_i}\right),\ \ (1-\alpha^{s_i})(1-\pi)\left(\bar{r}^b - c_b\right)\right\}.$$

In a symmetric equilibrium, the highest rate in the support must be one of the two reservation rates, $\bar{r}^g$ or $\bar{r}^b$. At the highest rate in the support, a bank loses every shopper to the rival and earns its equilibrium profit on captive firms alone. If the highest rate were strictly inside either the pooling or the separating region, a bank could raise its rate slightly, continue to trade with the same captive firms, whose reservation rate exceeds the posted rate, and earn a higher margin. Hence, such a rate cannot be optimal. At the two reservation rates, however, raising the rate further entails losing a set of borrowers: above $\bar{r}^g$, good types stop accepting, while above $\bar{r}^b$, no borrowers accept. Therefore, the highest rate in the support must be either $\bar{r}^g$ or $\bar{r}^b$, depending on which reservation rate yields higher expected profits on captive firms alone. The proof is in Appendix C.1.

The top of the support pins down the equilibrium profit, which is earned on captive firms alone. This, with the equal profit condition, pins down the equilibrium distribution on the support.

Lemma 4.3 (The support is disconnected when a bank is lending on both regions)**.** *In a symmetric equilibrium F^{s_i} may be defined on both regions. In that case, the support of F^{s_i} is disconnected, with a gap between the two regions. The lower bound of the separating region is $\bar{r}^g$, and the upper bound of the pooling region is $\underline{r}^{s_i}$, which is strictly below $\bar{r}^g$.*

The intuition behind the lemma is that the support of F^{s_i} cannot be connected across $\bar{r}^g$. If it were, banks could offer rates just above $\bar{r}^g$, where shoppers accept either bank with probability 1/2. A bank could then profitably undercut its rival by offering exactly $\bar{r}^g$: its margin would remain unchanged, while the mass of firms accepting the offer would increase as good firms re-enter the market. Hence, the support of F^{s_i} must be disconnected, with a gap between the pooling and separating regions. The lower bound of the separating region is $\bar{r}^g$, while the upper bound of the pooling region, $\underline{r}^{s_i}$, must lie strictly below $\bar{r}^g$. The proof is in Appendix C.1.

4.2.1 Equilibrium under perfect signal (q=1)

The bank perfectly recognizes the type when it meets a firm. Basically, this is equivalent to solving independently two type-specific maximization problems.

Profit maximization. Banks' expected profits are as follows:

$$\Pi^i(r) = \begin{cases} \mu_i[(1-\pi) + \pi(1-F(r))](r-c_i) & \text{if } r \leq \bar{r}, \\ 0 & \text{otherwise.} \end{cases}$$

That is, the bank trades off higher profits and lower acceptance probability: every firm i is met with probability μ_i; the bank gets $(r-c_i)$ if the firm accepts; the firm accepts with probability 1 if it is captive, and if it observes a higher offer from the other bank when it is a shopper, with probability $(1-F(r))$. $F(r)$ is the equilibrium rate distribution.

In limit cases of monopoly ($\pi = 0$) or perfect competition ($\pi = 1$), the distribution is degenerate. The bank offers to each type the reservation rate $\bar{r}^i$ under monopoly, as all firms will always sample only one offer, and the competitive rate c_i under perfect competition, as all firms will visit both banks and get two rate quotes. However, under imperfect competition $\pi \in (0, 1)$, a dispersed rate equilibrium arises, as in BJ.

Exactly like in BJ, the type-specific offered distribution is derived from the equal profit condition:

$$\mu_i(1-\pi)(\bar{r}^i - c_i) = \mu_i[1 - \pi + \pi(1-F^i(r))](r-c_i).$$

Proposition 4.1. *For $\pi \in (0,1)$ and under perfect information (or perfect signal), there exists a unique symmetric mixed-strategy equilibrium*[7]*. For each firm type i, banks randomize their rate offers according to the cumulative distribution function $F^i(r)$ on the support $[\underline{r}^i, \bar{r}^i]$:*

$$F^i(r) = \begin{cases} 0 & \text{if } r < \underline{r}^i, \\ 1 - \dfrac{1-\pi}{\pi}\left[\dfrac{\bar{r}^i - r}{r - c_i}\right] & \text{if } \underline{r}^i \leq r \leq \bar{r}^i, \\ 1 & \text{if } r > \bar{r}^i, \end{cases}$$

where the upper bound $\bar{r}^i$ is the reservation rate, and the lower bound, from $F^i(\underline{r}^i) = 0$,

[7] Dispersion arises for the same reason as in Burdett and Judd (1983). With two perfectly symmetric banks, no rate strictly between the cost c_i and the reservation rate $\bar{r}^i$ can be sustained as a pure-strategy equilibrium: a bank posting it is undercut by ε by its competitor, who thereby captures the entire population of shoppers. Undercutting continues until the rate reaches the competitive level, where profits are zero; but there any bank can raise its rate and earn strictly positive profits on its captive firms, who accept any offer below $\bar{r}^i$. No pure strategy survives, so banks randomize, and the equilibrium distribution is pinned down by the requirement that expected profits be equal across the support. The same argument underlies each of the mixed-strategy equilibria characterized below.

is $\underline{r}^i = c_i + (1-\pi)(\bar{r}^i - c_i)$.

Posted rates across types mix the two distributions by population share, $\mu_g F^g(r) + \mu_b F^b(r)$, on the support $\underline{r}^g < r \leq \bar{r}^b$. Rates on the loans actually made follow a different distribution: a shopper pays the lower of two draws, so low rates are over-represented among the loans that get signed. Appendix D.1 derives the map from the posted distribution to the observed one, and the calibration of Section 5 matches moments of the observed one.

Dispersion and predictive power of residuals. Interestingly, the total dispersion of rates, measured as the total variance of the observed distribution, can be decomposed into a within-type component, driven by the search friction only, and a between-type component, driven by the separation of the two types:

$$Var(R) = \mathbb{E}[Var(r \mid i)] + Var(\mathbb{E}[R \mid i]) = \sum_{i \in \{g,b\}} \mu_i Var(r \mid i) + \mu_g \mu_b (\mathbb{E}_g[r] - \mathbb{E}_b[r])^2.$$

Secondly, the correlation between rates and default is positive, as the expected rate of bad types is higher than that of good types:

$$\rho(r,p) = \sqrt{\mu_g \mu_b} \cdot \frac{E[r|g] - E[r|b]}{\sqrt{Var(r)}}.$$

4.2.2 Equilibrium under an imperfect signal ($0.5 < q < 1$)

Between the two extremes the bank reads a signal that is informative but not conclusive. It cannot tell the type, but the signal moves its belief, so it treats the two signal groups as two separate markets: a share σ_g of firms carries a good signal and a good-type share α^{s_g}, and a share σ_b carries a bad signal and a good-type share $\alpha^{s_b} < \alpha^{s_g}$. Within a group the problem is the one of Section 4.2: the bank posts F^{s_i}, lends on the pooling region below $\bar{r}^g$ at the expected cost c^{avg,s_i}, and lends on the bad-only region above $\bar{r}^g$ at the cost c_b, where it meets a borrower with probability $(1 - \alpha^{s_i})$.

The adverse-selection index. By Lemma 4.2 the highest posted rate is one of the two reservation rates, and the bank earns its profit there on captive firms alone. It therefore compares the two corners: $\bar{r}^g - c^{avg,s_i}$ from pooling at the good types' reservation rate, and $(1 - \alpha^{s_i})(\bar{r}^b - c_b)$ from lending to bad types only at theirs. The difference between them is the adverse-selection index of the group,

$$\boxed{\phi^{s_i} \equiv (\bar{r}^g - c^{avg,s_i}) - (1 - \alpha^{s_i})\left(\bar{r}^b - c_b\right)}, \tag{4}$$

and the equilibrium profit is $\Pi^{s_i} = (1-\pi)\max\left\{\bar{r}^g - c^{avg,s_i},\ (1-\alpha^{s_i})(\bar{r}^b - c_b)\right\}$. A positive ϕ^{s_i} says adverse selection is mild in the group: the bank earns more by keeping the good types in. A negative ϕ^{s_i} says it is severe, and the bank earns more by pricing for bad types and letting the good types leave. A more precise signal raises α^{s_g} and lowers α^{s_b}, so it pushes the good-signal group towards pooling and the bad-signal group towards bad-only lending.

Three regimes. The sign of ϕ^{s_i} and the size of the search friction place the group in one of three regimes. Write $\underline{\phi}^{s_i} \equiv -\frac{\pi}{1-\pi}\left(\bar{r}^g - c^{avg,s_i}\right)$ for the threshold at which the pooling region disappears:

$$\begin{aligned} \text{full pooling:} \quad & \phi^{s_i} > 0, \\ \text{hybrid:} \quad & \underline{\phi}^{s_i} < \phi^{s_i} \leq 0, \\ \text{bad-only:} \quad & \phi^{s_i} \leq \underline{\phi}^{s_i}. \end{aligned}$$

Under full pooling both types accept every posted rate and the top of the support is $\bar{r}^g$. In the other two the top is $\bar{r}^b$, so the bank always posts rates that only bad types accept; it posts rates in the pooling region as well, and the good types in the group still get credit, as long as ϕ^{s_i} stays above $\underline{\phi}^{s_i}$. The threshold follows from requiring that the pooling branch carry positive mass.

Proposition 4.2 (Equilibrium within a signal group)**.** *For $\pi \in (0,1)$ and any signal precision $q \in [1/2, 1]$ there exists a unique symmetric mixed-strategy equilibrium. Conditional on receiving signal s_i, banks randomize their rate offers according to the cumulative distribution function $F^{s_i}(r)$ below.*

Full pooling, $\phi^{s_i} > 0$. The support is $[\underline{r}^{s_i}, \bar{r}^g]$ and

$$F^{s_i}(r) = \begin{cases} 0 & \text{if } r < \underline{r}^{s_i}, \\ 1 - \dfrac{1-\pi}{\pi}\left[\dfrac{\bar{r}^g - r}{r - c^{avg,s_i}}\right] & \text{if } \underline{r}^{s_i} \leq r \leq \bar{r}^g, \\ 1 & \text{if } r > \bar{r}^g, \end{cases}$$

with $\underline{r}^{s_i} = c^{avg,s_i} + (1-\pi)\left(\bar{r}^g - c^{avg,s_i}\right)$.

Hybrid and bad-only, $\phi^{s_i} \leq 0$. The support is $[\underline{r}^{s_i}, \bar{r}^b]$ and

$$F^{s_i}(r) = \begin{cases} 0 & \text{if } r < \underline{r}^{s_i}, \\ F^{s_i}_{pool}(r) & \text{if } \underline{r}^{s_i} \leq r \leq \bar{r}^g, \\ \max\left\{F^{s_i}_{pool}(\bar{r}^g),\ F_{bad}(r)\right\} & \text{if } \bar{r}^g < r \leq \bar{r}^b, \\ 1 & \text{if } r > \bar{r}^b, \end{cases}$$

where the equal-profit condition on each region gives

$$F^{s_i}_{pool}(r) = \max\left\{1 - \frac{1-\pi}{\pi}\left[\frac{(1-\alpha^{s_i})\left(\bar{r}^b - c_b\right)}{r - c^{avg,s_i}} - 1\right],\ 0\right\},$$

$$F_{bad}(r) = 1 - \frac{1-\pi}{\pi}\left[\frac{\bar{r}^b - c_b}{r - c_b} - 1\right].$$

In the hybrid regime the support starts at $\underline{r}^{s_i} = c^{avg,s_i} + (1-\pi)(1-\alpha^{s_i})\left(\bar{r}^b - c_b\right)$ *and has a gap on* $\left(\bar{r}^g,\ c_b + \left(\bar{r}^g - c^{avg,s_i}\right)/(1-\alpha^{s_i})\right)$*, where the first term of the maximum binds (Lemma 4.3). In the bad-only regime* $F^{s_i}_{pool}$ *is zero throughout and the support is* $\left[c_b + (1-\pi)\left(\bar{r}^b - c_b\right),\ \bar{r}^b\right]$.

One functional form. Every branch above is the Burdett and Judd (1983) distribution of a one-type economy,

$$F(r) = 1 - \frac{1-\pi}{\pi}\left[\frac{c+m-r}{r-c}\right],$$

with c the bank's expected cost of lending in the region and $m = \Pi^{s_i}/(1-\pi)$ the margin it earns on a captive firm, so that the branch reaches one at $c+m$. Under full pooling $c = c^{avg,s_i}$ and $c+m = \bar{r}^g$; on the bad-only region $c = c_b$ and $c+m = \bar{r}^b$; under a perfect signal $c = c_i$ and $c+m = \bar{r}^i$. The hybrid regime is the one case in which the margin and the cap of the region come apart: the pooling branch has $c = c^{avg,s_i}$ but would reach one only at $c^{avg,s_i} + (1-\alpha^{s_i})\left(\bar{r}^b - c_b\right)$, which lies above $\bar{r}^g$, so it is cut off at $\bar{r}^g$ and the remaining mass is posted on the bad-only region. The regimes differ in which region sets the margin, not in the shape of the distribution.

Aggregate distribution. Posted rates across the whole population mix the two groups,

$$F(r) = \sigma_g F^{s_g}(r) + \sigma_b F^{s_b}(r).$$

By **??** F^{s_g} dominates F^{s_b}, so the mixture adds dispersion between the groups to the dispersion inside each one, and the rate a firm pays carries information about its type. The one exception is the pair of regimes in which both groups lend to bad types only: the factor $(1-\alpha^{s_i})$ cancels from the equal-profit condition above $\bar{r}^g$, so F_{bad} does not depend on the signal, the two groups post the same distribution, and the signal is not priced.

The two extremes. At $q = 1/2$ the signal carries no information: $\alpha^{s_g} = \alpha^{s_b} = \mu_g$, the two groups merge into one, and ϕ^{s_i} becomes the population-wide index ϕ that governs the uninformative case below. As $q \to 1$ the posteriors go to certainty, $\alpha^{s_g} \to 1$ and $\alpha^{s_b} \to 0$, so $\phi^{s_g} \to \bar{r}^g - c_g > 0$ and $\phi^{s_b} \to \bar{r}^g - \bar{r}^b < 0$: the good-signal group pools at the good types' own cost, the bad-signal group lends to bad types only, and the mixture is

the perfect-signal equilibrium above. Appendix C.2 derives the two branches and both limits.

4.2.3 Equilibrium under uninformative signal (q=0.5)

The bank operates under information asymmetry, namely it cannot distinguish the firm type upon meeting. I still assume that the bank knows the population composition and the type-specific costs and reservation rates. Consequently, it faces a type-specific demand on two separate intervals: if $r < \bar{r}^g$, both types would accept an offer (*pooling region*), while if $\bar{r}^g < r < \bar{r}^b$ the bank knows that only bad types would accept (*bad-only region*) - a dispersed version of the lemons outcome of Akerlof (1970). The equilibrium may feature a mix of lending on the pooling and bad-only region depending on the severity of the adverse selection problem; mild adverse selection would incentivize pooling, while in the severe adverse selection case banks are more likely to reduce or avoid pooling and offer only high rates.

Limit cases of monopoly/competition $\pi = 0, 1$. I begin by analyzing the equilibrium in the extreme cases of monopoly and perfect competition to build the intuition for the general case.

Monopoly $\pi = 0$. The bank makes profits

$$\Pi = \begin{cases} (r - c^{avg}) & \text{if } \underline{r} < r \leq \bar{r}^g, \\ \mu_b(r - c_b) & \text{if } \bar{r}^g < r \leq \bar{r}^b, \end{cases}$$

where $c^{avg} = \mu_g c_g + \mu_b c_b$ is the weighted average of the type-specific costs. The bank compares the maximum profit it can extract in each region:

$$\max\{(\bar{r}^g - c^{avg}), \mu_b(\bar{r}^b - c_b)\}.$$

Profits rise with the rate within each interval, so the optimal strategy is a corner solution: the bank compares the maximum profit from pooling (at $\bar{r}^g$) against the maximum profit from screening (at $\bar{r}^b$).

Lemma 4.4. *Write ϕ for the index (4) under an uninformative signal, $\phi \equiv (\bar{r}^g - c^{avg}) - \mu_b(\bar{r}^b - c_b)$. Under monopoly ($\pi = 0$) and with an uninformative signal, the unique equilibrium is $\bar{r}^g$ if $\phi > 0$, and $\bar{r}^b$ if $\phi \leq 0$.*

Under monopoly, the unique profit-maximizing interest rate depends on the sign of ϕ. **I can interpret ϕ as an indicator of the strength of asymmetric information.** If $\phi \leq 0$, adverse selection is severe, and banks extract more profits from trading with bad types only. If $\phi > 0$, adverse selection is mild, and banks extract more profits from pooling good and bad types.

Perfect competition $\pi = 1$. In this case, the bank makes zero profits, offering the competitive rate in the population.

Lemma 4.5. *Under perfect competition ($\pi = 1$) and asymmetric information, the unique equilibrium is a pooling contract at the competitive rate $r^c = c^{avg}$, the break-even rate given average costs. Given the assumption $c^b < \bar{r}^g$, this rate is strictly below the reservation rate of both types and is accepted by the entire population.*

When $\pi = 1$, firms always compare offers. Bertrand competition drives the interest rate down to the break-even point. Since the competitive rate satisfies $r^c < \bar{r}^g$ by assumption, both types participate, and the bank faces the average cost of the population. The equilibrium profit is therefore $\Pi = (\mu_b + \mu_g)(r^c - c^{avg}) = 0$.

General case: Imperfect Competition. Bank profits depend on the interval.

$$\Pi = \begin{cases} [1 - \pi + \pi(1 - F(r))](r - c^{avg}) & \text{if } r \leq \bar{r}^g, \\ \mu_b[1 - \pi + \pi(1 - F(r))](r - c_b) & \text{if } \bar{r}^g < r \leq \bar{r}^b, \end{cases}$$

where $c^{avg} = \mu_g c_g + \mu_b c_b$. As in the perfect information case, the bank trades off profits and acceptance probability. However, the bank knows that both types would accept a rate in the pooling region, while a rate on the bad-only region would be accepted by bad types and rejected by good types, which would exit the market. As in the perfect information case with one type, the equilibrium CDF will be derived as the isoprofit function given profits at the maximum rate charged. However, now the maximum rate depends on the severity of adverse selection, as captured by the value of ϕ.

Severe adverse selection: $\phi \leq 0$

Banks surely trade with bad types, and possibly offer some rates in the pooling region. When trading with captive firms, the bank makes more profits from trading with bad types than from pooling. Therefore, the maximum rate observed in the market is $\bar{r}^b$. The bank offers a distribution of rates on the bad-only region to extract profits from bad types, and may also offer some rates in the pooling region to extract profits from good types, such that:

$$\begin{aligned} &[1 - \pi + \pi(1 - F^a(r))](r - c^{avg}) = \mu_b(1 - \pi)(\bar{r}^b - c_b) && \text{if } r \leq \bar{r}^g, \\ &\mu_b[1 - \pi + \pi(1 - F^a(\bar{r}^g) - F^b(r))](r - c_b) = \mu_b(1 - \pi)(\bar{r}^b - c_b) && \text{if } \bar{r}^g < r \leq \bar{r}^b. \end{aligned}$$

Proposition 4.3. *For $\pi \in (0, 1)$, $\phi \leq 0$ and with an uninformative signal, there exists a unique symmetric mixed-strategy equilibrium. Banks randomize their rate offers according*

to the cumulative distribution function $F(r)$ *on the support* $[\underline{r}^a, \bar{r}^b]$*:*

$$F(r) = \begin{cases} 0 & \text{if } r < \underline{r}^a, \\ F^a(r) & \text{if } \underline{r}^a \leq r \leq \bar{r}^g, \\ F^a(\bar{r}^g) + F^b(r) & \text{if } \bar{r}^g < r \leq \bar{r}^b, \\ 1 & \text{if } r > \bar{r}^b, \end{cases}$$

where the equal profit condition on each interval gives $F^a(r), F^b(r)$*:*

$$F^a(r) = \max\left\{1 - \frac{1-\pi}{\pi}\left[\frac{\mu_b(\bar{r}^b - c_b)}{(r - c^{avg})} - 1\right], 0\right\},$$
$$F^b(r) = 1 - F^a(\bar{r}^g) - \frac{1-\pi}{\pi}\left[\frac{(\bar{r}^b - c_b)}{(r - c_b)} - 1\right],$$

and the lower bounds $\underline{r}^a$ *and* $\underline{r}^b$ *solve* $F^a(\underline{r}^a) = 0$ *and* $F^b(\underline{r}^b) = 0$*.*

In particular, $\underline{r}^a = c^{avg} + \mu_b(1-\pi)(\bar{r}^b - c_b), \underline{r}^b = c_b + \frac{\bar{r}^g - c^{avg}}{\mu_b}$. By construction, $F^a(\bar{r}^g) + F^b(\bar{r}^b) = 1$. My equilibrium differs from Burdett and Judd (1983) in accounting for demand heterogeneity, preserving the mixed-strategy equilibrium result. On the existence of a pooling region, I find that it depends on the severity of adverse selection, as captured by ϕ.

Lemma 4.6. *The pooling region exists when adverse selection is relatively mild,* $-\frac{\pi}{1-\pi}(\bar{r}^g - c^{avg}) < \phi \leq 0$*, while for values of* ϕ *below this threshold the bank lends to bad types only.*

I find the threshold from $F^a(\bar{r}^g) > 0$. Intuitively, when adverse selection is severe, the bank prefers to offer only bad-only contracts, as the profits from pooling are too low. When adverse selection is mild, the bank offers some pooling contracts to extract profits from good types, while still offering some bad-only contracts to extract profits from bad types.

Weak Adverse Selection: $\phi > 0$

Banks trade with both types (Pooling Equilibrium). Banks make more profits from pooling both firms at the reservation rate of good types when meeting captive firms. Consequently, the maximum rate observed in the market is $\bar{r}^g$, trades will only happen in the pooling region and all firms will receive credit. The CDF of rates is found by setting the equal profit condition on the pooling region:

$$[1 - \pi + \pi(1 - F^a(r))](r - c^{avg}) = (1-\pi)(\bar{r}^g - c^{avg}).$$

Proposition 4.4. *For $\pi \in (0,1)$, $\phi > 0$ and with an uninformative signal, there exists a unique symmetric mixed-strategy equilibrium. Banks randomize their rate offers according to the cumulative distribution function $F(r)$ on the support $[\underline{r}, \bar{r}^g]$:*

$$F(r) = \begin{cases} 0 & \text{if } r < \underline{r}, \\ 1 - \dfrac{1-\pi}{\pi}\left[\dfrac{(\bar{r}^g - r)}{(r - c^{avg})}\right] & \text{if } \underline{r} \le r \le \bar{r}^g, \\ 1 & \text{if } r > \bar{r}^g, \end{cases}$$

where the upper bound $\bar{r}^g$ is the reservation rate of the good type, and the lower bound, from $F(\underline{r}) = 0$, is $\underline{r} = c^{avg} + (1-\pi)(\bar{r}^g - c^{avg})$.

With mild adverse selection, the equilibrium is closest to the BJ dispersed price equilibrium, as banks offer only pooling contracts.

4.3 From the model to the data

The calibration of Section 5 matches three moments of the observed distribution of rates: the mean, the variance, and the correlation with the borrower's default probability. Writing the three in the model's parameters shows what each one measures. Search sets the level of rates and the width of the distribution; precision sets how much of that width lines up with default.

Rates and dispersion within a signal group. Take a group that posts on a single branch, with cost c and captive margin $m = \bar{r} - c$, where $\bar{r}$ is the rate at the top of its support. Reweighting the posted density by the probability of winning the firm, as in Appendix D.1, gives

$$\mathbb{E}[r \mid s] = \bar{r} - \omega(\pi)\, m, \qquad Var(r \mid s) = v(\pi)\, m^2,$$

where

$$\omega(\pi) = \frac{\pi}{2-\pi}, \qquad v(\pi) = [1 - \omega(\pi)]\left[\frac{1-\pi}{\pi}\ln\frac{1}{1-\pi} - 1 + \omega(\pi)\right].$$

I call ω the *pass-through* and v the *dispersion factor*. Both depend on search alone. The pass-through is the share of the captive margin that competition hands back to the firm: firms pay the reservation rate at $\pi = 0$, the cost of lending at $\pi = 1$, and ω rises from zero to one in between. It is also the share of firms that hold two offers. A bank's clientele of measure one consists of $1-\pi$ captives and π contacts with shoppers; each shopper contacts two banks, so those contacts come from $\pi/2$ firms, and $\pi/2$ of the $1-\pi/2$ firms in the market compare two rates. The dispersion factor is zero at both ends,

since a monopolist posts the reservation rate and perfect competition posts the cost, and positive in between: all dispersion inside a group comes from search. Three branches of Proposition 4.2 take this form, with the margin measured from a different cost and a different cap in each: full pooling, with $c = c^{avg,s_i}$ and $\bar{r} = \bar{r}^g$; a bad-only region, with $c = c_b$ and $\bar{r} = \bar{r}^b$; and the perfect signal, with $c = c_i$ and $\bar{r} = \bar{r}^i$. Only the hybrid regime posts on two branches at once.

Expected rates by type. Under full pooling the rate a firm pays depends on its signal and not on its own type, so each type's mean rate is a mixture of the two group means. The groups differ only in the cost they price on, which gives

$$\mathbb{E}[r \mid s_b] - \mathbb{E}[r \mid s_g] = \omega(\pi)\,(c_b - c_g)\,(\alpha^{s_g} - \alpha^{s_b})\,, \qquad \alpha^{s_g} - \alpha^{s_b} = \frac{\mu_g \mu_b (2q-1)}{\sigma_g \sigma_b}.$$

A firm's own type predicts its signal, so the gap between the types is the gap between the groups scaled by $2q-1$ once more,

$$\mathbb{E}[r \mid b] - \mathbb{E}[r \mid g] = (2q-1)\,(\mathbb{E}[r \mid s_b] - \mathbb{E}[r \mid s_g]) = (2q-1)^2\,\omega(\pi)\,(c_b - c_g)\,\frac{\mu_g \mu_b}{\sigma_g \sigma_b}.$$

Precision enters twice: a sharper signal pulls the two groups' costs further apart, and it makes the group a firm lands in a better guide to that firm's type. At $q = 1/2$ both factors vanish, the two groups coincide, and the rate a firm pays says nothing about its type. At $q = 1$ the groups are the types, and the gap splits into the two reasons a risky firm pays more,

$$\mathbb{E}[r \mid b] - \mathbb{E}[r \mid g] = [1 - \omega(\pi)]\left(\bar{r}^b - \bar{r}^g\right) + \omega(\pi)\,(c_b - c_g)\,:$$

it is willing to pay more, and it costs more to lend to. Competition moves the weight from the first reason to the second.

The correlation between rates and default. Let $d_\theta = 1 - p_\theta$ be a type's default probability and n_θ the share of loans made to type θ. Default takes two values, so its correlation with the rate depends on the two type means alone,

$$\rho(r, d) = \sqrt{n_g n_b}\,\frac{\mathbb{E}[r \mid b] - \mathbb{E}[r \mid g]}{\sqrt{Var(r)}},$$

and this holds in every regime. The variance splits into the dispersion inside the groups and the gap between them,

$$Var(r) = v(\pi)\left[\sigma_g\,(m^{s_g})^2 + \sigma_b\,(m^{s_b})^2\right] + \sigma_g \sigma_b\,\omega(\pi)^2\,(m^{s_g} - m^{s_b})^2\,,$$

writing $m^{s_i} = \bar{r}^g - c^{avg,s_i}$ for the margin of group s_i. The first term is search and the

second information: the second is zero at $q = 1/2$ and grows with precision. Under full pooling the rate depends on the type only through the signal, and the correlation factors into what the rate reveals about the signal and what the signal reveals about the type,

$$\rho(r, d) = \rho(r, S)\,\rho(S, \theta), \qquad \rho(S, \theta) = (2q - 1)\sqrt{\frac{\mu_g \mu_b}{\sigma_g \sigma_b}},$$

in which S is the indicator of a bad signal. The margins do not depend on π, so search enters the correlation only through the ratio $v(\pi)/\omega(\pi)^2$, which falls from $1/3$ at $\pi \to 0$ to zero at $\pi \to 1$. Competition therefore raises the correlation: it compresses the dispersion within each group faster than the gap between them. A high correlation between rates and default is not by itself evidence of a precise signal, and the calibration separates the two by matching the variance at the same time.

When good types are priced out. In the hybrid and bad-only regimes a group posts rates above $\bar{r}^g$, which good types refuse. Two things change. The share of loans to good types falls below μ_g, which lowers $\sqrt{n_g n_b}$; and every loan above $\bar{r}^g$ goes to a bad type, so the rate and the type covary inside the group and the factorization fails, although the formula for $\rho(r, d)$ still holds. In the extreme case both groups lend to bad types only: the two groups then post the same distribution, no good type borrows, and the correlation is zero whatever the precision. An uninformative signal gives zero for the opposite reason, that the two groups never differ. Either way the correlation carries no information about q, and Appendix D reports that q is the direction the data pin down least in the classes whose observed correlation is near zero.

4.4 Implications for the value of information and surplus sharing

Total surplus (per bank). Under perfect information or perfect signal, per-trade surplus between a bank and a firm of type i is $(\bar{r}^i - c_i)$. In the symmetric equilibrium, only half of the π shoppers will effectively accept the bank's offer, while the others will prefer the second offer. The potential surplus attainable per bank if all firms are matched is $TS^{q=1} = (1 - \frac{\pi}{2})(\mu_b(\bar{r}^b - c_b) + \mu_g(\bar{r}^g - c_g))$. However, under asymmetric information total surplus depends on the severity of adverse selection: if adverse selection is severe, at least a fraction of good firms obtains a rate quote beyond their reservation rate, thus refuses and exits the market; the gains from trade remain unrealized.

Proposition 4.5 (Banks' share of total surplus)**.** *Banks' share of total surplus is decreasing in competition and increasing in the precision of the signal.*

Under perfect information, each bank gets a share of realized surplus per trade exactly equal to $1 - \pi$, while firms get the remaining π share. Under asymmetric information,

banks' profits are lower: in pooling, the bank charges bad types the pooling rate and loses $(1-\pi)\mu_b(\bar{r}^b - \bar{r}^g)$ with respect to the perfect information case; under bad-only lending and the mixed equilibrium, each bank loses a share of surplus equal to the unrealized trade with good firms $(1-\pi)\mu_g(\bar{r}^g - c_g)$.

Regime at $q = 0.5$	Π gain at $q = 1$	The gain is the surplus ...
Full pooling, $\phi > 0$	$(1-\pi)\,\mu_b\,(\bar{r}^b - \bar{r}^g)$	... not extracted from *bad* types
Hybrid, $\underline{\phi} < \phi \leq 0$	$(1-\pi)\,\mu_g\,(\bar{r}^g - c_g)$	... mixed, equal to the *good* types that exit
Bad-only, $\phi \leq \underline{\phi}$	$(1-\pi)\,\mu_g\,(\bar{r}^g - c_g)$	... of the *good* types that exit

Table 2: Bank's gain from moving to a perfect signal, $\Delta\Pi \equiv \Pi^{q=1} - \Pi^{q=0.5}$, per firm approached by one bank.

Proposition 4.6 (Firms' surplus and information)**.** *Information reduces (increases) costs borne by good (bad) types.*

(i) Under no information, when adverse selection is mild ($\phi > 0$), good types trade at the higher pooling rates; when adverse selection is severe ($\phi \leq 0$), they may not trade at all. These gains are increasing in the signal precision q and decreasing in the level of competition.

(ii) On the contrary, bad types benefit from pooling when adverse selection is mild ($\phi > 0$), as they trade at rates below the good types' reservation rate; when adverse selection is severe $\underline{\phi} < \phi \leq 0$, they partially gain from cross-subsidization, but less than in pooling; they do not gain if adverse selection is very severe and banks lend only on the high-rate region. These benefits are decreasing in the signal precision q.

Good types bear two costs of asymmetric information: a cross-subsidy to bad types where the bank pools, and exclusion where the bank lends to bad types only. Both fall with signal precision. Under no information and $\phi > 0$ the bank pools good and bad types. Total cross-subsidy to bad types is $\pi\mu_b(c_b - c_g)$. Under no information and $\phi < 0$, a fraction $1 - F(\bar{r}^g)$ of good types are offered too high a rate, and exit the market. If $F(\bar{r}^g) > 0$, the remaining good types, if any, subsidize bad types. Table 3 summarizes the gain/loss in surplus going from the no information to the full information equilibrium.

Regime at $q = 0.5$	ΔS_g	ΔS_b
Full pooling, $\phi > 0$	$\mu_g \frac{\pi}{2} \mu_b (c_b - c_g) > 0$	$-\mu_b \left[(1-\pi)(\bar{r}^b - \bar{r}^g) + \frac{\pi}{2}\mu_g(c_b - c_g)\right] < 0$
Hybrid, $\underline{\phi} < \phi \leq 0$	no closed form, > 0	no closed form, ≤ 0
Bad-only, $\phi \leq \underline{\phi}$	$\mu_g \frac{\pi}{2} (\bar{r}^g - c_g) > 0$	0

Table 3: Firms' gain from moving to a perfect signal, $\Delta S_i \equiv S_i^{q=1} - S_i^{q=0.5}$, per firm approached by one bank.

5 Calibration

The next step is to give a quantitative assessment of the relative contribution of search and information to explained variance, along with the surplus-sharing trade-offs. Because the structural parameters, π for competition and q for the signal precision, are inherently latent, I employ a calibration exercise to recover these values by matching the model to specific moments of the residual distribution across Banque de France rating classes. For each rating class, I calibrate three parameters: the shopper share π, which indexes the intensity of competition, the signal precision q, and the reservation rate for good types $\bar{r}_g$. The central bank funding cost r_{CB} is fixed at 1, consistent with the average policy rate over the sample period. Four additional parameters are calibrated directly from the data: $\mu_b, p_g, p_b, \bar{r}_b$. I observe the evolution of firms across rating classes and label as good or bad those that transition to a better or worse class. Firms that default outright are not counted as bad under this definition; Appendix E reports the alternative that includes them, and shows that the estimates below are unchanged for ratings 3+ through 5+. From this classification I derive the population share of bad types μ_b and the average success probabilities p_g and p_b in each population (computed as one minus the three-year-ahead default frequencies from Banque de France historical data), and the upper bound $\bar{r}_b$ (set at the 99th percentile of the observed rate distribution in each rating class, a more outlier-robust analogue of the literal maximum). For each rating class, I recenter the residuals on the class mean rate, so that all moments are expressed in rate levels rather than in deviations from the class mean.

I view this exercise as illustrative rather than as a literal structural estimation of the full cross-sectional distribution of borrower types. The model of Section 5 characterizes equilibrium for a stylized environment with exactly two borrower types, good and bad; the data, by contrast, reflect a continuum of borrower heterogeneity within each rating class. I do not expect, nor aim for, the two-type model to replicate every empirical moment exactly: my goal is to show that the search-and-screening mechanism generates rate dispersion and a rate–default correlation of the right sign and the right qualitative pattern

across the credit-quality spectrum, using the two-type model as a tractable benchmark for the richer, continuum-type environment that plausibly governs the data. Accordingly, I treat the correlation and mean moments as the primary targets: the correlation is matched essentially exactly in every rating class, and the mean to within roughly 15% (Table 5); the variance moment, which is mechanically the most sensitive to how many latent types the model allows for, is matched only partially, and should be read as a lower bound on the dispersion the search-and-screening channel alone can generate rather than a claim that these two channels are the only source of within-class rate dispersion in the data. I return to this point after presenting the results.

5.1 Calibration Strategy

The calibration proceeds in two stages. First, I conduct a grid search over the parameter space, evaluating 20^3 combinations of parameter values within economically plausible bounds: $\pi \in [0.05, 0.95]$, $q \in [0.5, 1.0]$, and $\bar{r}_g \in [0.5\,\bar{r}_{\text{mean}}, 0.98\bar{r}_b]$, where $\bar{r}_{\text{mean}}$ is the class mean rate. For each combination, I compute the model-implied moments - mean rate, variance, and correlation with subsequent default probability changes. Because a fraction π of firms compare offers from both banks and accept the lower one, the equilibrium distribution $F(r)$ derived in Section 5 – each bank's own posted-rate strategy – is not the same object as the distribution of *realized*, originated rates: a firm that shops around ends up paying the better of two draws from F, not a raw draw from it. I therefore compute all targeted moments under the properly reweighted realized-rate distribution $g(r) \propto f(r)\,(1 - \pi F(r))$, derived in Appendix D.1, rather than under $F(r)$ directly. I keep the combination that minimizes the objective function, which is the weighted sum of squared percentage errors of targeted moments. Second, I refine the best grid point using local optimization (L-BFGS-B algorithm) to minimize the weighted sum of squared percentage deviations between model and data moments.

The objective function to minimize is the sum of squared proportional errors across the three moments[8]:

$$Q(\theta) = \left(\frac{m_{\text{mean}}^{\text{model}} - m_{\text{mean}}^{\text{data}}}{m_{\text{mean}}^{\text{data}}}\right)^2 + \left(\frac{m_{\text{var}}^{\text{model}} - m_{\text{var}}^{\text{data}}}{m_{\text{var}}^{\text{data}}}\right)^2 + \left(\frac{m_{\text{corr}}^{\text{model}} - m_{\text{corr}}^{\text{data}}}{m_{\text{corr}}^{\text{data}}}\right)^2 \tag{5}$$

With three moments and three free parameters (r_{CB} fixed), the order condition for exact identification is satisfied. The parameters must also generate a feasible equilibrium (costs below reservation rates) and satisfy the mixed-strategy conditions, which impose additional structure beyond the three explicit moments matched. Because the equilibrium is piecewise in $(\pi, q, \bar{r}_g)$, however, a formal rank condition for global identification is not

[8]Proportional errors are used to ensure that moments of different scales are weighted comparably in the estimation.

analytically tractable; Section 5.4 reports numerical evidence in its place.

5.2 Counterfactual Analysis

To decompose the relative contributions of search frictions and information acquisition, I compute counterfactual equilibria holding recovered parameters $(\hat{\pi}, \hat{\bar{r}}_g)$ and calibrated parameters $\mu_b, p_g, p_b, \bar{r}_b, r_{CB}$ fixed while varying the signal precision q. In particular, I compare the percentage increase in variance given the calibrated parameters against the counterfactual with $q = 0.5$ (no information) isolates the role of search frictions:

$$\text{Information Contribution} = \frac{\text{Var}(\hat{q}) - \text{Var}(q = 0.5)}{\text{Var}(q = 0.5)} \times 100\% \tag{6}$$

5.3 Calibration Results

Table 4 presents the calibrated parameters for each rating class; Table 5 reports the corresponding model fit. The correlation moment is matched closely across all classes, and the mean is matched to a reasonable approximation. Variance is matched only partially in every rating class; consistent with the two-type-versus-continuum-heterogeneity gap discussed above, I do not read this as a failure of the search-and-screening mechanism, and return to it below and in Section 5.4.

Table 4: Search Frictions and Information by Rating Class

Rating	$\hat{\pi}$	$\hat{q}$	$\hat{\bar{r}}_g$	Info Contr. (%)	Obj. Value $Q(\theta)$
3+	0.772	0.997	3.461	4.89	0.41
3	0.759	1.000	3.706	4.56	0.35
4+	0.738	0.999	4.609	3.94	0.15
4	0.757	1.000	5.056	5.06	0.19
5+	0.749	1.000	5.403	3.05	0.20
5	0.729	0.944	6.662	0.04	0.04
6	0.721	0.526	7.057	0.00	0.04
7	0.711	0.526	9.350	0.00	0.00

Notes: $\hat{\pi}$ is the search-friction parameter, $\hat{q}$ the signal precision, $\hat{\bar{r}}_g$ the reservation rate of good borrowers, and "Information Contribution" measures the percentage increase in rate dispersion attributable to screening relative to the no-information benchmark.

The calibration results reveal three key findings. First, search frictions are relatively low across all rating classes, with $\hat{\pi}$ estimated between $[0.71, 0.77]$, increasing in ex-ante safety. This suggests that the French corporate lending market is relatively competitive, with most firms sampling multiple lenders - consistent with the prevalence of multi-bank relationships in the sample: on average about 60% of firms across classes maintain at least two active bank relationships (Table D.9).

Table 5: Model Fit: Observed vs. Model-Implied Moments

	Mean		Variance		Correlation	
Rating	Data	Model	Data	Model	Data	Model
3+	1.67	1.94	0.406	0.154	0.025	0.025
3	1.83	2.08	0.443	0.188	0.033	0.033
4+	2.40	2.53	0.553	0.338	0.032	0.032
4	2.37	2.63	0.730	0.424	0.048	0.048
5+	2.57	2.80	0.867	0.492	0.048	0.048
5	3.36	3.44	0.985	0.794	0.017	0.017
6	3.63	3.67	1.137	0.913	0.000	0.000
7	4.79	4.78	1.843	1.725	0.000	0.000

Notes: Mean is the class mean rate and variance the variance of rate residuals within the class, both in percentage points. Correlation is between the rate residual and the three-year-ahead change in the borrower's default probability. For ratings 6 and 7 the estimated correlation is not significantly different from zero ($p = 0.65$ and $p = 0.54$) and is set to zero as a calibration target.

Second, screening intensity exhibits heterogeneity across the credit quality distribution. Good borrowers (ratings 3+ through 5+) face near-perfect screening ($\hat{q}$ close to 1), while poor borrowers (ratings 6 and 7) face essentially no screening ($\hat{q} \approx 0.53$).

Third, despite the high screening intensity implied by the data moments for good ratings, information contributes minimally to observed rate dispersion. The information contribution to variance never exceeds 5.1%, indicating that search frictions account for most of the unexplained dispersion even in segments with near-perfect screening. This partly reflects the mild severity of the adverse selection problem within each rating class, for example, the relatively low distance of default probabilities between types mechanically limits between-type variance.

The objective values in Table 4 are highest for the safest classes and fall with credit quality, from 0.41 at rating 3+ to essentially zero at rating 7. The pattern is mechanical: in the safest classes the two types differ hardly at all in cost - $c_b - c_g$ is about 0.003 at rating 3+, against an observed standard deviation of rate residuals of 0.64 - so between-type dispersion is negligible and the model has to generate nearly all observed variance from search alone. The estimated $\hat{\bar{r}}_g$ sits just below the class upper bound $\bar{r}_b$ in every class, so the pooling region covers most of the support. Appendix E reports the same two tables under the alternative bad-type definition that counts outright defaults as bad.

5.4 Robustness of the Estimated Equilibrium

The response to each posterior signal realization depends endogenously on the estimated parameters through the threshold conditions governing ϕ_{good} and ϕ_{bad} - an analytical proof that the three-parameter GMM estimator $(\pi, q, \bar{r}_g)$ is globally identified from the three targeted moments (mean, variance, and the rate–default correlation) is not tractable.

I therefore rely on three complementary numerical checks, run separately for each of the eight rating classes, to build confidence that the parameter vector reported by my estimation procedure is the unique - or at least an isolated, dominant - minimizer of the GMM objective. Appendix D documents each of these three checks, and their complete numerical results, in full.

First, I assess local identification via the standard rank condition (Rothenberg, 1971): at the estimated parameter vector $\hat{\theta}$ I compute the 3×3 Jacobian of the three matched moments with respect to $(\pi, q, \bar{r}_g)$ by numerical differentiation. In every rating class the Jacobian has full rank (3/3), ruling out exact local non-identification, as summarized in Table 6. The associated condition numbers are not uniform, however: they range from roughly 5 to over 400 across rating classes, with the weakest direction consistently loading on q (screening precision) - most pronounced for the safest classes, whose near-perfect estimated precision ($\hat{q}$ close to 1) leaves the moments comparatively insensitive to q. I report this pattern transparently rather than treating identification as uniformly strong: q is comparatively less sharply pinned down than π and $\bar{r}_g$ in some classes, even though the local rank condition is always satisfied.

Table 6: Local Identification Diagnostics: Elasticity-Scaled Jacobian of (mean, var, corr) w.r.t. $(\pi, q, \bar{r}_g)$

Rating	Rank	Condition Number	Weakest Direction
3+	3/3	431.7	q
3	3/3	209.4	q
4+	3/3	134.2	q
4	3/3	82.1	q
5+	3/3	65.9	q (mixed with $\bar{r}_g$)
5	3/3	4.8	π (mixed with $\bar{r}_g$)
6	3/3	21.6	q
7	3/3	14.1	q

Notes: Rank and condition number computed from the singular value decomposition of the elasticity-scaled Jacobian (percentage change in moment per percentage change in parameter) at each rating class's estimated $\hat{\theta}$. "Weakest direction" identifies the parameter with the largest loading on the smallest singular vector; a second parameter is named when its loading also exceeds 0.45 in absolute value.

Second, because local curvature alone cannot rule out a second, disconnected minimum elsewhere in the parameter space, I complement it with a dense search over the full admissible cube of $(\pi, q, \bar{r}_g)$ for each rating class, visualized directly as three-dimensional plots of the parameter combinations whose implied moments come within a chosen tolerance of the data. Within every equilibrium regime realized on the grid, the set of parameter combinations that adequately fit the data forms a single connected neighborhood around one point rather than several disjoint clusters - visual evidence against multiple, unrelated local optima within a given regime.

Third, I exploit the fact that the equilibrium's regime-switching structure (as derived in Section 5) is itself governed by two threshold conditions - one for the pool of borrowers assigned a favorable signal, one for the pool assigned an unfavorable signal - each admitting three possible states (full pooling, hybrid, or bad-only lending), for nine theoretically possible regime pairs. Rather than assume ex ante which pairs are empirically relevant, I determine this data-drivenly: for each rating class I tag every point of the parameter grid with its actually realized (good-pool, bad-pool) regime pair, and find that only a subset - between three and six, depending on the rating class - are ever realized within the admissible parameter region, always respecting a simple ordering in which the good-signal pool is never in a "less pooled" regime than the bad-signal pool. I re-solve the GMM objective separately within each empirically realized regime and compare the best fit achieved across all of them. In every rating class, the best-fitting parameter vector across all admissible regimes coincides with the parameter vector reported by my baseline, regime-unconstrained estimation. This confirms that the baseline optimizer converges to the true global optimum of the objective function and is not trapped in a locally optimal equilibrium regime.

Taken together, these three checks - local curvature (rank and condition number of the moment Jacobian), a dense global search of the parameter space visualized directly, and an exhaustive comparison across every equilibrium regime the model can produce - provide converging numerical evidence that the reported parameter estimates for each rating class correspond to a well-isolated global optimum of the GMM criterion. They do not constitute a formal identification proof, which the model's regime-switching structure precludes in closed form, but they substantially reduce the concern that my point estimates are an artifact of the numerical optimization routine rather than a genuine feature of the data.

6 Can the calibrated precision be an equilibrium?

The calibration puts signal precision at the two extremes of the admissible range: close to perfect in ratings 3+ through 5, and uninformative in ratings 6 and 7. In this section, I check whether the calibrated precision can be the equilibrium of an information acquisition game, where each bank can buy its own precision at a cost $C_s(q)$, holding its rival at the calibrated $\hat{q}$, and ask whether the estimate is what a bank would choose.

I show that a necessary condition is to introduce an additional punishment to being uninformed. In fact, *a signal that only sorts borrowers does not incentivize information acquisition*, and a full information equilibrium $\hat{q} \approx 1$ is not sustainable. I provide the example of a monitoring effect: a better informed bank not only sorts types, but also reduces their riskiness via monitoring. I find that the estimated full information equilib-

rium in the safest classes is sustained if the monitoring effect is strong enough, much less in the riskiest classes.

I focus on the two extremes of full and no information and check if they are robust to deviations. I start from full information, $q = 1$, and I assume that one bank deviates to stop acquiring the signal. Its rivals keep posting the two type-specific distributions of **??**, whose supports overlap on $[\underline{r}^b, \bar{r}^g]$. The deviator maximizes its profits by posting a single rate in that overlap; the equal-profit condition holds on the overlap, so it earns exactly the same profits as informed banks: $(1-\pi)(\bar{r}^g - c_g)$ on every good type and $(1-\pi)(\bar{r}^b - c_b)$ on every bad type.[9] While it makes the same profits as informed banks, the deviator saves C_s, so $q = 1$ is not an equilibrium for any strictly positive cost.

I also consider a deviation from the no information equilibrium, $q = 0.5$. A deviating bank that acquires a signal always increases its profits. The deviation is not profitable if the cost of acquiring the signal is higher than the benefit.

6.1 Example: monitoring

Suppose that an informed bank does not only recognize types, but also monitors them; for example, an informed bank knows which covenant to write in the contract to incentivize repayment. The clients of an informed bank repay more often, and the cost of lending to them is lower. I specify a functional form:

$$c_\theta(1) = (1 - \delta_j)\, c_\theta(0.5), \qquad \theta \in \{g, b\}, \tag{7}$$

so that full precision cuts the risk-adjusted cost of lending to each type by the fraction δ_j, the *monitoring strength* of rating class j. I assume that the effect is the same for the two types inside a class, while I allow it to vary across classes. Write $\kappa_\theta = \delta_j\, c_\theta(0.5)$ for the saving per unit lent. I derive the two conditions under which $q = 0.5$ and $q = 1$ can be equilibria in Appendix F.

Condition A: $G_j \leq C_s$**.** $\hat{q} = 0.5$ survives if a bank that buys a perfect signal gains less than the cost, $G_j \leq C_s$, where:

$$G_j = \underbrace{\pi\mu_g\mu_b(c_b - c_g)}_{\text{cream-skimming}} + \underbrace{(1-\pi)\mu_b(\bar{r}^b - \bar{r}^g)}_{\text{captive bad types}} + \underbrace{\left[(1-\pi) + \pi\mu_g\right](\mu_g\kappa_g + \mu_b\kappa_b)}_{\text{monitoring}}. \tag{8}$$

Condition B: $C_s \leq L_j$**.** $\hat{q} = 1$ survives if a bank that drops the signal loses more than

[9] The overlap is non-empty when $\pi \geq (\bar{r}^b - \bar{r}^g)/(\bar{r}^b - c_b)$, which holds in all eight rating classes at the calibrated parameters. Where it fails the deviator loses $\mu_b\left[(1-\pi)(\bar{r}^b - c_b) - (\bar{r}^g - c_b)\right] > 0$, and the argument below goes through unchanged.

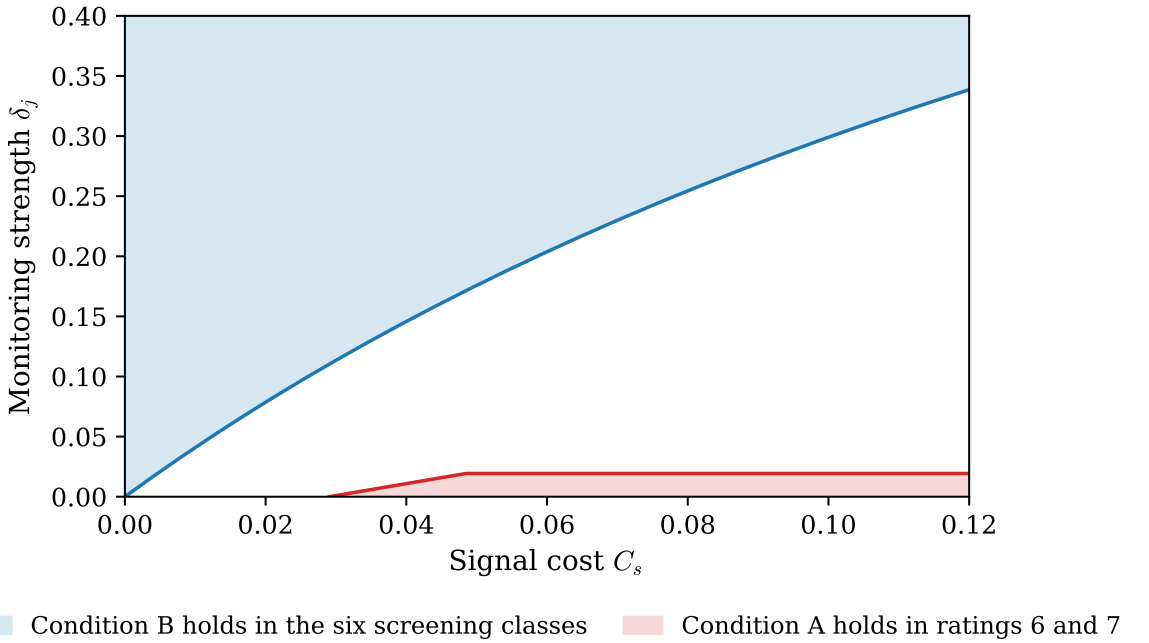


Figure 5: **Monitoring strength that sustains the calibrated precision, against the cost of the signal.** Horizontal axis: the price of a perfect signal over no signal, $C_s \equiv C_s(q{=}1) - C_s(0.5)$, for the mass one of firms a bank screens, in the units of the calibration ($r^{CB} = 1$). Vertical axis: the monitoring strength δ_j of (7), the fraction by which full precision cuts the risk-adjusted cost of lending to each type. The upper blue band is where the full information equilibrium of the safest classes is stable; the lower red band is where the no information equilibrium of the worse classes is stable; the feasibility constraint that monitoring cannot increase the success probability beyond one is binding.

the cost it saves, $C_s \leq L_j$, where:

$$L_j = \mu_g \kappa_g (1 - \pi) + \mu_b \kappa_b \frac{(1 - \pi)(\bar{r}^b - c_b)}{\bar{r}^g - c_b}, \tag{9}$$

where the bank holds a rate on the overlap, so the loss depends on the monitoring penalty. Both G_j and L_j rise with δ_j: stronger monitoring makes the uninformed equilibrium easier to break and the informed one easier to hold. Condition A therefore caps δ_j in ratings 6 and 7, and Condition B puts a floor under it in the other six.

Figure 5 evaluates the two conditions at the calibrated parameters, given a cost C_s. When the cost of acquiring the signal is too low, below $C_s = 0.029$, the gain of being informed is always higher than the cost. As the cost increases, deviating becomes less profitable, and for low enough monitoring effects the equilibrium is stable. For the safest classes, the deviation to the no information equilibrium is not profitable only if the punishment is large enough, that is, if monitoring ensures a considerable reduction in firms' riskiness. The two bands never meet, however. At the cost that brings them closest, $C_s = 0.048$, or between 2 and 8 percent of a bank's profit, the six screening classes need $\delta_j \geq 0.17$

while ratings 6 and 7 admit at most $\delta_j = 0.019$. A monitoring strength of 0.17 raises the bad type's repayment probability by 15 to 17 points; one of 0.019 raises it by one to two points. Feasibility caps the riskiest classes as tightly as Condition A does: their good types already repay with probability 0.98, which leaves monitoring almost nothing to add. The estimated precisions are therefore an equilibrium of the acquisition game in all eight classes at one common signal cost, but only under a technology whose effectiveness is concentrated where borrowers are safest, while it is ineffective for borrowers that are already very risky.

7 Surplus sharing and cross-subsidies

I return to the calibrated model of Section 5 to ask a surplus-sharing question raised, but not yet quantified, by Tables 2 and 3: how the split of gains from trade between banks and firms, and the cross-subsidy from good to bad types within firms, vary with banks' signal precision q.

Pooling and cross-subsidization at low signal precision. I simulate the model over the full admissible range of signal precision, $q \in [0.5, 1]$, holding all other parameters - π, $\bar{r}^g$, $\bar{r}^b$, μ_b, p_g, p_b - fixed at their rating-class-specific calibrated values (Table 4); $q = 0.5$ corresponds to an uninformative signal ($\alpha^{s_g} = \alpha^{s_b} = \mu_g$), and $q = 1$ to a perfect one ($\alpha^{s_g} = 1$, $\alpha^{s_b} = 0$). At $q = 0.5$, given the calibrated parameters, every rating class lies in a pooling equilibrium ($\phi^{s_i} > 0$): banks cannot condition on the signal and quote a common rate distribution to both types. Fig. 6 shows this directly - at $q = 0.5$ the expected rate paid by good and bad types agrees to four decimals in every rating class (3.6660 for both types at rating 6, 4.7805 at rating 7) - exactly the pooling outcome anticipated in Table 3: good types cross-subsidize bad types by $\pi\mu_b(c_b - c_g)$ in expectation, and banks do not manage to separate types.

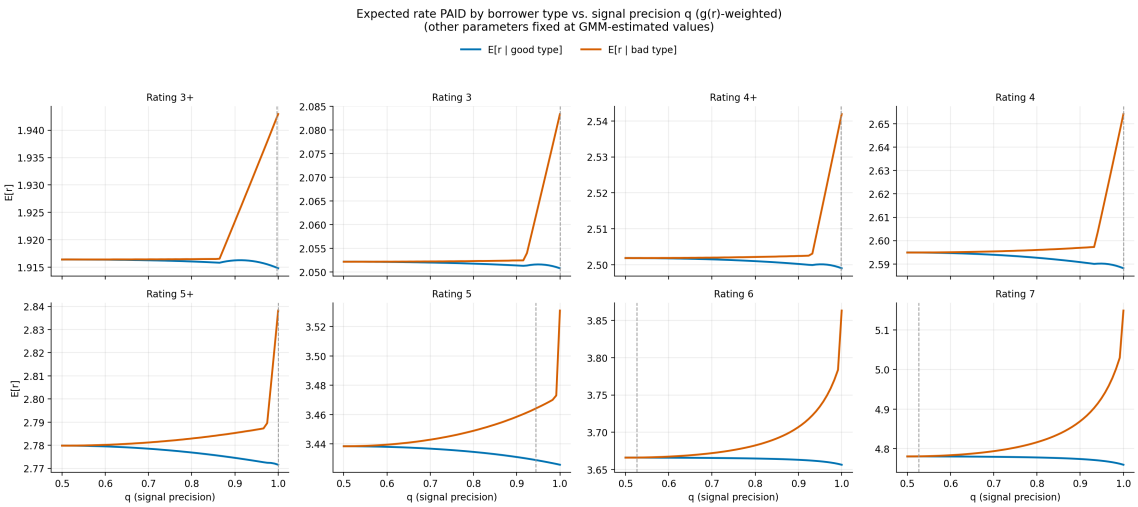


Figure 6: Expected interest rate paid by good vs. bad types as a function of signal precision q, by rating class. For each rating class, the panel holds π, $\bar{r}^g$, $\bar{r}^b$, μ_b, p_g, p_b fixed at their calibrated values (Table 4) and varies q over $[0.5, 1]$; the dashed vertical line marks the class's own estimated $\hat{q}$. Rates are computed under the realized, transaction-weighted rate distribution $g(r)$ of Section 5 (Appendix D.1) - the rate actually paid by a borrower who obtains a loan, not the bank's posted-quote distribution.

Rising rates for bad types and the shift out of pooling. As q increases, the signal grows more informative, the posterior belief α^{s_b} that a bad-signal firm is actually good falls further, and conditioning on the signal becomes more valuable to the bank. Fig. 6 shows the rate paid by bad types rising, in every rating class, essentially monotonically in q, while the rate paid by good types is flat or mildly declining - so the type-conditional rate gap widens. This widening is concentrated late in the $[0.5, 1]$ range rather than gradual throughout: once q crosses the (calibration-specific) threshold at which ϕ^{s_b} turns negative, the bad-signal group leaves full pooling for the hybrid regime, in which a bad-only segment opens above the pooling region and the bank sacrifices the good types whose offer now exceeds $\bar{r}^g$ in exchange for extracting more surplus from the bad types it continues to serve. At the calibrated parameters ϕ^{s_b} stays above its own lower threshold throughout, reaching only -0.07 at rating 3+ and -0.19 at rating 7 at $q = 1$, so the pooling region shrinks but never disappears. Because this threshold depends on the calibrated parameters, its location differs sharply by rating class, mirroring the screening-intensity heterogeneity already documented in Section 5: for ratings 3+ through 5+, whose own estimated $\hat{q}$ is already close to 1 (Table 4), the bad-signal group has already entered the hybrid regime at the calibrated point, so for these classes the low-q region of Fig. 6 is more a counterfactual than a description of the estimated status quo - and q is, in any case, the least precisely pinned-down of the three calibrated parameters for these classes (Section 5.4). Ratings 6 and 7, in contrast, are estimated at $\hat{q} \approx 0.53$, one grid step above the uninformative floor of the domain, consistent with the near-zero screening intensity found in Section 5; ratings 5 through 7 are still fully pooling on both

signals at their own $\hat{q}$, so for these classes Fig. 6 traces the whole arc out of pooling as an in-sample description rather than a counterfactual, with the two types paying rates that agree to four decimals at $\hat{q}$ (3.6660 against 3.6661 at rating 6) and separating only in the immediate vicinity of $q = 1$.

Full separation and profit extraction at $q = 1$. At $q = 1$ the signal is perfect and banks fully separate types, so the type-conditional rate gap in Fig. 6 is at its widest in every rating class (0.028 for rating 3+, 0.207 for rating 6, 0.390 for rating 7). Fig. 8 shows the corresponding effect on the bank's bottom line: total expected profit per loan - the population-share-weighted average of the type-specific profits shown in Fig. 7 - is essentially flat while the bank pools, and rises once the bank starts separating types, reaching its maximum at $q = 1$ in every rating class. Acquiring a (near-)perfect signal therefore lets the bank extract the maximum profit from both types simultaneously; the cross-subsidy documented above disappears, at the cost, as anticipated in Table 2, of some good types exiting the market once their equilibrium offer exceeds $\bar{r}^g$.

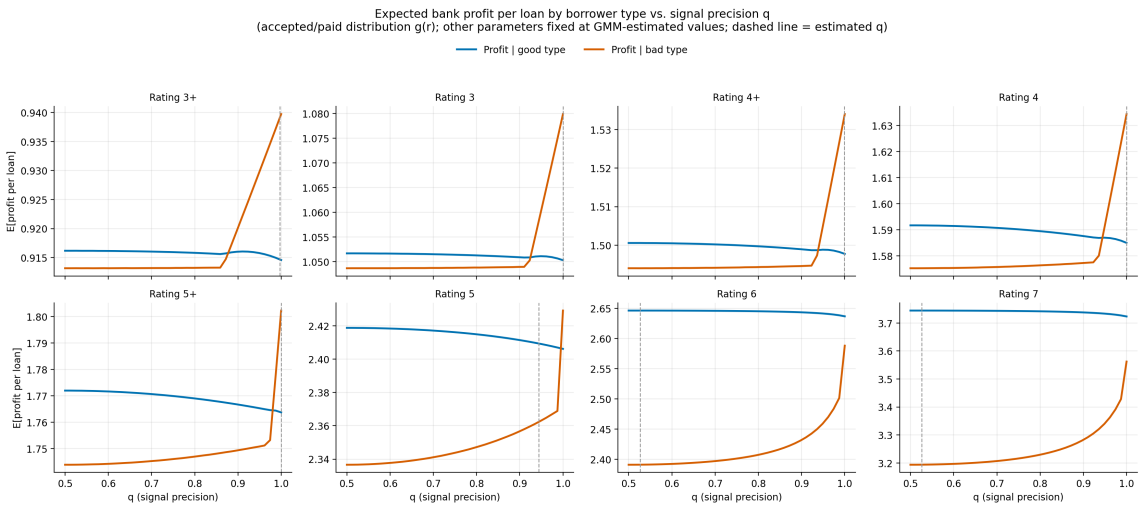


Figure 7: Expected bank profit per loan, by true borrower type, as a function of signal precision q, by rating class. Profit is $E[r \mid \text{type}] - c_{\text{type}}$ on the same realized-rate basis as Fig. 6; the dashed vertical line again marks the class's own estimated $\hat{q}$.

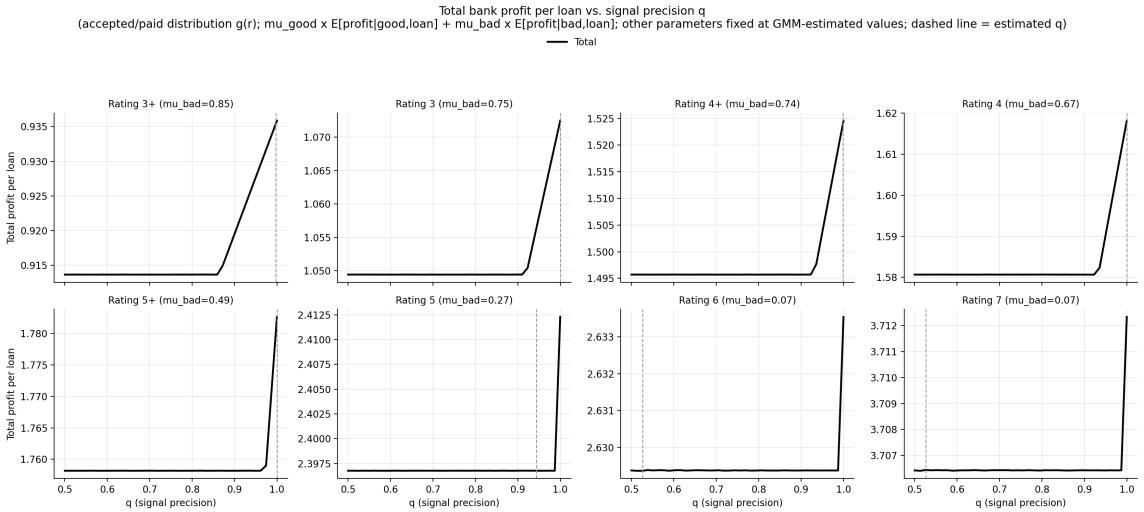


Figure 8: Total expected bank profit per loan as a function of signal precision q, by rating class: the population-share-weighted average $\mu_g E[\text{profit} \mid \text{good}] + \mu_b E[\text{profit} \mid \text{bad}]$ of the two series in Fig. 7. The dashed vertical line marks the class's own estimated $\hat{q}$.

Table 7: Redistribution from a perfect signal, baseline: percent of the rating class's total surplus

Rating	Bad types	Banks	Good types
3+	-0.892	0.881	0.010
3	-0.847	0.833	0.013
4+	-0.806	0.786	0.020
4	-0.958	0.907	0.054
5+	-0.646	0.550	0.095
5	-0.438	0.274	0.163
6	-0.217	0.069	0.148
7	-0.308	0.071	0.236

Notes: change in each party's expected surplus when signal precision moves from $q = 0.5$ to $q = 1$, as a percentage of the class's maximum total surplus. Rows sum to zero: every class is in full pooling at $q = 0.5$, so no firm is rationed at either end and information redistributes without creating surplus.

Table 8: Redistribution from a perfect signal, baseline: percent of each party's own surplus

Rating	Bad types	Banks	Good types
3+	-1.65	2.43	0.10
3	-1.80	2.19	0.08
4+	-1.82	1.93	0.13
4	-2.27	2.37	0.27
5+	-2.13	1.39	0.31
5	-2.76	0.65	0.39
6	-5.58	0.16	0.28
7	-7.75	0.16	0.46

Notes: the same changes, each divided by that party's own surplus at $q = 0.5$ rather than by the class total. Rows do not sum. This is the denominator relevant to a bank weighing the cost of acquiring precision, and to a borrower judging the size of the repricing.

8 Conclusion

This paper asks how much of the dispersion in loan rates across otherwise similar firms reflects unobserved differences in borrower risk, and how much reflects market inefficiencies. Using French loan-level data, I document substantial rate dispersion among firms with similar credit risk and contract, even when they are clients of the same bank. I show that part of it reflects information acquired by the bank, as rate residuals carry forward information about the future evolution of firms' creditworthiness, and this is strongest precisely where information is most valuable - for young and small firms, for longer-maturity loans, and during the financial crisis. The remainder is attributable to market inefficiencies that allow banks to depart from the law of one price.

To separate the two, I propose a novel equilibrium model of loan pricing in which dispersion is jointly determined by imperfect competition, modelled through search frictions, and a costly, imperfect public signal about observationally equivalent borrowers, whose precision is a feature of the market rather than a choice of the banks. Search frictions induce *within-type* dispersion, as banks' find it profitable to randomize to avoid being undercut by competitors, while pricing on the signal generates *between-type* dispersion, as banks charge on average higher rates to borrowers, together with the positive rate–default correlation I observe in the data. Because both forces are latent by construction, I recover them by calibrating the model to the variance of rate residuals, their correlation with default, and the maximum rate observed, separately for each Banque de France rating class.

The calibration delivers a clear quantitative message. The signal available in the market is highly precise, yet the quantitative role of information in accounting for observed dispersion is rather small: it accounts for at most about 5% relative to a no-information

benchmark, and for essentially nothing in the riskiest classes. Imperfect competition is therefore the prevailing force behind the dispersion I measure, even where the market's information is close to perfect.

Information nonetheless matters for surplus sharing. Screening substantially benefits safer borrowers by limiting the two costs that adverse selection imposes on them: in a pooling equilibrium good types cross-subsidize bad types, and when adverse selection is severe they are priced above their reservation rate and exit the market altogether. More precise signals shrink both, lowering the borrowing costs of safer firms and reducing credit rationing, while bad types lose the subsidy they were implicitly receiving. A small aggregate contribution to dispersion and large consequences for how the surplus is shared are thus fully compatible. For policy, the two findings point in the same direction: since dispersion is driven mainly by imperfect competition rather than by risk-based pricing, a policy maker seeking to maximise firms' surplus and limit capital misallocation should aim at making the banking market more contestable, and at widening the information about borrowers that the market makes available, rather than at curbing banks' use of it.

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A Additional tables and figures

Back to main↑

LA COTE DE CRÉDIT
(capacité de l'entreprise à honorer ses engagements financiers à 3 ans)

3++	Excellente
3+	Très forte
3	Forte
4+	Assez forte
4	Correcte
5+	Assez faible
5	Faible
6	Très faible
7	Appelant une attention spécifique présence d'au moins un incident de paiement significatif
8	Menacée
9	Compromise
P	Procédure collective redressement ou liquidation judiciaire
0	Pas de documentation comptable analysée et absence d'informations défavorables

Figure A.1: Source: Banque de France. Credit rating scale until 2022.

B Additional facts on rate dispersion

Back to main↑

Table B.1: Data: Nouveaux Credits (BdF), 2006q1-2024q2. Excludes credit lines and undefined maturity products. All credit and balance-sheet variables are expressed in terms of reported assets. Multiple products (dummy) equals one if the firm takes more than one product type in the quarter, counting the excluded products; Multiple loans, one product (dummy) equals one if it takes several loans in the quarter, all of the same type.

	Total effective rate					
log.Loan quantity (K-Eur)	-0.2762***	-0.1994***	-0.1969***	-0.1826***	-0.1859***	-0.1242***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Maturity (months)	-0.0006**	0.0020***	0.0029***	0.0019***	0.0020***	0.0019***
	(0.044)	(0.003)	(0.002)	(0.004)	(0.004)	(0.005)
Age	-0.0108***	-0.0076***	-0.0065***	-0.0057***	-0.0055***	-0.0033***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Age squared	0.0000***	0.0000***	0.0000***	0.0000***	0.0000***	0.0000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Young firm 0-3y (dummy)	-0.0484***	-0.0482	-0.0539	-0.0288	-0.0259	-0.0654**
	(0.006)	(0.244)	(0.200)	(0.367)	(0.397)	(0.017)
New bank-firm relationship (dummy)	0.1549***	0.0027	-0.0044	-0.0317**	-0.0321**	-0.0455***
	(0.000)	(0.942)	(0.907)	(0.027)	(0.023)	(0.003)
Variable rate (dummy)	0.3917***	0.4994***	0.2026**	0.1477	0.1455	0.1603*
	(0.000)	(0.000)	(0.022)	(0.112)	(0.112)	(0.085)
Personal loan (dummy)	-0.3016**	-0.0067	0.0758	0.0329	0.0374	0.0733
	(0.044)	(0.958)	(0.391)	(0.686)	(0.637)	(0.309)
Individual entrepreneur (dummy)	0.4292***	0.4879***	0.4569***	0.3821***	0.3708***	0.2313***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.003)
Loan over assets	2.2482***	1.6201***	1.5301***	1.3391***	1.3166***	0.7238***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)
Pledged guarantees	1.6770***	0.2564	0.0107	-0.4193***	-0.3806***	-0.3332***
	(0.000)	(0.117)	(0.931)	(0.000)	(0.000)	(0.000)
Revenues	-0.0940***	-0.0426***	-0.0342***	-0.0133***	-0.0113**	-0.0116**
	(0.000)	(0.000)	(0.001)	(0.010)	(0.019)	(0.021)
Bank Debt	0.4944***	0.2907***	0.2538***	0.2188***	0.2368***	0.0177
	(0.000)	(0.000)	(0.001)	(0.000)	(0.000)	(0.580)
Non-Bank Debt	0.0157	-0.0572	-0.0504	0.0206	0.0212	-0.0339
	(0.782)	(0.729)	(0.773)	(0.859)	(0.848)	(0.665)
Mobilised credit	-0.2899***	-0.1095***	-0.0607	-0.0315	-0.0233	0.0185
	(0.000)	(0.007)	(0.125)	(0.377)	(0.483)	(0.526)
Short Term Debt	0.3258***	0.0071	-0.0315	0.0779	0.0745	-0.1557***
	(0.000)	(0.952)	(0.803)	(0.272)	(0.282)	(0.004)
Undrawn credit lines	-0.0385	-0.0994	-0.1246	-0.2140**	-0.1931**	-0.1339*
	(0.762)	(0.410)	(0.329)	(0.013)	(0.025)	(0.085)
Multiple products (dummy)	0.0296	0.0339	0.0025	-0.0024	-0.0012	0.0196*
	(0.122)	(0.120)	(0.897)	(0.869)	(0.933)	(0.055)
Multiple loans, one product (dummy)	-0.1562**	-0.0357	-0.0565	-0.0224	-0.0227	0.0029
	(0.017)	(0.506)	(0.285)	(0.471)	(0.459)	(0.909)
Quarter FE	✓					
Lender × Quarter FE		✓				
Lender × Quarter × Loan Type FE			✓			
Lender × Quarter × Loan Type × 2digit Sector FE				✓	✓	✓
Region FE					✓	✓
Group indicator, Rating FE						✓
Observations	449,973	449,973	449,973	449,973	449,973	449,973
R^2	0.57	0.72	0.76	0.82	0.82	0.83

p-values in parentheses

* p< 0.10, ** p< 0.05, *** p< 0.01

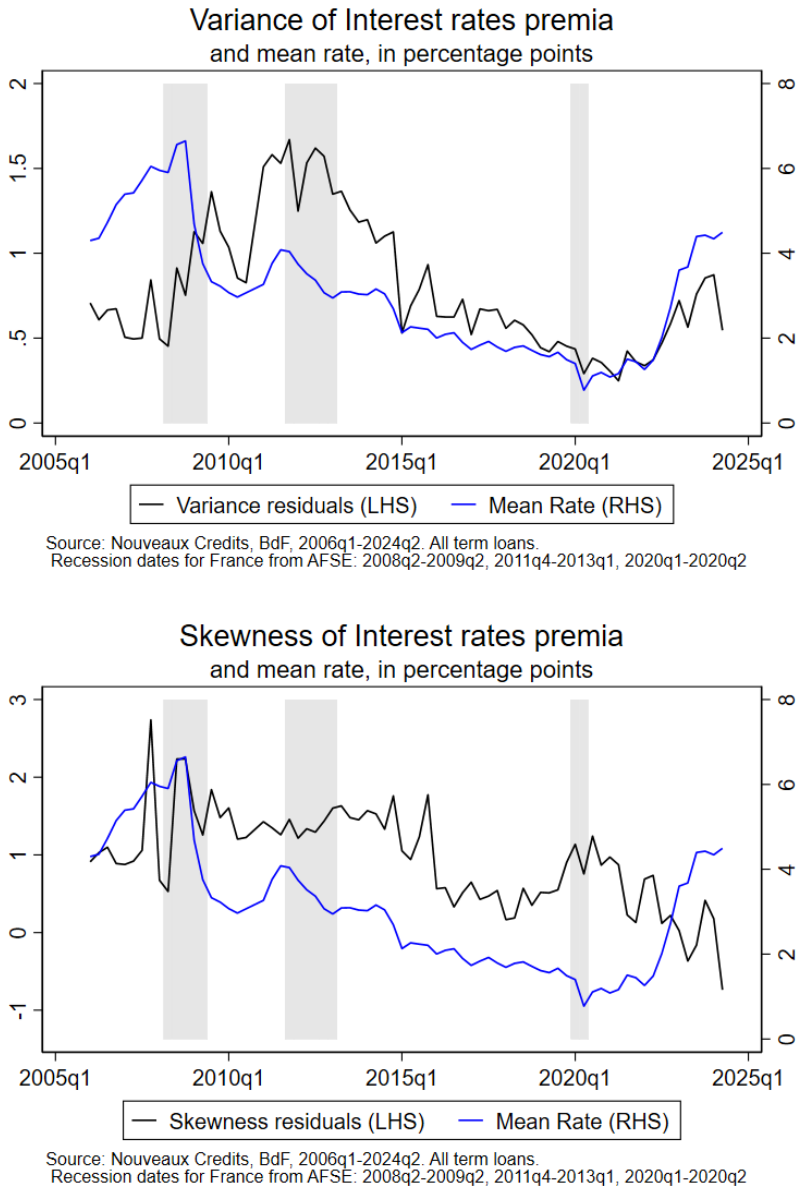
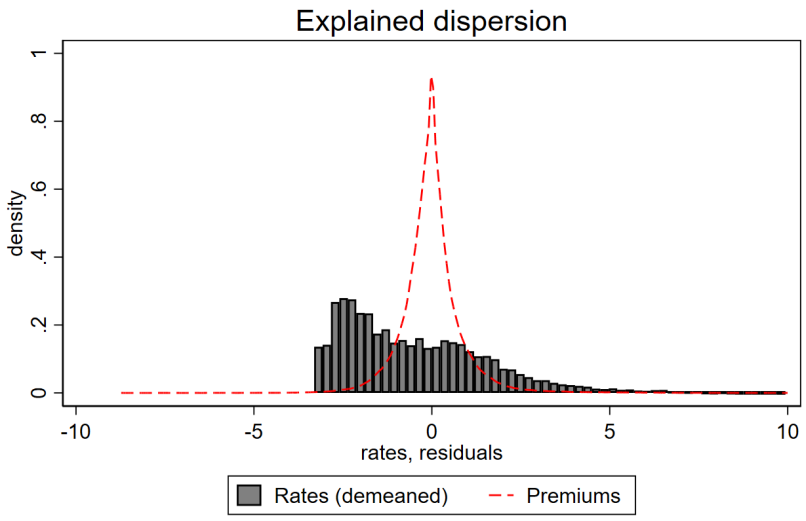


Figure B.2: Variance (top) and skewness (bottom) of interest rate residuals by quarter, black line, left axis, against the mean interest rate, blue line, right axis, both in percentage points. Shaded bands are French recessions (AFSE): 2008q2–2009q2, 2011q4–2013q1 and 2020q1–2020q2. Data: Nouveaux Crédits (BdF), 2006q1–2024q2, all term loans.



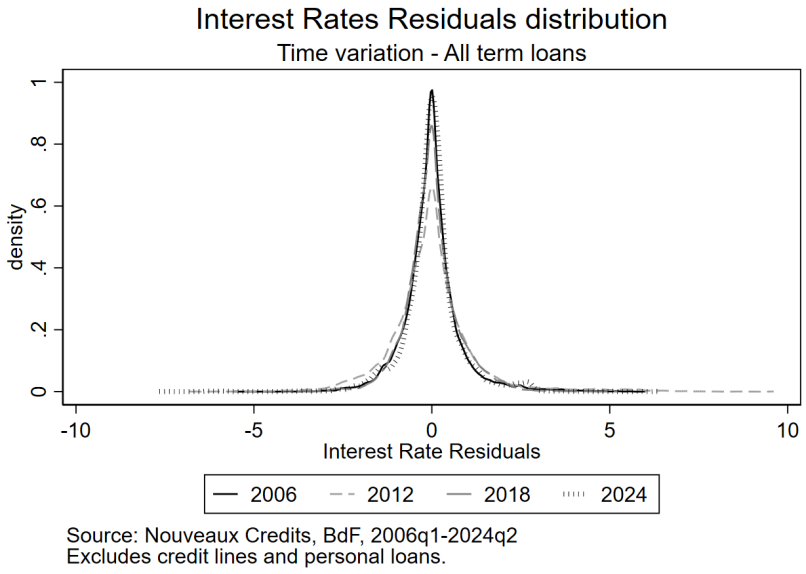


Figure B.3: Regression outcome. Top: residuals versus actual rates, all sample. Bottom: residuals distribution for selected years.

B.1 Additional specification 1: Firm fixed effects

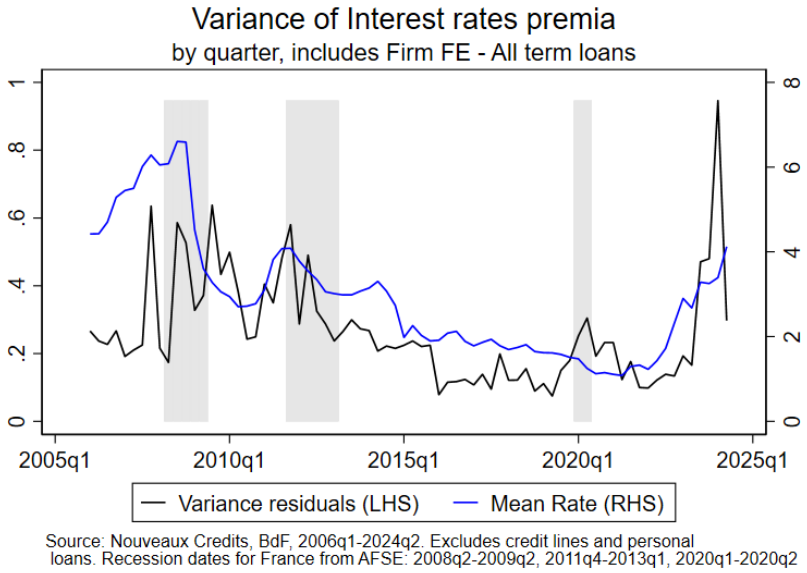
	Total effective rate					
Loan quantity (K-Eur)	-0.0000***	-0.0000***	-0.0000	-0.0000	-0.0000**	-0.0000
	(0.000)	(0.006)	(0.486)	(0.143)	(0.044)	(0.117)
Maturity (months)	-0.0016***	0.0013***	0.0015***	0.0015***	0.0014***	0.0014***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Variable rate (dummy)	0.2381***	0.1357***	0.2176***	0.2368***	0.2113***	0.1995***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Loan over assets	0.0524***	-0.0068**	-0.0076***	-0.0070**	-0.0069**	-0.0074***
	(0.000)	(0.025)	(0.009)	(0.012)	(0.012)	(0.007)
Quarter FE	✓					
Firm × Quarter FE		✓	✓	✓	✓	
Lender FE			✓			
Lender × Quarter FE				✓	✓	
Firm × Lender × Quarter FE						✓
Loan Type FE					✓	✓
Observations	590,621	197,343	197,328	196,048	196,042	156,211
R^2	0.51	0.92	0.93	0.94	0.94	0.94

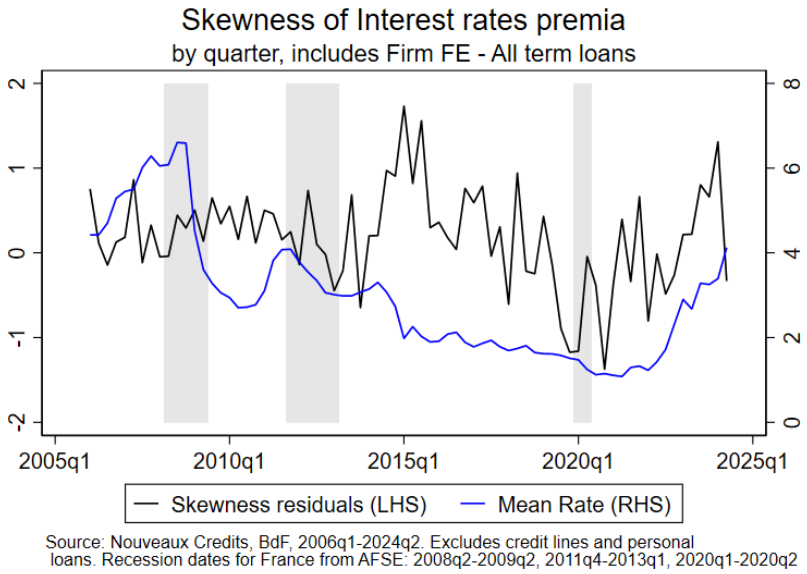
p-values in parentheses

* p< 0.10, ** p< 0.05, *** p< 0.01

Table B.2: "Data: Nouveaux Credits (BdF), 2006q1-2024q2""Excludes credit lines and personal loans. All credit and"" balancesheet variables are expressed in terms of reported assets."

Figure B.4: Variance and skewness of residuals from the firm FE specification.





B.2 Additional specification 2: truncated regression approach

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One possible concern is that it is not possible to observe rates below 0. Regression approaches assuming normality may be downward biased, predicting negative rates even when they are impossible to observe. Two approaches can help address this issue. Firstly, I verify to what extent the OLS is predicting negative values and I find that only 0.4% of predictions fall slightly below zero; this is probably due to the fact that most of the variation is absorbed by fixed effects. Secondly, the same analysis can be repeated using Tobit regression, which accounts for this lower bound; this approach is more limited in

the inclusion of many interactions of fixed effects, thus it forces a more parsimonious approach, resulting in a worse performance.

B.3 Rate dispersion and bank heterogeneity

One question that naturally arises is: is this dispersion hiding heterogeneity across banks? The answer is yes. There is consistent heterogeneity between banks in the explained variation, as shown below by repeating the above estimation of rates by bank, and collecting the R^2.

$$\text{Rate}_{i,b,t} = \alpha + \beta \cdot \text{LoanChar}_{i,b,t} + \gamma \cdot \text{FirmChar}_{i,t} + \lambda_t + \varepsilon_{i,b,t}. \tag{10}$$

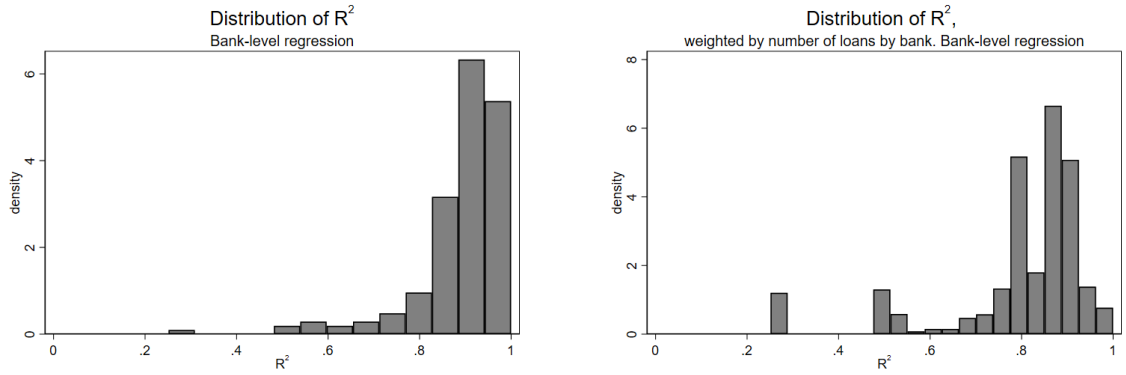


Figure B.5: The right image reports the distribution of R^2 weighted by the number of loans issued.

In the following section the credit rating of firms will be used as a source of public information available to banks and to study the evolution of firms' outcomes in terms of their default probability. Consequently, it is interesting at this stage to report to what extent banks use these ratings. Firstly, the same estimation is repeated by including only the rating as the explanatory variable, along with time and loan type FE and by lender, and collecting the R^2.

$$\text{Rate}_{i,b,t} = \alpha + \beta \cdot \text{Firm Rating}_{i,b,t} + \lambda_t + \varepsilon_{i,b,t}, \tag{11}$$

where the only fixed effects kept are by loan type and month.

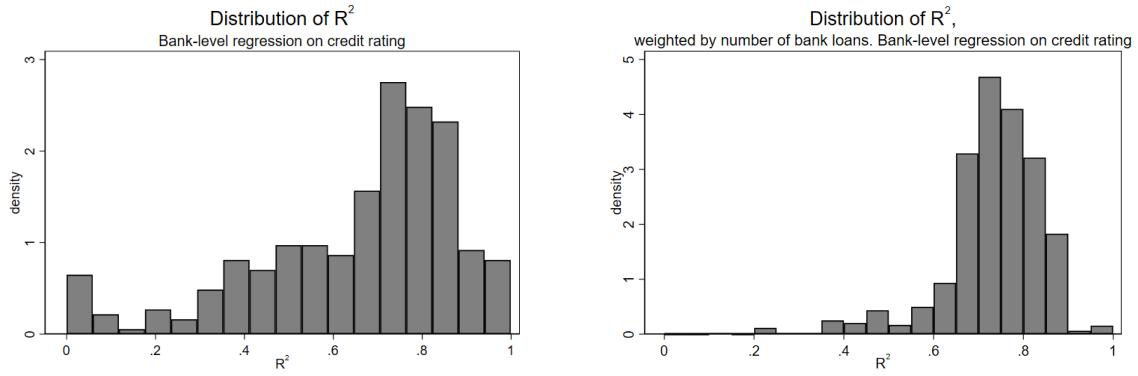


Figure B.6: The right image reports the distribution of R^2 weighted by the number of loans issued.

Secondly, one can look at how much of the variation is explained when including the public rating, in particular by collecting the increment in the R^2.

$$\begin{aligned} \text{Rate}_{i,b,t} &= \alpha + \lambda_t + \varepsilon_{1,i,b,t}, && \rightarrow \text{Get } R_1^2 \\ \text{Rate}_{i,b,t} &= \alpha + \beta \cdot \text{Firm Rating}_{i,b,t} + \lambda_t + \varepsilon_{2,i,b,t}, && \rightarrow \text{Get } R_2^2 \\ \Delta R^2 &= R_2^2 - R_1^2, \end{aligned}$$

where the only fixed effects kept are by loan type and month.

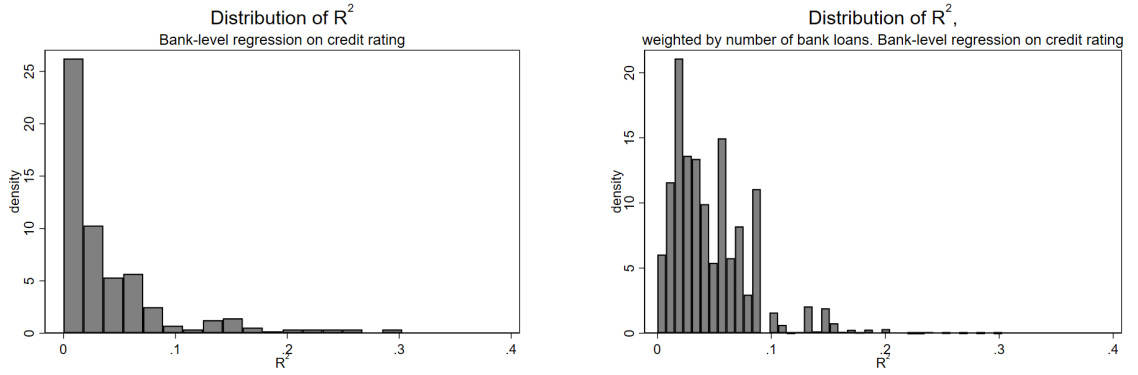


Figure B.7: The right image reports the distribution of R^2 weighted by the number of loans issued.

B.4 Residuals and market competition

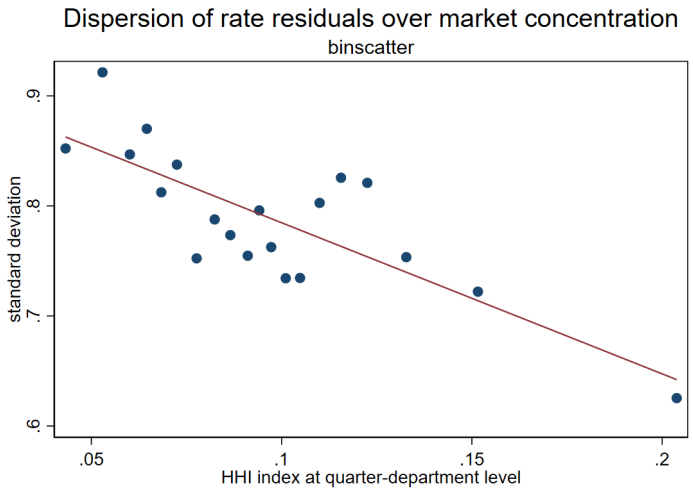


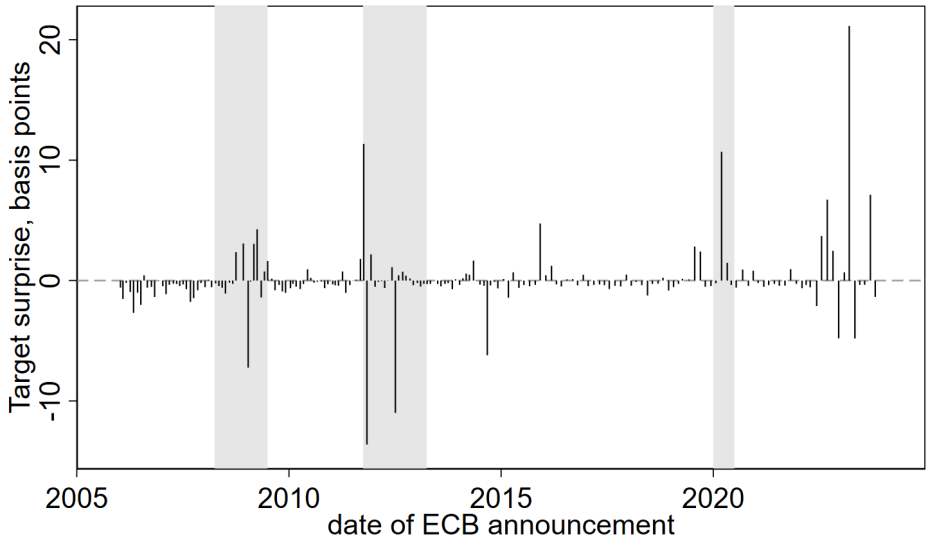
Figure B.8: Unexplained rate residuals' dispersion over HHI by department-quarter.

B.5 Rate dispersion and monetary policy

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Monetary policy shocks. I use the monetary policy surprises of Altavilla et al. (2019), updated to October 2023. For each ECB announcement, the Euro Area Monetary Policy Event-Study Database records the change in the 1-, 3- and 6-month and 1-, 2-, 5- and 10-year OIS rates in a narrow window around the press release and around the press conference. In each window I extract three latent factors by principal components and rotate them so that they can be read as policy surprises: the second and third factors do not load on the one-month OIS rate, and the third has the smallest variance before the 2008 crisis. The identifying assumption is that policy decisions do not respond to asset prices at this frequency, and given the narrow window the movement in asset prices measures the surprise component of the decision. I use the Target factor, from the press-release window, which captures surprises in the current policy rate; it is standardized such that it has a unit effect on the one-month OIS rate. Figure B.9 plots the Target surprise at each announcement.

Figure B.9: Target surprise of Altavilla et al. (2019) at each ECB announcement, January 2006 to October 2023, in basis points: the factor is scaled to have a unit effect on the one-month OIS rate. It is estimated from changes in OIS rates in a narrow window around the ECB press release. Shaded bands are French recessions (AFSE): 2008q2–2009q2, 2011q4–2013q1 and 2020q1–2020q2.



Specification. For each lender and quarter I compute the standard deviation and the skewness of the rate residuals from Eq. (1) across that lender's loans, and regress these moments on the lagged policy surprises, aggregated at the quarterly level. On 7189 lender-quarter observations, I estimate a specification in levels, which controls for the lagged moment because dispersion is persistent,

$$Y_{b,t} = \beta_0 + \beta_1 \cdot \text{MPshock}_{t-1} + \beta_2 Y_{b,t-1} + \text{FE}_b + \varepsilon_{b,t}, \tag{12}$$

where $Y_{b,t}$ is either the standard deviation or the skewness. The coefficient β_1 measures the effect of a surprise on the level of dispersion, holding its own lag fixed. I report it with one lag of the shock and with three.

	Dispersion		Skewness	
	(1)	(2)	(3)	(4)
L1.rate shock	-0.0096***	-0.0085***	-0.0135***	-0.0099*
	(0.000)	(0.003)	(0.010)	(0.093)
L2.rate shock		0.0001		-0.0227***
		(0.979)		(0.000)
L3.rate shock		-0.0017		-0.0208***
		(0.568)		(0.000)
L1.SD / L1.Skew	0.2813***	0.2799***	0.1294***	0.1171***
	(0.000)	(0.000)	(0.000)	(0.000)
Lender FE	✓	✓	✓	✓
Observations	7,189	7,077	7,189	7,077
R^2	0.40	0.41	0.23	0.24

p-values in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table B.3: Response of the distribution of rate residuals to monetary policy. Each observation is a lender-quarter. Dispersion and Skewness are the standard deviation and the skewness of the lender's rate residuals from Eq. (1) in the quarter. Each column estimates Eq. (12), with one lag of the shock in columns (1) and (3) and three lags in columns (2) and (4). The shock is the Target factor of Altavilla et al. (2019) summed over the ECB announcements in the quarter, in standard deviations of the announcement-level factor (one standard deviation is 2.7 basis points of the one-month OIS rate). Data: Nouveaux Crédits (BdF), 2006q1–2024q2, quarterly shocks.

Results. Table B.3 reports the estimates. A monetary tightening reduces the dispersion of the unexplained component of loan rates. The effect on the level of the standard deviation is negative and significant at 1% in both specifications (-0.0096 and -0.0085). It is concentrated in the quarter after the shock: the second and third lags are small and not significant. The lagged dependent variable enters at 0.28, confirming that dispersion is persistent, which is why the specification controls for it; with this persistence, the effect of a shock fades within about two quarters. Since one standard deviation of the shock is a surprise of 2.7 basis points in the one-month OIS rate, a surprise of 25 basis points lowers the standard deviation of a lender's rate residuals by about 0.09 percentage points.

The skewness moves in the same direction. A tightening reduces the level of skewness, significantly at 1% with one lag and at 10% with three, and the second and third lags of the shock are individually significant and negative, suggesting that the tail adjusts more slowly than the dispersion does.

One concern is that dispersion falls because the composition of borrowers changes after a tightening. The construction of the residual limits this: it already nets out the loan

contract, the firm's observables, its rating class and the lender's schedule in that month, so a shift towards safer or larger borrowers is absorbed by the controls rather than showing up in the residual.

B.6 Alternative specifications: Rate residuals and changes in default probability

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This section reports four alternatives to Eq. (2). The first replaces the rating revision with the default itself. The other three turn the regression around and put the revision in default probability on the left, so that the coefficient of interest is the one on the rate r_{ilbt} itself. The reversal changes the normalization, not the question: a positive coefficient says that firms charged a higher rate at issuance are revised towards a higher default probability afterwards, which is what $\beta_k > 0$ says in Eq. (2). Magnitudes are not comparable across the two directions - one is measured in percentage points of rate per percentage point of default probability, the other the other way round.

B.6.1 Actual defaults

I repeat the test replacing the rating revision with the default itself; for each horizon k, I run:

$$r_{ilbt} \;=\; \alpha \;+\; \sum_{j=1}^{k} \beta_j^D \, D_{i,t+j} \;+\; \mathbf{x}'_{ilbt}\,\gamma \;+\; \mu_{bt\tau s} \;+\; \varepsilon_{ilbt}, \tag{13}$$

in which $D_{i,t+j}$ equals 100 if firm i defaults in the j-th year after issuance and 0 otherwise. This test uses the annual rating file, which records a default whenever it occurs, so it needs no balanced panel; it adds a dummy for a new bank-firm relationship to $\mathbf{x}_{ilbt}$ and clusters standard errors by lender and by quarter. I estimate Eq. (13) once per horizon $k = 1, \ldots, 10$, each regression carrying the default indicators up to that horizon, and report β_k^D.

Rating revisions on the left-hand side. Three choices distinguish the three specifications that follow: whether the revision is the year-on-year change or the cumulative change since issuance, whether the earlier revisions enter as controls, and whether the sample is balanced across horizons. Everything else is held fixed. All three carry the same controls $\mathbf{x}_{ilbt}$ and fixed effects $\mu_{bt\tau s}$ as Eq. (2), cluster standard errors by lender, drop loans to firms already in default at issuance ($P_{i,t} \neq 1$), and run one regression per horizon $k = 1, \ldots, 10$, reporting the coefficient on the rate. The rating-class split is the one used in Fig. B.18: classes 3++ to 4+ for the good group and classes 5 to 9 for the bad one, with classes 4 and 5+ left out.

Figure B.10: Regression coefficient of actual default, controlling for lags and observables, on interest rates.

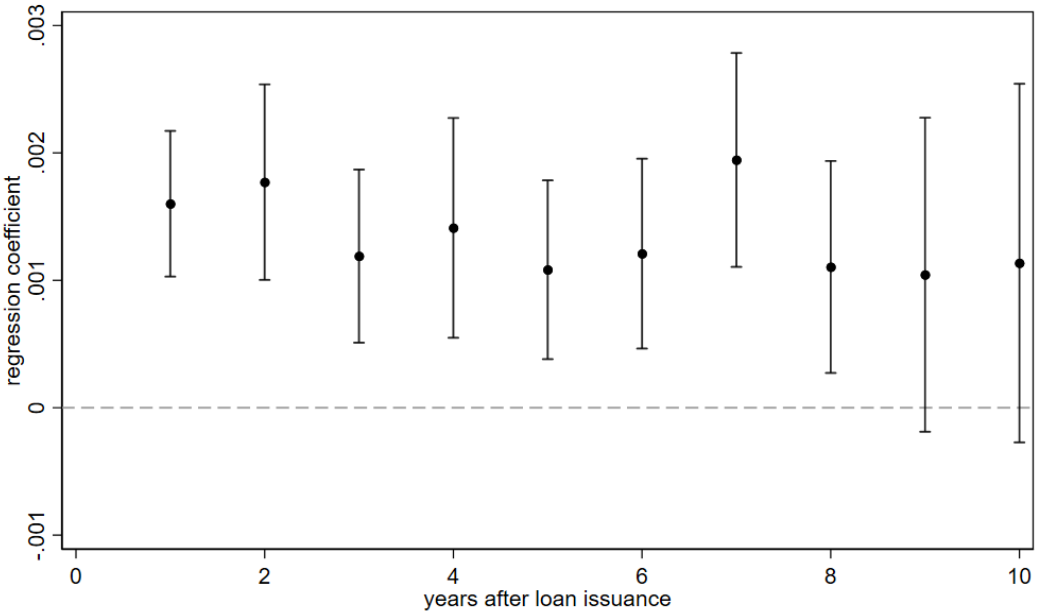
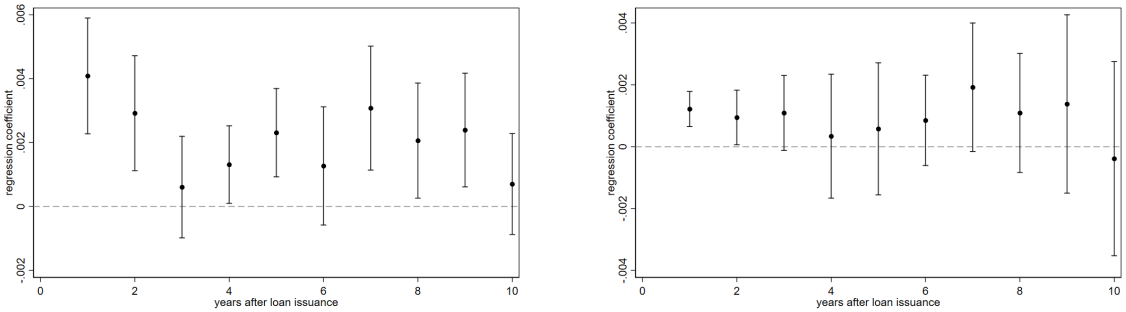


Figure B.11: Correlation computed separately for firms with a good ex ante rating, classes 3++ to 4+ (left), and a bad one, classes 5 to 9 (right). Classes 4 and 5+ are left out.



B.6.2 Year-on-year revisions, past revisions controlled for, balanced sample

For each horizon k I estimate

$$\Delta P_{i,t+k} \;=\; \alpha_k \;+\; \theta_k\, r_{ilbt} \;+\; \sum_{j=1}^{k-1} \delta_{kj}\, \Delta P_{i,t+j} \;+\; \mathbf{x}'_{ilbt}\, \gamma_k \;+\; \mu_{bt\tau s} \;+\; \varepsilon_{ilbt}, \tag{14}$$

in which $\Delta P_{i,t+k}$ is the change in default probability between years $k-1$ and k after issuance, in percentage points, as in Eq. (2). The sample is balanced: a loan enters only if every revision $\Delta P_{i,t+1}, \ldots, \Delta P_{i,t+10}$ is observed, so the same firms identify every θ_k. This is the closest counterpart to Eq. (2), which is also estimated on the balanced panel. The rate predicts the first-year revision, $\theta_1 = 0.064$ (s.e. 0.017), and the coefficient falls towards zero by the tenth year. The pattern holds for the good rating classes and is too noisy to sign for the bad ones.

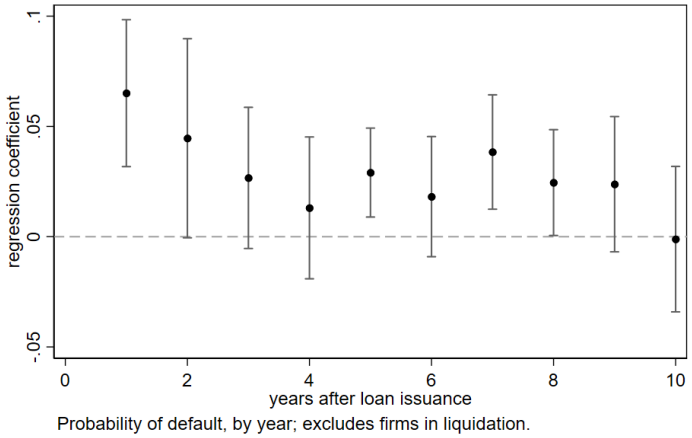
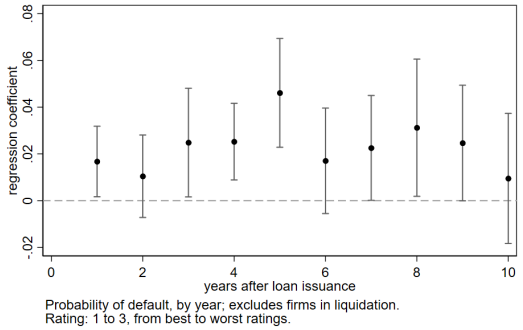
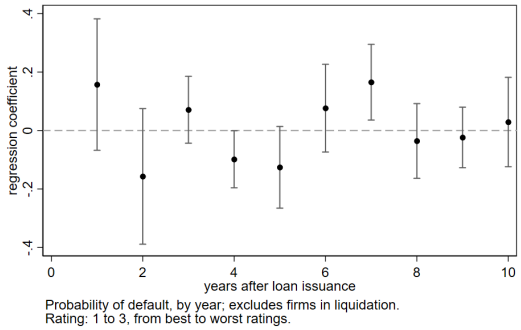


Figure B.12: Coefficient θ_k on the loan rate in Eq. (14), by horizon k. The dependent variable is the year-on-year change in the firm's default probability between years $k-1$ and k after issuance, in percentage points. Balanced sample; past revisions controlled for; bars are 95% confidence intervals, standard errors clustered by lender.

Figure B.13: Coefficient θ_k in Eq. (14), estimated separately for firms with a good ex ante rating, classes 3++ to 4+ (left), and a bad one, classes 5 to 9 (right). Classes 4 and 5+ are left out. Note the different vertical scales.





B.6.3 Year-on-year revisions, past revisions controlled for, unbalanced sample

Eq. (14) keeps only firms followed for the full ten years, which selects on survival in the rating file. Dropping that requirement, I estimate the same equation horizon by horizon on whatever firms are observed at that horizon:

$$\Delta P_{i,t+k} \;=\; \alpha_k \;+\; \theta_k\, r_{ilbt} \;+\; \sum_{j=1}^{k-1} \delta_{kj}\, \Delta P_{i,t+j} \;+\; \mathbf{x}'_{ilbt}\, \gamma_k \;+\; \mu_{bt\tau s} \;+\; \varepsilon_{ilbt}, \qquad (15)$$

so the sample shrinks with k and differs across horizons. The coefficients are larger than on the balanced panel - $\theta_1 = 0.14$ (s.e. 0.022) - and decline to zero by the tenth year. The gap between the two samples says that the firms that stay in the rating file for ten years are the ones whose rates carry less information about their own revisions.

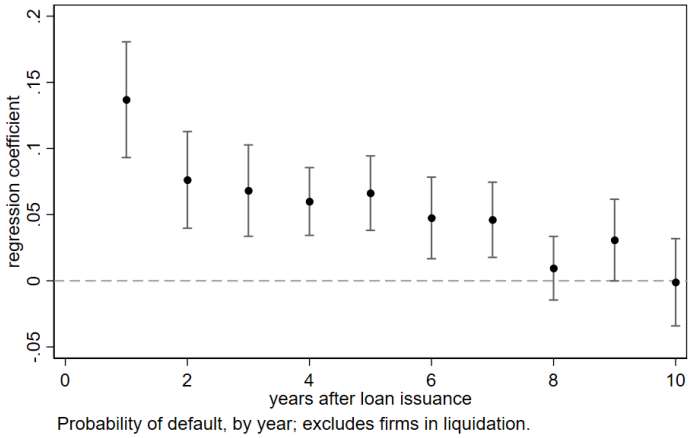
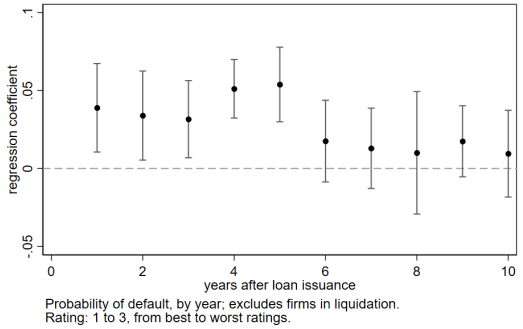
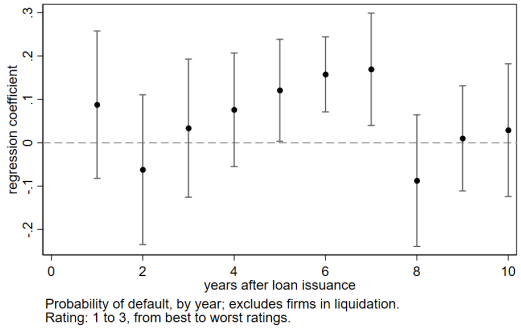


Figure B.14: Coefficient θ_k on the loan rate in Eq. (15), by horizon k. The dependent variable is the year-on-year change in the firm's default probability between years $k-1$ and k after issuance, in percentage points. Unbalanced sample; past revisions controlled for; bars are 95% confidence intervals, standard errors clustered by lender.

Figure B.15: Coefficient θ_k in Eq. (15), estimated separately for firms with a good ex ante rating, classes 3++ to 4+ (left), and a bad one, classes 5 to 9 (right). Classes 4 and 5+ are left out. Note the different vertical scales.





B.6.4 Cumulative revision since issuance

The year-on-year revision at horizon k is an increment, so θ_k in Eqs. (14) and (15) measures how much of the revision arriving in year k alone the rate anticipates. Measuring the revision from issuance instead asks how much of the firm's total drift in default probability the rate anticipates by year k:

$$\Delta P_{i,t\to t+k} \;=\; \alpha_k \;+\; \theta_k\, r_{ilbt} \;+\; \mathbf{x}'_{ilbt}\,\gamma_k \;+\; \mu_{bt\tau s} \;+\; \varepsilon_{ilbt}, \tag{16}$$

in which $\Delta P_{i,t\to t+k} = 100 \times \big(P_{i,t+k} - P_{i,t}\big)$, the same cumulative object used for the demographic splits in Fig. 4. Past revisions no longer enter, since the dependent variable already contains them. At $k = 1$ the two measures coincide and Eq. (16) reproduces Eq. (15), $\theta_1 = 0.14$. Beyond that the coefficient stays positive at every horizon, peaking at $\theta_5 = 0.18$ (s.e. 0.023) and still 0.11 at ten years. The rate anticipates a level shift in the firm's default probability that does not unwind, and the decline in Fig. B.14 reflects the exhaustion of new information, not a reversal of the old.

Splitting by rating class sharpens the reading. For the good classes the cumulative coefficient rises from 0.039 at one year to 0.14 at five and is positive at every horizon. For the bad classes it is two to three times larger at the longer horizons but is not distinguishable from zero in the first three years: rates predict where bad borrowers end up, not when they get there.

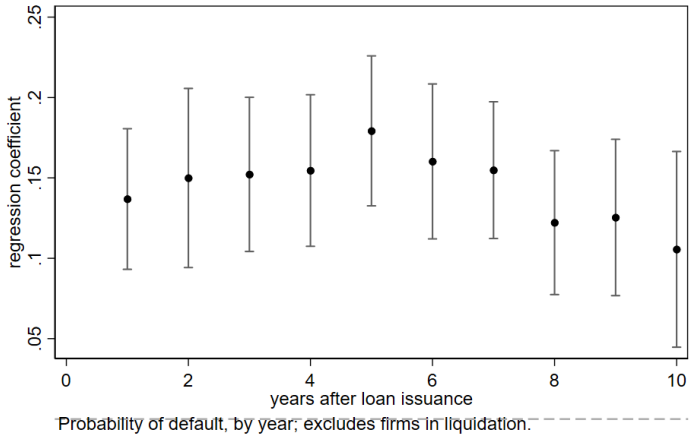
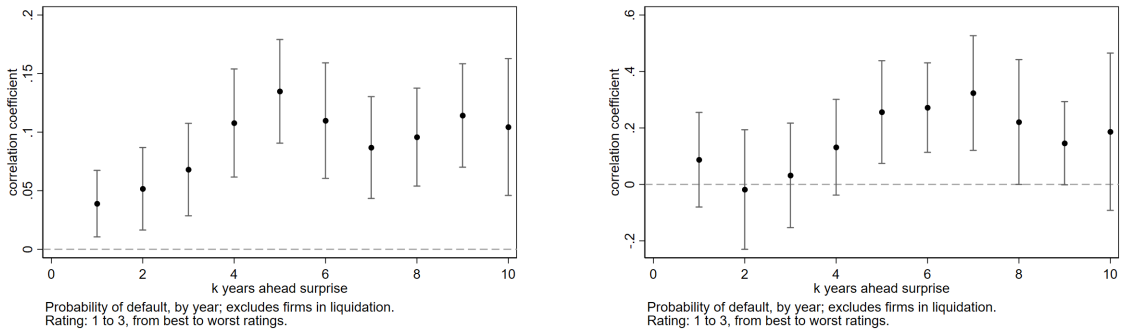


Figure B.16: Coefficient θ_k on the loan rate in Eq. (16), by horizon k. The dependent variable is the cumulative change in the firm's default probability between issuance and year k, in percentage points. Bars are 95% confidence intervals, standard errors clustered by lender.

Figure B.17: Coefficient θ_k in Eq. (16), estimated separately for firms with a good ex ante rating, classes 3++ to 4+ (left), and a bad one, classes 5 to 9 (right). Classes 4 and 5+ are left out. The horizontal axis is the horizon k of the cumulative revision. Note the different vertical scales.



B.7 Correlation across firms' demographics

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Figure B.18: Correlation computed separately for firms with a good ex ante rating, classes 3++ to 4+ (left), and a bad one, classes 5 to 9 (right). Classes 4 and 5+ are left out.

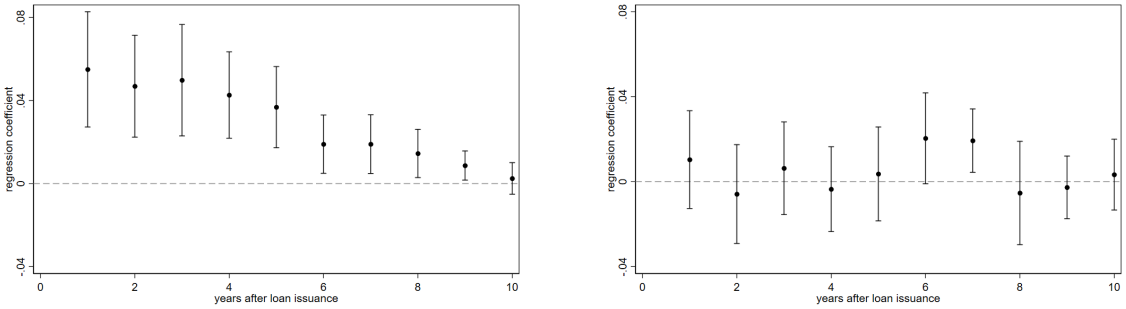
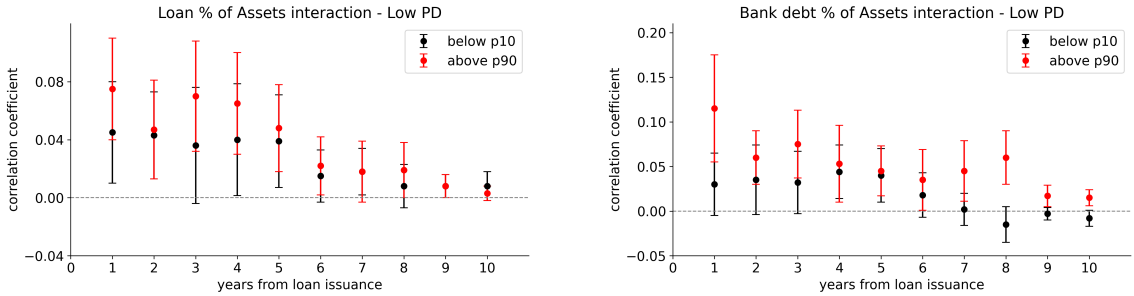


Figure B.19: Interactions of ex-post default probability and loan-size over assets (left) and bank debt over assets (right)



The figure compares the correlation coefficients of residual rates and unexpected changes in ex-post default probability across percentiles of loan-size over assets (left) and size of bank debt over assets (right) for firms with a good ex ante rating (classes 3++ to 4+).

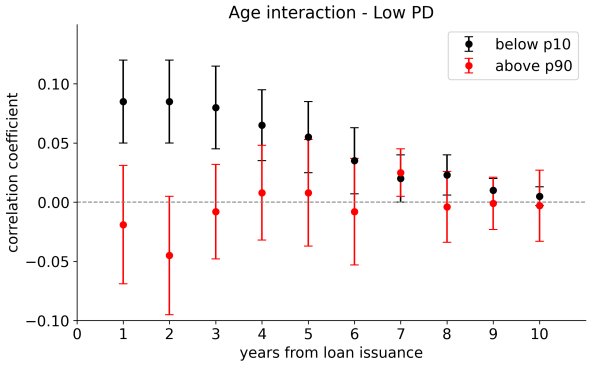


Figure B.20: The figure shows the correlation coefficients of the interactions of ex-post default probability and age (below 10th vs above 90th age percentile). Sample of firms with a good ex ante rating (classes 3++ to 4+).

B.7.1 Fixed vs variable rate test

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	Default Probability						
Variable rate (dummy)	0.1176**		0.1407	0.1039**			
	(0.014)		(0.127)	(0.026)			
cum dRate end y2		0.0433			0.0374		
		(0.229)			(0.254)		
annual dRate end y1			0.0812			0.0524	0.0425
			(0.600)			(0.711)	(0.773)
annual dRate end y2			0.0567			0.0363	0.0128
			(0.330)			(0.551)	(0.879)
Variable rate (dummy) × Loan over assets				0.1932			
				(0.370)			
cum dRate end y2 × Loan over assets					0.0711		
					(0.264)		
annual dRate end y1 × Loan over assets						-0.1330	
						(0.763)	
annual dRate end y2 × Loan over assets						0.0905	
						(0.148)	
annual dRate end y1 × Bank Debt share							0.0058
							(0.980)
annual dRate end y2 × Bank Debt share							0.0986
							(0.623)
Loan over assets	0.2250**	0.2002**	0.2048**	0.1808*	0.1894*	0.1849*	0.1929*
	(0.030)	(0.035)	(0.035)	(0.094)	(0.054)	(0.059)	(0.054)
Bank Debt share	0.0846**	0.1139**	0.1134**	0.0850**	0.1118**	0.1114**	0.1127**
	(0.040)	(0.019)	(0.019)	(0.034)	(0.022)	(0.023)	(0.025)
Lender × Month × Loan Type × 2digit Sector FE	✓	✓	✓	✓	✓	✓	✓
Region FE	✓	✓	✓	✓	✓	✓	✓
All other observables	✓	✓	✓	✓	✓	✓	✓
Observations	102,463	96,414	96,414	102,463	96,414	96,414	96,414
R^2	0.42	0.40	0.40	0.42	0.40	0.40	0.40

p-values in parentheses

* p< 0.10, ** p< 0.05, *** p< 0.01

Table B.4: Data: Nouveaux Crédits (BdF), 2006q1-2024q2. Term loans, at least 30-month maturity, only firms with a rating. Dependent variable: change_PD3. Exclude loans with non-specified reference rate.

Identifying assumptions. The exercise rests on three assumptions.

(i) *Exogeneity.* Reference-rate changes are set by monetary policy and aggregate conditions, and it is realistic to assume they are not endogenous to expectations about firm i's outcomes.

(ii) *No sorting on private information or rate expectations.* The main concern is that

firms may self-select into fixed- or variable-rate contracts based on their expectations about the future evolution of rates. As explained below, I mitigate this issue in two ways.

(iii) *Parallel trends.* Absent the rate increase, default probabilities of fixed- and floating-rate borrowers would have evolved in the same way.

The rest of this section gives evidence on (ii) and (iii), and then repeats the exercise on the sample for which (ii) is least likely to bind.

Self-selection and parallel trends. I mitigate concern (ii) by showing that the contract choice is predominantly determined by supply-side lender preferences rather than firm-specific unobservables. Focusing on firms with a rating at the time of the loan, I estimate a linear probability model of the form:

$$Floating_{loan} = \beta X^{observables} + \delta_{t,b,p,s} + \varepsilon_{loan}, \tag{17}$$

where $Floating_{loan}$ is a dummy for variable-rate loans and $\delta_{t,b,p,s}$ is the fixed effects interaction term for month, bank, loan purpose and firm sector. Following a variance decomposition approach, I gradually saturate the model with fixed effects. As shown in Table B.5, the lender-month dimension alone accounts for approximately 60% of the variation, with the full specification explaining up to 91% of the total variance. This suggests that the contract choice is predominantly driven by supply-side and sectoral factors rather than idiosyncratic firm risk; in fact, banks may prefer to offer floating rate contracts to match the rate structure of their own liabilities (see Kirti (2020)).

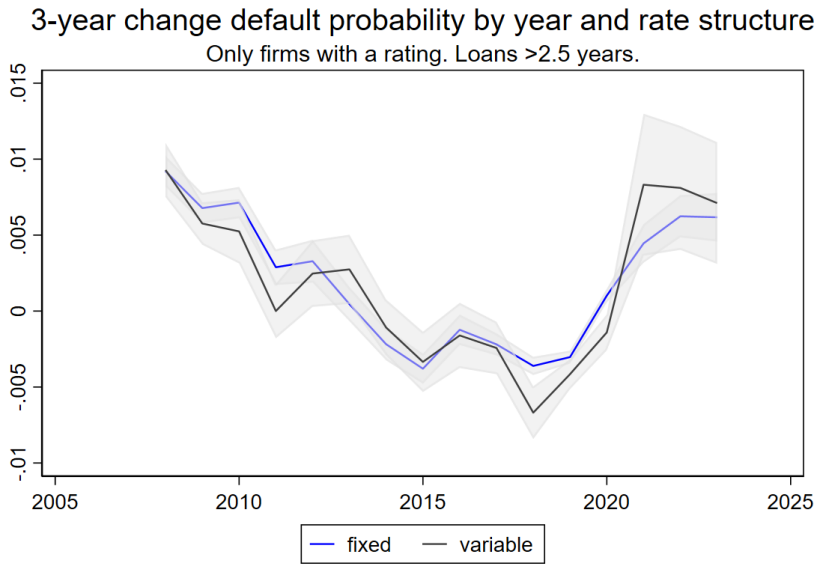
Table B.5: Variance Decomposition of Variable Rate Choice

Fixed Effects Included	**Adjusted R^2**
Month	0.04
+ Lender	0.63
+ Loan Type	0.86
+ Firm Sector	0.91

I check the balance of observable characteristics between firms taking fixed and variable rate loans. While I find that the two groups differ in some characteristics (variable: fewer firms, larger size, older, riskier, larger financial debt), I include all those characteristics as controls in the following regression, to mitigate the issue of selection bias. I also check the parallel trends assumption by plotting the evolution of rating transitions for firms taking fixed vs variable rate loans. I find that rating transitions for firms taking variable rate loans do not differ significantly from those taking fixed rate loans during the period considered. Fig. B.21 plots the two series.

This evidence overall mitigates concerns about selection bias.

Figure B.21: Mean three-year change in the default probability by year, for fixed- and floating-rate borrowers separately. Term loans over 2.5 years, firms with a rating. Shaded: 95% confidence bands.



Restrict to loans issued in 2021q3–q4. I repeat the baseline exercise in Eq. (3) on loans issued in the second half of 2021: the policy rate was still at its floor, and the tightening that followed in 2022 and 2023 came as a surprise. The crucial point is that a firm choosing between a fixed- and a floating-rate contract in those two quarters likely was not anticipating a rate increase, so restricting the timing to the end of 2021 mitigates the possibility of sorting on rate expectations. This is the version of the exercise in which the leverage channel, if it operated, would be hardest to hide. Here I lower the maturity cut to 24 months: the 2021q3–q4 window leaves too few loans at the 30-month threshold for the estimator to report standard errors on several coefficients. I run the exercise including the cumulative reference rates increase in the increase to 2022 (Table B.6), and observing the change in probability of default of firms in 2023. For robustness, I also include the rate increase in 2023, although firms' rating revision might have happened well before the end of the year (Table B.7). I again find no significant effect of reference rate increases on rating worsening.

	Default Probability - 2021q3-2021q4 only						
Variable rate (dummy)	-0.5698		-6.1935	-0.5121			
	(0.169)		(0.193)	(0.186)			
annual dRate end y2			1.8860			-0.3828*	-0.4445
			(0.261)			(0.095)	(0.218)
Variable rate (dummy) × Loan over assets				-0.6271			
				(0.249)			
annual dRate end y2 × Loan over assets						-0.3410	
						(0.228)	
annual dRate end y2 × Bank Debt share							0.0555
							(0.915)
Loan over assets	-0.5329	-1.1142	-1.1243	-0.1584	-0.6316	-0.6318	-1.1395
	(0.374)	(0.168)	(0.216)	(0.740)	(0.224)	(0.223)	(0.177)
Bank Debt share	0.0986	-0.2614	-0.2672	0.1401	-0.2234	-0.2234	-0.2780
	(0.725)	(0.316)	(0.448)	(0.631)	(0.375)	(0.375)	(0.364)
Lender × Month × Loan Type × 2digit Sector FE	✓	✓	✓	✓	✓	✓	✓
Region FE	✓	✓	✓	✓	✓	✓	✓
All other observables	✓	✓	✓	✓	✓	✓	✓
Observations	4,324	4,089	4,089	4,324	4,089	4,089	4,089
R^2	0.41	0.41	0.41	0.41	0.41	0.41	0.41

p-values in parentheses

* p< 0.10, ** p< 0.05, *** p< 0.01

Table B.6: Data: Nouveaux Crédits (BdF), 2006q1-2024q2. Term loans, maturity at least 24 months, only firms with a rating. Dependent variable: change_PD3. Exclude loans with non-specified reference rate.

	Default Probability - 2021q3-2021q4 only						
Variable rate (dummy)	-0.5698		-19.6243	-0.5121			
	(0.169)		(0.479)	(0.186)			
annual dRate end y2			5.0816			0.0935	0.6439
			(0.477)			(0.943)	(0.316)
annual dRate end y3			2.6790			-0.7103	-1.5704
			(0.563)			(0.718)	(0.179)
Variable rate (dummy) × Loan over assets				-0.6271			
				(0.249)			
annual dRate end y2 × Loan over assets						5.0753	
						(0.781)	
annual dRate end y2 × Bank Debt share							-0.6506
							(0.424)
annual dRate end y3 × Loan over assets						-7.6615	
						(0.769)	
annual dRate end y3 × Bank Debt share							0.9643
							(0.155)
Loan over assets	-0.5329	-1.1153	-1.1298	-0.1584	-0.6351	-0.6550	-1.1379
	(0.374)	(0.168)	(0.233)	(0.740)	(0.216)	(0.287)	(0.283)
Bank Debt share	0.0986	-0.2617	-0.2685	0.1401	-0.2242	-0.2431	-0.2795
	(0.725)	(0.313)	(0.569)	(0.631)	(0.369)	(0.431)	(0.674)
Lender × Month × Loan Type × 2digit Sector FE	✓	✓	✓	✓	✓	✓	✓
Region FE	✓	✓	✓	✓	✓	✓	✓
All other observables	✓	✓	✓	✓	✓	✓	✓
Observations	4,324	4,089	4,089	4,324	4,089	4,089	4,089
R^2	0.41	0.41	0.41	0.41	0.41	0.41	0.41

p-values in parentheses

* p< 0.10, ** p< 0.05, *** p< 0.01

Table B.7: Data: Nouveaux Crédits (BdF), 2006q1-2024q2. Term loans, maturity at least 24 months, only firms with a rating. Dependent variable: change_PD3. Exclude loans with non-specified reference rate.

B.8 Rates and loan quantity

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In Section 4 I deviate from the assumption of menu contracts. Table B.8 shows the evidence behind this choice. Conditional on all observable characteristics in my sample, there is a negative relationship between rates and quantity: a loan twice as large pays about 0.07 percentage points less (column 2). Loan size relative to firm assets has no significant effect on the rate (column 1).

Table B.8: Loan rates and loan size. Each column reruns the baseline regression (Table B.1, column 6) keeping one measure of loan size: loan over assets in column (1), log loan quantity in column (2). Data: Nouveaux Crédits (BdF), 2006q1–2024q2, excluding credit lines and undefined maturity products; variables defined as in Table B.1.

	Total effective rate	
	(1)	(2)
log.Loan quantity (K-Eur)		-0.0960***
		(0.000)
Loan over assets	-0.0437	
	(0.735)	
Maturity (months)	0.0012**	0.0020***
	(0.034)	(0.005)
Age	-0.0044***	-0.0038***
	(0.000)	(0.000)
Age squared	0.0000***	0.0000***
	(0.000)	(0.000)
Young firm 0-3y (dummy)	-0.0799***	-0.0653**
	(0.006)	(0.015)
New bank-firm relationship (dummy)	-0.0422***	-0.0379**
	(0.009)	(0.010)
Variable rate (dummy)	0.0907	0.1479
	(0.335)	(0.107)
Personal loan (dummy)	0.1023	0.0741
	(0.136)	(0.309)
Individual entrepreneur (dummy)	0.2617***	0.2434***
	(0.001)	(0.001)
Pledged guarantees	-0.4114***	-0.3477***
	(0.001)	(0.000)
Revenues	0.0054	-0.0030
	(0.464)	(0.630)
Bank Debt	-0.0270	-0.0249
	(0.400)	(0.411)
Non-Bank Debt	-0.0301	-0.0296
	(0.689)	(0.701)
Mobilised credit	0.0413	0.0714**
	(0.158)	(0.013)
Short Term Debt	-0.2244***	-0.1596***
	(0.000)	(0.002)
Undrawn credit lines	-0.1253	-0.0255
	(0.146)	(0.833)
Multiple products (dummy)	0.0016	0.0024
	(0.887)	(0.843)
Multiple loans, one product (dummy)	0.0124	-0.0074
	(0.608)	(0.781)
Lender × Quarter × Loan Type × 2digit Sector FE	✓	✓
Region FE	✓	✓
Group indicator, Rating FE	✓	✓
Observations	449,973	451,305
R^2	0.82	0.83

p-values in parentheses
* p< 0.10, ** p< 0.05, *** p< 0.01

C Model appendix

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C.1 Proofs of the properties of equilibria

This appendix collects the proofs of the three properties stated in Section 4.2.

Proof of Lemma 4.1. Suppose banks post some rate r with probability $m > 0$. A shopper observing two quotes equal to r splits its business between the two banks, so a bank posting r wins that firm half of the time, with probability $m/2$. If one bank reduces the rate by an arbitrarily small amount ε, it wins the remaining $m/2$ fraction of shoppers, at the lower profit for trade $r - \varepsilon$. As ε goes to zero, the deviation raises the acceptance probability by $\pi m/2$, while the profit per trade it gives up vanishes. Therefore, each bank has an incentive to lower the rate by an infinitesimally small ε and gain an additional $\pi m/2$ without giving up profits. Such a distribution cannot be an equilibrium. The argument holds also at the lowest point of the separating region, where lowering the rate just above $\bar{r}^g$ brings the good types back in the market and lowers the expected cost of lending from c_b to c^{avg,s_i}, so the deviation pays there too. ■

Proof of Lemma 4.2. Write $\bar{r}^{s_i}$ for the highest rate in the support. Because no rate is offered beyond it, $F^{s_i}(\bar{r}^{s_i}) = 1$, and by Lemma 4.1 a bank posting it loses every shopper to the rival: it sells to captive firms only, and earns $(1 - \pi)$ times its margin. Assume $\bar{r}^{s_i}$ sits in the interior of one of the regions, pooling or separating. The per-trade margin rises with the rate inside each region, so a bank can raise its rate to $\bar{r}^{s_i} + \varepsilon$, trade with the same $(1 - \pi)$ captive firms, that will accept as $\bar{r}^{s_i}$ is below the reservation rate, and earn a higher per-trade margin. At $\bar{r}^g$ or $\bar{r}^b$, on the contrary, raising the rate means losing a set of borrowers: $\bar{r}^g$, above which good types stop accepting, and $\bar{r}^b$, above which nobody accepts. The choice between the two reservation rates will depend on banks' profit expectations. ■

Proof of Lemma 4.3. Assume that the two supports were not disconnected. Take two banks offering $\bar{r}^g + \varepsilon$, with probability of acceptance of a shopper equal to $1/2$. For infinitesimally small ε, each bank could undercut its rival by offering $\bar{r}^g$: the margin would not change, but the mass of firms suddenly jumps as good firms would re-enter the market. Such deviation would be profitable. Therefore, the support of F^{s_i} must be disconnected, with a gap between the two regions. ■

C.2 Derivations for the signal-group equilibrium

This appendix derives Proposition 4.2: the two branches of the rate distribution within a signal group, the bounds of their supports, and the threshold between the hybrid and the bad-only regime. Throughout, α^{s_i} is the share of good types among firms carrying

signal s_i, $c^{avg,s_i} \equiv \alpha^{s_i} c_g + (1-\alpha^{s_i}) c_b$ is the bank's expected cost of lending to such a firm, and

$$A^{s_i} \equiv \bar{r}^g - c^{avg,s_i}, \qquad B^{s_i} \equiv (1-\alpha^{s_i})\left(\bar{r}^b - c_b\right)$$

are the two corner profits per unit of demand, so that $\phi^{s_i} = A^{s_i} - B^{s_i}$ and $\Pi^{s_i} = (1-\pi)\max\{A^{s_i}, B^{s_i}\}$.

Bank profits within a group. For firms carrying signal s_i,

$$\Pi^{s_i}(r) = \begin{cases} \left[1 - \pi + \pi(1 - F^{s_i}(r))\right](r - c^{avg,s_i}) & \text{if } r \le \bar{r}^g, \\ (1-\alpha^{s_i})\left[1 - \pi + \pi(1 - F^{s_i}(r))\right](r - c_b) & \text{if } \bar{r}^g < r \le \bar{r}^b. \end{cases}$$

Both types accept a rate in the first region, and the bank lends at the expected cost c^{avg,s_i}; only bad types accept in the second, and the bank lends at c_b to the fraction $(1-\alpha^{s_i})$ of the group that is of the bad type.

The two branches. Equilibrium requires the same profit Π^{s_i} at every posted rate. On the pooling region this gives

$$\left[1 - \pi F^{s_i}(r)\right](r - c^{avg,s_i}) = \Pi^{s_i} \qquad \Longrightarrow \qquad F^{s_i}(r) = \frac{1}{\pi}\left[1 - \frac{\Pi^{s_i}}{r - c^{avg,s_i}}\right],$$

and on the bad-only region

$$(1-\alpha^{s_i})\left[1 - \pi F^{s_i}(r)\right](r - c_b) = \Pi^{s_i} \qquad \Longrightarrow \qquad F^{s_i}(r) = \frac{1}{\pi}\left[1 - \frac{\Pi^{s_i}/(1-\alpha^{s_i})}{r - c_b}\right].$$

Under full pooling the top of the support is $\bar{r}^g$ by Lemma 4.2, so $\Pi^{s_i} = (1-\pi)A^{s_i}$ and the first expression is the pooling branch of Proposition 4.2. When $\phi^{s_i} \le 0$ the top of the support is $\bar{r}^b$ and $\Pi^{s_i} = (1-\pi)B^{s_i}$; substituting it in the first expression gives $F^{s_i}_{\text{pool}}$, and in the second it gives F_{bad}, because $\Pi^{s_i}/(1-\alpha^{s_i}) = (1-\pi)(\bar{r}^b - c_b)$ no longer involves α^{s_i}. The bad-only branch is therefore the same function in both signal groups: above $\bar{r}^g$ the bank prices against bad types only, and the probability of meeting one scales profit and the required margin in the same proportion.

Bounds and the gap. Setting each branch to zero gives the bottom of the support: $c^{avg,s_i} + (1-\pi)A^{s_i}$ under full pooling, $c^{avg,s_i} + (1-\pi)B^{s_i}$ in the hybrid regime, and $c_b + (1-\pi)(\bar{r}^b - c_b)$ in the bad-only regime. In the hybrid regime the bad-only branch starts where it reaches the mass already posted below $\bar{r}^g$, that is at the rate solving $F_{\text{bad}}(r) = F^{s_i}_{\text{pool}}(\bar{r}^g)$, which is $r = c_b + A^{s_i}/(1-\alpha^{s_i})$. The support therefore has a gap just

above $\bar{r}^g$, of width

$$c_b + \frac{A^{s_i}}{1-\alpha^{s_i}} - \bar{r}^g = \frac{\alpha^{s_i}}{1-\alpha^{s_i}}\left(\bar{r}^g - c_g\right) > 0,$$

which closes only as $\alpha^{s_i} \to 0$. Raising the rate from $\bar{r}^g$ costs the bank the good types in the group, and it posts again only once the rate compensates for that loss.

The hybrid threshold. The pooling branch carries positive mass if and only if $F^{s_i}_{\text{pool}}(\bar{r}^g) > 0$, that is $A^{s_i} > (1-\pi)B^{s_i}$, which rearranges to

$$\phi^{s_i} > -\frac{\pi}{1-\pi}\left(\bar{r}^g - c^{avg,s_i}\right) \equiv \underline{\phi}^{s_i},$$

the threshold between the hybrid and the bad-only regime of Section 4.2.2. The friction is what keeps the good types in: a bank posts in the pooling region only because a shopper may bring it the whole surplus of a good type, and as $\pi \to 0$ the threshold rises to zero and the hybrid regime disappears.

Convergence to perfect information. As the signal becomes conclusive ($q \to 1$) the posteriors converge to certainty,

$$\alpha^{s_g} = \frac{q\mu_g}{q\mu_g + (1-q)\mu_b} \to \frac{1\cdot\mu_g}{1\cdot\mu_g + 0\cdot\mu_b} = 1,$$
$$\alpha^{s_b} = \frac{(1-q)\mu_g}{(1-q)\mu_g + q\mu_b} \to \frac{0\cdot\mu_g}{0\cdot\mu_g + 1\cdot\mu_b} = 0,$$

and the two groups separate:

- **Good signal group:** $c^{avg,s_g} \to c_g$, so

  $$\phi^{s_g} \to \bar{r}^g - c_g > 0,$$

  and $F^{s_g}(r)$ converges to $F^g(r)$, the good-type distribution under a perfect signal.

- **Bad signal group:** $c^{avg,s_b} \to c_b$, so

  $$\phi^{s_b} \to \left(\bar{r}^g - c_b\right) - \left(\bar{r}^b - c_b\right) = \bar{r}^g - \bar{r}^b < 0,$$

  and $F^{s_b}(r)$ converges to $F^b(r)$, the bad-type distribution under a perfect signal.

Since $\sigma_g \to \mu_g$ and $\sigma_b \to \mu_b$, the aggregate distribution converges to

$$F(r) \to \mu_g F^g(r) + \mu_b F^b(r),$$

the perfect-signal equilibrium of **??**. At the other end, $q = 1/2$ gives $\alpha^{s_g} = \alpha^{s_b} = \mu_g$, the two groups merge, and ϕ^{s_i} collapses to the population index ϕ of the uninformative case.

D Numerical Identification Checks for the Calibration Exercise

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D.1 From the offered distribution to the observed distribution

The equilibrium distribution $F(r)$ characterized in Section 5 is each bank's own symmetric mixed-strategy *posting* rule: the distribution from which a bank draws the rate it quotes to a given firm. It is not, in general, the same object as the distribution of rates actually *paid* by borrowers, once search is taken into account. A captive firm (mass $1-\pi$) samples a single bank and pays a raw draw from $F(r)$. A shopper (mass π) samples both banks – each an independent draw from the same F, by the symmetry of the equilibrium – and accepts the lower of the two quotes. Consequently, conditional on transacting with a given bank, a shopper's paid rate is not a raw draw from F: winning requires that the offer beat an independent competing draw from F, which is more likely at low r than at high r.

Define $w(r)$ as the probability that a firm quoted rate r by a given bank actually transacts with it. A captive always accepts (contributing $1-\pi$); a shopper transacts with this bank only if the rival's draw exceeds r, an event of probability $1-F(r)$ (contributing $\pi\,(1-F(r))$). Hence

$$w(r) \;=\; (1-\pi)\cdot 1 \;+\; \pi\,(1-F(r)) \;=\; 1-\pi F(r). \tag{18}$$

At the bottom of the support $F(r)=0$ and $w(r)=1$: the cheapest possible quote always wins. At the top of the support, $F(r)=1$ and $w(r)=1-\pi$: only captives remain, as every shopper has by then found a strictly better rival quote.

Aggregating captives (who pay a raw draw from F) and shoppers (who pay the minimum of two independent draws from F, with CDF $1-(1-F(r))^2$) across both banks gives the density of *realized*, originated rates,

$$g(r) \;\propto\; f(r)\,w(r) \;=\; f(r)\,(1-\pi F(r))\,, \tag{19}$$

where $f = F'$. The normalizing constant is, remarkably, always $1-\pi/2$ regardless of the equilibrium regime (pooling, hybrid, or bad-only) governing F: because $F(r)$ is Uniform$(0,1)$-distributed whenever $r\sim F$ (the probability integral transform), $\mathbb{E}[F(r)] = 1/2$ identically, so $\int f(r)\,(1-\pi F(r))\,dr \;=\; 1-\pi/2$. Two limiting cases confirm the construction: at $\pi=0$, $g=f$ exactly (with no search, offered and realized rates coincide); at $\pi=1$, $g(r)\propto f(r)(1-F(r))$, the density of the minimum of two draws, since every

firm then shops.

For any moment $\mathbb{E}[r^k]$, this reweighting can be written as a correction to the moment computed directly under F:

$$\int r^k f(r)\, w(r)\, dr \;=\; \underbrace{\int r^k f(r)\, dr}_{\text{moment under } F} \;-\; \pi \underbrace{\int r^k f(r) F(r)\, dr}_{\text{correction term}}, \tag{20}$$

and the correction term admits a closed form in every equilibrium regime of the model (full pooling, hybrid, and bad-only), obtained by symbolic integration and verified against direct numerical quadrature. Throughout the calibration exercise of this section, the three targeted moments (mean, variance, and the rate–default correlation) are computed under $g(r)$ – properly reweighted by $w(r)$ and renormalized by $1-\pi/2$ – rather than under $F(r)$ directly, since it is $g(r)$, not $F(r)$, that corresponds to the distribution of rates actually observed in loan-level data.

Section 5.4 summarizes three complementary numerical checks used to assess whether the parameter vector recovered by my calibration procedure (Table 4) for each rating class is a well-isolated global optimum of the GMM objective, given that the model's regime-switching structure precludes a closed-form identification proof. This appendix documents each check, and its complete findings, in detail.

D.2 Local identification: rank and condition number of the moment Jacobian

At the estimated parameter vector $\hat{\theta} = (\hat{\pi}, \hat{q}, \hat{\bar{r}}_g)$ for each rating class, I compute the 3×3 Jacobian of the three matched moments (mean, variance, and rate–default correlation) with respect to the three parameters by centered finite differences, with a one-sided fallback near a parameter bound (e.g. when $\hat{q}$ is at or close to 1). I report both the raw Jacobian and an elasticity-scaled version – entry (i, j) rescaled by $\hat{\theta}_j$ divided by the observed value of moment i (or by 0.01 when the observed correlation is within 0.01 of zero, to avoid dividing by a near-zero target) – so that condition numbers are comparable in units across moments of very different natural scale (correlations of order 10^{-2} against rates and variances of order 1). Table 6 reports, for the elasticity-scaled Jacobian at each rating class's $\hat{\theta}$, the rank, the condition number (ratio of largest to smallest singular value), and the parameter with the largest loading on the singular vector associated with the smallest singular value – the "weakest identified direction."

The Jacobian has full rank (3/3) in every one of the eight rating classes, ruling out exact local non-identification. Condition numbers, however, range from 3.9 (rating 7) to 1097.0 (rating 3+), and the weakest direction loads overwhelmingly on q in six of the eight classes; the exceptions are rating 5, where π dominates, and rating 5+, where the

loading is split between q and $\bar{r}_g$. The pattern is most pronounced for rating classes whose observed rate–default correlation is close to zero (ratings 6 and 7, and to a lesser extent the safest classes): correlation is the moment most informative about screening precision q, so a target correlation close to zero leaves q comparatively unpinned by the data, even though the model remains exactly locally identified in the formal rank sense.

D.3 Global search: dense grid visualization of the parameter space

Local curvature alone cannot rule out a second, disconnected minimum elsewhere in the admissible parameter cube $\pi \in [0.05, 0.95]$, $q \in [0.5, 1.0]$, $\bar{r}_g \in [0.5\,\bar{r}_{\text{mean}}, 0.98\,\bar{r}_b]$. I complement the Jacobian check with a dense grid search (tens of thousands of parameter combinations per rating class) evaluating the GMM objective at every point, and visualize, for each rating class, the subset of the cube whose implied moments fall within a chosen multiple of the best-achieved objective, as an interactive three-dimensional scatter colored by objective value, with the reported estimate marked separately. In every rating class, this "acceptable fit" region forms a single connected neighborhood around the reported estimate rather than several disjoint clusters – visual evidence against multiple, unrelated local optima, and a small fraction of the cube in every case (between 0.02% and 0.49% of grid points). Consistent with the Jacobian findings above, the matched region is elongated primarily along the q dimension for the rating classes with high condition numbers (3+ through 4); $\bar{r}_g$ is comparatively tightly pinned in every rating class, while q remains comparatively loosely pinned even for ratings 6 and 7, whose low condition numbers reflect a well-conditioned, but locally flat, objective in the q direction – consistent with their near-zero observed correlation leaving screening precision the least informative parameter to match.

As a complementary exercise, I also compute, for each rating class, the three surfaces in $(\pi, q, \bar{r}_g)$-space along which the model exactly matches one observed moment at a time (mean, variance, or correlation individually). With three free parameters, each single scalar equation generically defines a two-dimensional surface; two such surfaces crossing is generically a one-dimensional curve, and all three crossing is generically an isolated point – the geometric counterpart of the Jacobian's rank condition. This exercise reveals that for rating 3+, the observed variance itself lies outside the model's achievable range anywhere in the admissible cube: the maximum variance the model can generate there, at any $(\pi, q, \bar{r}_g)$, is approximately 0.28, against an observed variance of 0.40. More generally, model-implied variance falls short of the data in every rating class (Table 5), so the three moment-matching surfaces do not intersect exactly anywhere in the admissible region. This is the numerical counterpart of the two-type-versus-continuum discussion in Section 5: with only two latent borrower types, the maximum rate dispersion the model

can generate is mechanically bounded, and this bound binds most severely for the safest rating classes.

D.4 Cross-checking the global optimum across equilibrium regimes

Finally, I exploit the fact that the equilibrium's regime-switching structure is fully characterized by two threshold conditions, ϕ_{good} and ϕ_{bad} – one for the pool of borrowers who receive a favorable screening signal, one for the pool who receive an unfavorable signal – each independently placing that pool in one of three regimes (full pooling, hybrid, or bad-only lending), for nine theoretically possible regime pairs. Rather than assume ex ante which pairs are empirically relevant, I tag every point of the parameter grid with its actually realized $(\phi_{\text{good}}, \phi_{\text{bad}})$ regime pair. Only a subset of the nine pairs is ever realized within the admissible parameter region for a given rating class – six of nine for ratings 3+ through 5+, and only three of nine (all with the good-signal pool fully pooling) for ratings 6 and 7 – and the realized pairs always respect a simple ordering in which the good-signal pool is never in a strictly "less pooled" regime than the bad-signal pool. This is itself a useful by-product of the exercise: the theoretically possible nine-way regime space collapses, empirically, to a much smaller and highly structured set.

For each rating class and each empirically realized regime pair, I re-solve the GMM objective with the search restricted to that regime pair, implemented by penalizing, within the local optimizer, any candidate parameter vector whose realized regime pair differs from the target. In every rating class, the best objective value achieved across *all* admissible regime pairs coincides with the parameter vector reported by my baseline, regime-unconstrained estimation (Table 4), confirming that the baseline optimizer converges to the true global optimum of the objective function rather than a locally optimal point within a single equilibrium regime.

D.5 An alternative specification: fixing π at an observable proxy for search intensity

As a further check on the calibration, I re-estimate q and $\bar{r}_g$ for each rating class holding π fixed at an external, model-free proxy for search intensity, rather than estimating it jointly: specifically, I set π equal to the share of firms in that rating class holding at least two active banking relationships, averaged over the sample period – the same underlying statistic cited in Section 5 as broadly consistent with the freely estimated $\hat{\pi}$ (on average 60% of firms across classes maintain at least two bank relationships). With π fixed at this value and r_{CB} still fixed at 1, I re-estimate $(q, \bar{r}_g)$ by the same grid-search-plus-L-BFGS-B GMM procedure used for the baseline (Table 4).

Table D.9 reports the results. The externally measured π is lower, and more compressed

across rating classes, than the freely estimated $\hat{\pi}$ (0.40–0.68 against 0.71–0.77). Fit deteriorates in all eight rating classes: the objective value rises from 0.41 to 0.62 at rating 3+, and from essentially zero to 0.47 at rating 7.

Where the deterioration falls depends on the class. In the five safest classes the variance error barely moves and the extra loss lands on the mean: the model overshoots the class mean rate by 21–33% here, against 5–16% in the baseline. In the three riskiest classes both moments worsen at once - the share of observed variance the model reproduces falls from 81%, 80% and 94% to 67%, 30% and 45%, and the mean overshoots by 33–42%. Because rate dispersion in this model scales with $\pi \times (\bar{r}_g - c_{\text{avg}})$, a lower π compresses the dispersion the model can generate for a given $\bar{r}_g$, and the optimizer compensates by raising $\bar{r}_g$, which lifts the mean.

The constraint binds hardest at rating 3+, the one class in which the observed variance lies outside the model's achievable range anywhere in the admissible $(q, \bar{r}_g)$ box at this fixed π. There $\hat{q}$ falls to its lower bound of 0.5 - no screening at all - against 1.00 in the baseline: at a low π the model buys what dispersion it can from pooling rather than from separation. In the other seven classes $\hat{q}$ stays within 0.05 of the baseline. The observed mean is unreachable in the four safest classes (3+, 3, 4+ and 4).

I read this exercise as informative rather than as grounds to prefer one specification over the other. The two-bank-relationship share is plausibly a lower bound on effective search intensity: maintaining a second banking relationship is necessary but not obviously sufficient for the kind of active rate shopping that π represents in the model, and the freely estimated $\hat{\pi}$ being somewhat higher is consistent with that interpretation. The exercise does, however, sharpen the point made throughout this appendix: the amount of rate dispersion the two-type model can generate is highly sensitive to π, and matching the variance moment closely requires π at or above roughly the level the baseline estimation recovers.

Table D.9: Alternative Specification: π Fixed at the Share of Firms with At Least Two Bank Relationships

Rating	π (fixed)	$\hat{q}$	$\hat{\bar{r}}_g$	Obj. (fixed π)	Obj. (baseline)	Var. reachable?
3+	0.600	0.500	3.055	0.62	0.41	No
3	0.651	0.985	3.709	0.45	0.35	Yes
4+	0.651	0.986	4.609	0.21	0.15	Yes
4	0.675	1.000	5.080	0.25	0.19	Yes
5+	0.670	0.999	5.417	0.25	0.20	Yes
5	0.562	0.991	6.662	0.22	0.04	Yes
6	0.399	0.552	6.347	0.64	0.04	Yes
7	0.426	0.554	8.895	0.47	0.00	Yes

Notes: π is fixed at the share of firms in each rating class with at least two active bank relationships, averaged over the sample period; q and $\bar{r}_g$ are then re-estimated with r_{CB} fixed at 1, as in the baseline. "Obj. (baseline)" reproduces the objective value from Table 4; both runs match the same moments, so the two columns are directly comparable. "Var. reachable?" indicates whether the observed variance falls inside the model's achievable range anywhere in the admissible $(q, \bar{r}_g)$ box at this fixed, lower π.

E Robustness: counting defaults as bad types

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The baseline calibration of Section 5 labels a firm bad when its Banque de France rating deteriorates over the three years following origination. Firms that default outright are not counted as bad. This appendix repeats the exercise under the alternative definition, in which the bad group also contains firms that default. Everything else - the rate regression, the residuals, the targeted mean and variance, $\bar{r}_b$, and p_g - is held fixed, so the two runs differ only through the three quantities the reclassification touches: the share of bad types μ_b, their success probability p_b, and the rate–default correlation.

Table E.10 shows how much those three move. For ratings 3+ through 5+ the reclassification is almost immaterial: μ_b shifts by at most 0.06 and p_b not at all. In the three riskiest classes it is large: μ_b roughly quintuples at rating 6 and sextuples at rating 7, and p_b collapses, because a firm labelled bad on the strength of its own default has, by construction, a default probability of one.

The estimates are unchanged where the definition is. For ratings 3+ through 5+, which account for the bulk of the sample, the two runs agree almost exactly. Comparing Table E.11 with Table 4, across those five classes $\hat{\pi}$ differs by at most 0.001, $\hat{q}$ by at most 0.021, and the objective value by at most 0.005; the fitted means and variances in Table E.12 reproduce Table 5 to two digits. The estimate of competition, which is the paper's main quantitative result, is therefore not an artifact of how the bad type is defined: $\hat{\pi}$ lies in $[0.74, 0.77]$ here against $[0.71, 0.77]$ in the baseline.

Table E.10: Inputs that change under the alternative bad-type definition

	μ_b		p_b		Rate–default corr.	
Rating	Baseline	With default	Baseline	With default	Baseline	With default
3+	0.846	0.846	0.997	0.997	0.025	0.023
3	0.749	0.751	0.996	0.996	0.033	0.034
4+	0.739	0.745	0.992	0.992	0.032	0.022
4	0.671	0.691	0.981	0.981	0.048	0.037
5+	0.491	0.549	0.965	0.965	0.048	0.044
5	0.268	0.415	0.908	0.630	0.017	0.049
6	0.066	0.344	0.784	0.200	0.000	0.068
7	0.069	0.446	0.630	0.200	0.000	0.000

Notes: "Baseline" is the definition used in Section 5, in which bad types are firms whose rating deteriorates; "With default" adds firms that default. For ratings 6 and 7 the raw data give a default frequency of exactly one in the bad group, which would imply an infinite cost of lending $c_b = r_{CB}/p_b$; p_b is therefore floored at 0.2, so the reported values for those two classes are the floor rather than an estimate.

Table E.11: Search frictions and information by rating class, defaults counted as bad

Rating	$\hat{\pi}$	$\hat{q}$	$\hat{\bar{r}}_g$	Info Contr. (%)	Obj. Value $Q(\theta)$
3+	0.772	0.988	3.461	4.58	0.41
3	0.759	1.000	3.703	4.76	0.35
4+	0.737	0.978	4.609	2.83	0.16
4	0.758	1.000	5.092	3.81	0.19
5+	0.750	1.000	5.416	2.99	0.19
5	0.737	0.757	6.662	1.01	0.06
6	0.767	0.613	7.057	12.79	0.24
7	0.754	0.526	9.350	0.37	0.20

Notes: Columns as in Table 4, which reports the same quantities under the baseline definition.

Table E.12: Model fit, defaults counted as bad

	Mean		Variance		Correlation	
Rating	Data	Model	Data	Model	Data	Model
3+	1.67	1.94	0.406	0.153	0.023	0.023
3	1.83	2.08	0.443	0.188	0.034	0.034
4+	2.40	2.53	0.553	0.334	0.022	0.022
4	2.37	2.63	0.730	0.426	0.037	0.037
5+	2.57	2.81	0.867	0.494	0.044	0.043
5	3.36	3.51	0.985	0.740	0.049	0.049
6	3.63	4.15	1.137	0.600	0.068	0.070
7	4.79	5.39	1.843	1.064	0.000	0.003

Notes: Moments defined as in Table 5.

The three riskiest classes are where the definitions part, and where the alternative is least reliable. At rating 5, $\hat{q}$ falls from 0.94 to 0.76; at rating 6 it rises from 0.53 to 0.61. The fit deteriorates sharply: the objective rises from 0.04 to 0.24 at rating 6 and from 0.00 to 0.20 at rating 7, the model mean overshoots the data by 14% and 13% against 1% and 0% in the baseline, and the share of observed variance the model reproduces falls from 80% and 94% to 53% and 58%.

Two features of the exercise explain this, and both argue for keeping the baseline definition in the text. First, the definition is partly circular in these classes: a firm counted as bad because it defaulted has a default probability of one by construction, so p_b measures the labelling rule rather than the risk of the bad type. Second, the floor imposed on p_b to keep c_b finite binds at exactly 0.2 for ratings 6 and 7, which fixes $c_b = r_{CB}/p_b = 5$ against an upper bound $\bar{r}_b$ of 7.2 and 9.5. That gap is wide enough for information to generate more dispersion than the data contain, which is what produces the 12.8% information contribution at rating 6 - the one figure in either run that exceeds the 5% ceiling reported in Section 5. Excluding rating 6, the largest information contribution here is 4.8%, in line with the baseline. The same floor also makes the implied cross-subsidy at rating 6 unusually large: bad types pay an expected rate of 4.2 against a cost of $c_b = 5$, so the bank runs a loss of 0.8 per bad-type loan, covered by the surplus it earns on the good types pooled with them. I therefore read the rating-6 figure as an artifact of the floor rather than as evidence that screening contributes more to dispersion than the baseline implies.

Table E.13: Redistribution from a perfect signal, defaults counted as bad: percent of the rating class's total surplus

Rating	Bad types	Banks	Good types
3+	-0.893	0.883	0.010
3	-0.881	0.868	0.012
4+	-0.817	0.795	0.020
4	-0.747	0.693	0.052
5+	-0.640	0.547	0.094
5	-1.901	0.430	1.471
6	-12.231	0.396	11.834
7	-9.450	0.506	8.942

Notes: change in each party's expected surplus when signal precision moves from $q = 0.5$ to $q = 1$, as a percentage of the class's maximum total surplus. Rows sum to zero: every class is in full pooling at $q = 0.5$, so no firm is rationed at either end and information redistributes without creating surplus.

Table E.14: Redistribution from a perfect signal, defaults counted as bad: percent of each party's own surplus

Rating	Bad types	Banks	Good types
3+	-1.65	2.43	0.10
3	-1.87	2.29	0.08
4+	-1.83	1.94	0.14
4	-1.73	1.81	0.28
5+	-1.89	1.39	0.35
5	-7.61	1.04	4.35
6	-55.09	1.06	29.28
7	-33.82	1.30	27.02

Notes: the same changes, each divided by that party's own surplus at $q = 0.5$ rather than by the class total. Rows do not sum. This is the denominator relevant to a bank weighing the cost of acquiring precision, and to a borrower judging the size of the repricing.

F Stability of the calibrated precision

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Here I derive the two conditions of Section 6 and I evaluate them numerically in each rating class j.

Let $C_s \equiv C_s(q{=}1) - C_s(0.5)$ be the price of a perfect signal over no signal. A bank is of measure zero, so a deviation leaves unchanged both the rivals' rate distributions and the population pool (μ_g, μ_b). I restrict to deviations to and from the extremes of perfect signal and no signal, as the calibration returns these two corners. Precision affects the cost of lending via borrowers' risk: $c_\theta(1) = (1 - \delta_j)c_\theta(0.5)$ as in (7), with saving $\kappa_\theta = \delta_j c_\theta(0.5)$ per unit lent, and $\Delta r \equiv \bar{r}^b - \bar{r}^g$.

F.1 Deviation from the uninformed market

The market sits at $q = 0.5$ and banks lend at the un-monitored pooled cost $c_0^{avg} \equiv \mu_g c_g(0.5) + \mu_b c_b(0.5)$. The equal-profit condition of **??** gives

$$\bar{\Pi} = (1 - \pi)\left(\bar{r}^g - c_0^{avg}\right), \qquad \underline{r} = (1 - \pi)\bar{r}^g + \pi c_0^{avg}, \qquad 1 - \pi F(r) = \frac{\bar{\Pi}}{r - c_0^{avg}}.$$

The deviator acquires a signal at cost C_s: it recognizes each type and lends at the monitored cost $c_\theta \equiv c_\theta(1)$. It chooses the rate distribution that maximizes its profits, given the firms' type and the rivals' offers.

Conditional on meeting a good type, the deviator makes maximum profits on the pooling region, where it earns $\Pi_g(r) = \mu_g \bar{\Pi}\,(r - c_g)/(r - c_0^{avg})$. Note that profits on the pooling region are not flat: the deviator has a different cost of lending $c_g < c_g(0.5)$ with respect to the uninformed rivals; I expect a corner solution. The slope in r carries the sign of

$c_g - c_0^{avg} = -\left[\mu_b(c_b - c_g) + \kappa^{avg}\right] < 0$, with $\kappa^{avg} \equiv \mu_g\kappa_g + \mu_b\kappa_b$: the deviator undercuts the whole support and posts $\underline{r}$, which every good type accepts.

For bad types, assume the deviator is also posting on the overlapping region, with profits $\Pi_b(r) = \mu_b\bar{\Pi}\,(r - c_b)/(r - c_0^{avg})$. The sign of the slope is ambiguous, so I compute it at the calibrated parameters, and it is positive. The deviator has an incentive to increase the rate and sells to captives only, and posts $\bar{r}^b$.[10] The deviator's profits are:

$$\Pi^{dev}(1) = \mu_g\,(\underline{r} - c_g) + \mu_b(1-\pi)\left(\bar{r}^b - c_b\right), \qquad \underline{r} - c_g = (1-\pi)(\bar{r}^g - c_g) + \pi\left[\mu_b(c_b - c_g) + \kappa^{avg}\right],$$

and defining G_j as the net $\Pi^{dev}(1) - \Pi^*(0.5) = \Pi^{dev}(1) - \bar{\Pi}$ gives the stability condition in (8) (Condition A). The gain is made of three terms: the surplus from winning more good types, the surplus lost from pooling bad types with good ones at low rates, and the monitoring saving. The monitoring effect reinforces the gain of the deviator with respect to the non-monitoring case.

In each class j, holding constant the calibrated parameters, Condition A is linear in δ_j:

$$G_j(\delta_j) = G_{0,j} + \delta_j S_j \leq C_s \quad \Longleftrightarrow \quad \delta_j \leq \frac{C_s - G_{0,j}}{S_j}, \tag{21}$$

where $G_{0,j} = \pi\mu_g\mu_b\Delta c_0 + (1-\pi)\mu_b\Delta r$ is the gain independent of monitoring, $\Delta c_0 = c_b(0.5) - c_g(0.5)$ and $S_j = c_0^{avg}\left[(1-\pi) + \pi\mu_g\right] - \pi\mu_g\mu_b\Delta c_0$.

F.2 Deviation from the informed market

The market sits at $q = 1$: banks are perfectly informed, they price off the monitored costs, so $\bar{\Pi}_\theta = (1-\pi)(\bar{r}^\theta - c_\theta)$ and the type-θ acceptance probability is $A_\theta(r) = 1 - \pi F^\theta(r) = \bar{\Pi}_\theta/(r - c_\theta)$, on $\underline{r}^\theta, \bar{r}^\theta$.

I consider a bank that stops acquiring the signal, and saves the cost C_s. Its clients being riskier, it lends at $c_\theta + \kappa_\theta$, and competes with the offers of the informed banks. If the deviator posts a rate on the overlapping region $\underline{r}^b, \bar{r}^g$, on type θ borrowers it earns:

$$A_\theta(r)\,(r - c_\theta - \kappa_\theta) = \bar{\Pi}_\theta - \kappa_\theta A_\theta(r).$$

The first term is profits before introducing monitoring, which are flat by the equal-profit condition: an uninformed bank posting inside the overlap collects the informed profit. The second is the monitoring penalty, paid on every loan actually made and therefore smallest where acceptance is lowest. The deviator cannot distinguish types and sets one rate for all. Due to the monitoring cost, profits are increasing in r and the deviator

[10] The slope is $\mu_g(c_b - c_g) - \kappa^{avg}$. The deviator prefers to undercut if monitoring is strong enough to push the deviator's own bad-type cost below the uninformed market average cost.

chooses either to price at the top of the overlap or in the bad-only region. At the top of the overlap, $r^d = \bar{r}^g$, where $A_g = 1 - \pi$ and $A_b = \bar{\Pi}_b / w$ with $w \equiv \bar{r}^g - c_b$; the deviator's profits are lower by the higher cost of lending. Otherwise, it posts anywhere on the bad-only region, where good types reject, and profits on bad types are the profits of informed banks net of the higher cost penalty. In general

$$L_j = \min \left\{ \mu_g \kappa_g (1-\pi) + \mu_b \kappa_b \frac{\bar{\Pi}_b}{w} \ , \quad \mu_g \bar{\Pi}_g + \mu_b (1-\pi) \kappa_b \right\}. \tag{22}$$

The former is the loss L_j of (9). With no monitoring both branches collapse to $L_j = 0$, which is the statement in Section 6 that dropping the signal is free.

In each class j, holding constant the calibrated parameters, I aggregate the terms that do not depend on δ_j and I write $t \equiv \delta_j / (1 - \delta_j)$, so (22) becomes $L_j = \min\{K_j t, \ M_j + K'_j t\}$ with

$$K_j = \mu_g c_g(1)(1-\pi) + \mu_b c_b(1) \frac{\bar{\Pi}_b}{w}, \qquad K'_j = \mu_b (1-\pi) c_b(1), \qquad M_j = \mu_g \bar{\Pi}_g.$$

The equilibrium is stable if the loss is larger than the cost saving. The calibration parameters are consistent with L_j on the pooling branch, where the constraint is written:

$$C_s \leq L_j(\delta_j) \quad \Longleftrightarrow \quad \delta_j \geq \frac{C_s}{K_j + C_s}. \tag{23}$$

This means that a high enough monitoring cost is necessary to satisfy the condition. At the calibrated parameters, as δ_j increases, the bad-only branch overtakes the pooling one - from $\delta_j = 0.68$ in class 3+ and 0.79 in class 4, well above the values below - and both branches rise in δ_j, so for any given cost there is one minimum threshold.

F.3 The two conditions at the calibrated parameters

Table F.15 collects the coefficients of the two conditions above, (21) and (23). The two binding classes read $\delta_7 \leq (C_s - 0.0291)/0.996$ and $\delta_{3+} \geq C_s/(0.234 + C_s)$.

Each class also faces a feasibility bound: a screening class needs $c_b(0.5) < \bar{r}^g$ for its pooling regime to survive, which caps δ_j at $1 - c_b(1)/\bar{r}^g$, and a non-screening class needs $p_g(1) = \hat{p}_g/(1-\delta_j) \leq 1$, which caps it at $1 - \hat{p}_g$. Table F.16 reports the resulting intervals at $C_s = 0.0483$, the cost at which the two bands of Fig. 5 come closest. Every interval is non-empty, so all eight estimates are stable at that cost, but the floors of the first six lie an order of magnitude above the ceilings of the last two.

$C_s = 0.0483$ is also where the ceiling on the two non-screening classes changes hands. Figure F.22 resolves that band: class 7's Condition A binds below 0.0483, class 6's probability ceiling above it, and the ratio between the two bands is smallest exactly at the

Table F.15: Coefficients of the two stability conditions, by rating class

Rating	Role	c_b	$\bar{\Pi}_b/w$	K_j	M_j
3+	screening	1.003	0.235	0.234	0.087
3	screening	1.004	0.248	0.247	0.164
4+	screening	1.008	0.269	0.269	0.246
4	screening	1.020	0.252	0.252	0.324
5+	screening	1.036	0.258	0.260	0.562
5	screening	1.102	0.278	0.284	1.120
				$G_{0,j}$	S_j
6	not screening	1.275	-	0.014	0.976
7	not screening	1.587	-	0.029	0.996

Notes: computed at the estimates of Table 4 and at the calibrated μ_b, p_g, p_b and $\bar{r}^b$, with $r^{CB} = 1$. "Screening" classes are those the calibration estimates at $\hat{q} \approx 1$, to which Condition B applies; the other two are estimated at $\hat{q} = 0.53$ and Condition A applies to them. c_b is the monitored cost $c_b(1)$ for a screening class and the un-monitored $c_b(0.5)$ for the other two, since each class reports the default probability of its own equilibrium. K_j and M_j are the coefficients of L_j in $t = \delta_j/(1-\delta_j)$; $G_{0,j}$ and S_j the intercept and the slope of G_j in δ_j. K'_j, the slope of the bad-only branch of (22), runs from 0.080 at rating 5 to 0.195 at rating 4+ and never binds in the relevant range.

Table F.16: Monitoring strength consistent with the estimated precision, at $C_s = 0.0483$

Rating	Role	From	To	Upper end set by
3+	screening	**0.171**	0.710	pooling regime, $c_b(0.5) < \bar{r}^g$
3	screening	0.164	0.729	pooling regime
4+	screening	0.152	0.781	pooling regime
4	screening	0.161	0.798	pooling regime
5+	screening	0.157	0.808	pooling regime
5	screening	0.145	0.835	pooling regime
6	not screening	0	**0.0193**	$p_g(1) \leq 1$; Condition A alone allows 0.0351
7	not screening	0	**0.0193**	Condition A; $p_g(1) \leq 1$ allows 0.0347

Notes: lower ends from (23), upper ends from (21) and from feasibility, at the estimates of Table 4. The binding pair is $\delta_j = 0.171$ in the six screening classes against $\delta_j = 0.0193$ in ratings 6 and 7, a ratio of 8.8. Putting the screening classes at their lower ends and the other two at their upper ends is what minimizes the spread, so these two endpoints answer both questions at once: which condition binds, and how uniform the technology can be. At those values the saving per unit lent is $\kappa_g \in [0.206, 0.210]$ and $\kappa_b \in [0.207, 0.227]$ in the screening classes, raising the bad type's repayment probability by 15.5 to 17.0 points, against $\kappa_g = 0.020$, $\kappa_b \in [0.025, 0.031]$ and a gain of 1.2 to 1.5 points in ratings 6 and 7.

handover. Raising C_s past it buys no further uniformity, because class 6's ceiling is flat in C_s. Lowering C_s below 0.0291 leaves class 7 with no admissible δ_j at all.

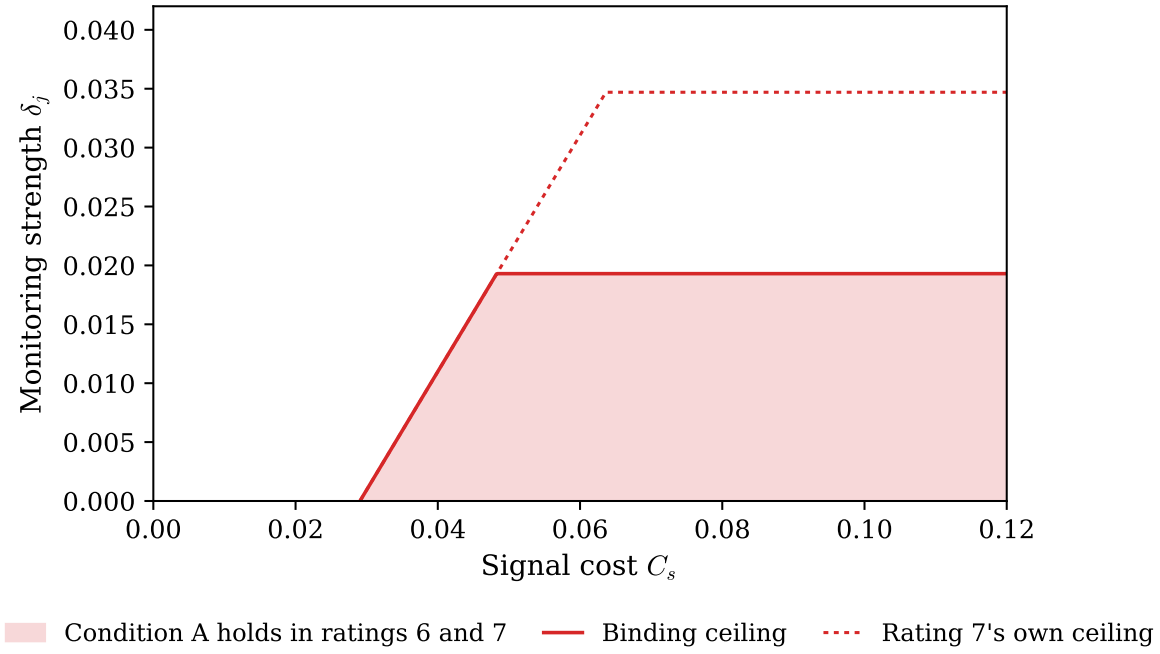


Figure F.22: **Ceiling on monitoring strength in the two non-screening classes, against the cost of the signal.** The lower band of Fig. 5, on a scale that resolves it; axes as there. The solid line is the binding ceiling, the lower of class 7's Condition A and class 6's probability cap $p_g(1) \leq 1$. The dotted line is class 7's own ceiling, which carries on above the band once class 6 takes over at $C_s = 0.0483$ and reaches its own probability cap of 0.0347 at $C_s = 0.0636$.

F.4 Deviations to intermediate precisions

The two conditions compare the corners, which is what the calibration delivers, but a cost *schedule* in the window they define also blocks every deviation in between. Let $\Delta_j(q)$ be the gain to a deviator at precision q against a market at $q = 0.5$, and $L_j(q)$ the loss from dropping to q against a market at $q = 1$, with Δ_j increasing and L_j decreasing. For any C_s satisfying both conditions,

$$C_s(q) - C_s(0.5) \;=\; \max\left\{ \max_{j \text{ not screening}} \Delta_j(q) \;,\;\; C_s - \min_{j \text{ screening}} L_j(q) \;,\;\; 0 \right\}$$

is increasing, equals 0 at $q = 0.5$ and C_s at $q = 1$, and by construction exceeds $\Delta_j(q)$ in every non-screening class while leaving less than $L_j(q)$ on the table in every screening class. One number, and one schedule through it, therefore block every deviation in every class.